GSSSB · STAFF NURSE EXAM

Fundamentals of Nursing

Complete Colourful Notes · 3D Diagrams · Mnemonics · Exam Points



20 Chapters | 3D Coloured Diagrams | Procedures step-by-step | Normal values | Drug calculations | Exam Points (PYQ style)

Hinglish explanation + English exam terms



Sister Asha
Aapki Nursing teacher

Index – Kya padhenge?

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Health concepts & models, levels of care, hospital & ward, admission, transfer, discharge, LAMA, safety, falls, restraints, fire

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Principles, CPR/BLS, choking, bleeding & haemorrhage control, shock, fractures & splints, burns, poisoning, bites, drowning, heat/cold, fainting, seizures, triage

19 **Care of Unconscious, Peri-operative, Terminally Ill & Dead Patient**
Unconscious patient care, pre & post-op basics, sensory alterations, stress & coping, loss & grief, dying patient, signs of death, last office, organ donation

20 **Quick Revision – Master Tables & 150 One-Liners**
All numbers in one place – temperatures, volumes, lengths, angles, times, positions, colours, persons & years; 150 exam one-liners

★ **Key Points** – sabse important facts

Asha Didi ki Tip – samjhane wali baat

! **Nursing Alert** – galti mat karna

✎ **Exam Point** – PYQ style, baar baar aata hai

♫ **Mnemonic** – yaad karne ki trick

D Definition – exact definition

1 Nursing as a Profession – History, Ethics, Law & Theories

1.1 Nursing – Definitions

D Famous definitions (exam me directly poochhe jaate hain)

- **Virginia Henderson (1966 – ICN accepted)**: "The unique function of the nurse is to **assist the individual, sick or well**, in the performance of those activities contributing to health or its recovery (or to peaceful death) that he would perform unaided if he had the necessary strength, will or knowledge, and to do this in such a way as to help him **gain independence as rapidly as possible**." → "**Mother of modern nursing definition**"; **14 basic needs**.
- **Florence Nightingale (1859, "Notes on Nursing")**: Nursing is "to put the patient in the **best condition for nature to act** upon him." (Environment theory)
- **ICN (2002)**: Nursing encompasses autonomous and collaborative care of individuals of all ages, families, groups and communities, sick or well, and in all settings.
- **ANA (American Nurses Association)**: Nursing is the **diagnosis and treatment of human responses** to actual or potential health problems.
- "Nurse" word – Latin **"Nutrix"** = to nourish / nurture.

Nursing as a profession – characteristics

- Well-defined **body of knowledge**, **long education**, **service orientation** (altruism), **code of ethics**, **autonomy**, **professional organization**, **licensure/registration**, research-based practice.
- **Registration** – State Nursing Council (Gujarat me – **Gujarat Nursing Council**) – **license to practice**; renewal periodically.
- **Accreditation** – institution ko recognition (INC).
- **Certification** – specialty me competence.

1.2 Florence Nightingale – The Lady with the Lamp

Florence Nightingale – must-know facts

- Born **12 May 1820, Florence (Italy)**; died **13 August 1910**, London. Parents – British.
- **Crimean War (1854–56)** – Scutari Barrack Hospital (Turkey) – mortality **42 % → 2 %** by sanitation & hygiene. Soldiers called her "**Lady with the Lamp**" (night rounds). Also "**Angel of Crimea**".
- **1860 – Nightingale Training School for Nurses, St. Thomas' Hospital, London** – first secular nursing school → "**Founder of Modern Nursing**".
- Books: **"Notes on Nursing: What it is and what it is not" (1859)**, "Notes on Hospitals" (1859).
- **Statistician** – invented **Polar area diagram (Coxcomb chart / rose diagram)**; first woman **Fellow of the Royal Statistical Society (1858)**.
- Theory: **Environmental theory** – 10 components (ventilation, warmth, light, noise, cleanliness, bed & bedding, personal cleanliness, variety, food, observation).
- **Order of Merit (1907)** – first woman. **Nightingale Pledge (1893)** – written by **Lystra Gretter** (Farrand Training School, Detroit) – not by Nightingale herself.
- **12 May = International Nurses Day** (ICN since 1974). **Florence Nightingale Award (India)** – by President of India on 12 May.
- Also helped Indian sanitation – **Royal Commission on Health of Army in India (1859)**.

Other pioneers

Person	Contribution
Hippocrates	**Father of Medicine**; Hippocratic oath
Phoebe	**First visiting nurse / deaconess** (Christian era)
Fabiola	Founded first **free hospital** in Rome (390 AD)
St. Vincent de Paul	Sisters of Charity (1633)
Theodor Fliedner	**Kaiserswerth Deaconess Institute (1836)**, Germany – Nightingale trained here
Clara Barton	Founded **American Red Cross (1881)**
Henry Dunant	Founded **International Red Cross (1863, Geneva)**; first Nobel Peace Prize (1901); birthday **8 May = World Red Cross Day**
Linda Richards	**First trained nurse of America**
Mary Seacole	Jamaican nurse in Crimean war
Isabel Hampton Robb	Founded American Nurses Association
Lillian Wald	Founder of **public health nursing** (Henry Street Settlement)
Edith Cavell	British WWI nurse, executed for helping soldiers escape
Virginia Henderson	"**First lady of nursing**", "Modern-day Nightingale"
Hildegard Peplau	"**Mother of psychiatric nursing**"
Dorothea Dix	Mental health reform (asylums)

1.3 History of Nursing in India

Year	Event
1664	**First hospital in India – Fort St. George, Madras (Chennai)** by East India Company (military)
1854	Madras Govt. started training of **midwives/ nurses** at Govt. General Hospital, Madras
1867	**St. Stephen's Hospital, Delhi** – first to train Indian women as nurses
1871	**First school of nursing** – Govt. General Hospital, **Madras** (6-month diploma, midwives)
1905	**Association of Nursing Superintendents of India** formed
1908	**TNAI formed** (Association of Nursing Superintendents + Trained Nurses Association)
1906	**Lady Minto's Indian Nursing Association**
1922	**TNAI became member of ICN**
1926	**Madras State** – first **State Registration Council**
1943	Bhore Committee appointed (report **1946**)
1946	**First B.Sc Nursing** – College of Nursing, Delhi (RAK College of Nursing) & **CMC Vellore**
1947	**Indian Nursing Council (INC) Act – 31 Dec 1947**; INC constituted **1949**
1959	First **M.Sc Nursing** – RAK College of Nursing, Delhi
1986	**Ph.D (Nursing)** programme started – RAK College of Nursing, Delhi
2023	**National Nursing & Midwifery Commission Act** – INC ki jagah naya commission (transition)

- **Indian Nursing Council (INC)** – **statutory body** under **INC Act 1947**, under Ministry of Health & Family Welfare; **uniform standards of nursing education**; prescribes syllabus (GNM, ANM, B.Sc, M.Sc); recognizes institutions. HQ – **New Delhi**. President – elected; ex-officio members include DGHS, Nursing Advisor.
- **TNAI** (Trained Nurses' Association of India) – **1908**, **professional organization**; journal – "**The Nursing Journal of India**" (since 1910); HQ New Delhi.
- **ICN (International Council of Nurses)** – **1899, Geneva**, first international organization of health professionals; founder – **Ethel Gordon Fenwick**; **ICN Code of Ethics** (first 1953, latest **2021**).
- **SNA** – Student Nurses Association (1929) – TNAI ka part.
- **Nursing Advisor to Govt. of India** – DGHS ke under. **Florence Nightingale Award** – instituted **1973**.

PYQ style

- First school of nursing in India – **Govt. General Hospital, Madras (1871)**; first hospital – **Fort St. George, Madras (1664)**.
- INC Act passed in – **1947**. TNAI formed in – **1908**. ICN – **1899**.
- "Notes on Nursing" written by – **Florence Nightingale (1859)**.
- Father of medicine – **Hippocrates**; father of modern medicine – **William Osler**; father of microbiology – **Louis Pasteur / Antonie van Leeuwenhoek**; father of surgery – **Sushruta** (India) / Ambroise Paré (modern).
- Red Cross symbol – red cross on **white** background (Swiss flag reversed).

1.4 Qualities, Roles & Functions of a Nurse

- **Qualities**: kindness, empathy (not sympathy), honesty, patience, tact, observation skill, sense of responsibility, punctuality, good health, neat appearance, confidentiality, good communication.
- **Empathy** = patient ki feelings ko **samajhna** ("uske jagah khud ko rakh ke") – therapeutic. **Sympathy** = dukh me **sath dukhi** hona – non-therapeutic.
- **Roles**: care-giver, **advocate** (patient ke rights ki raksha – **most important legal-ethical role**), educator, counsellor, communicator, manager, researcher, leader, change agent, coordinator, case manager, consultant.
- **Functions**: **Independent** (nurse ka apna – positioning, health teaching, hygiene), **Dependent** (doctor's order – drugs, IV), **Interdependent/collaborative** (team ke saath).
- **Nursing care delivery models**: **Functional** (task-oriented – one nurse = one task, e.g. medication nurse), **Team nursing** (team leader + members), **Primary nursing** (one nurse **24-hr responsibility** of patient from admission to discharge), **Total patient care / case method** (one nurse – all care during shift; oldest), Case management, Patient-focused care.

1.5 Ethics in Nursing

D Ethics

Ethics = moral principles – kya **sahi** hai aur kya **galat** (right vs wrong conduct). **Law** = legally binding rules. **Morals** = personal beliefs. **Values** = beliefs jo behaviour guide karte hain. **Bioethics** = health care ki ethics.

Ethical principles

Principle	Meaning	Example
Autonomy	Patient ka **apna decision lene ka adhikar** (self-determination)	Patient treatment refuse kar sakta hai; **informed consent**
Beneficence	**Bhalaai karna** (do good)	Pain relief dena
Non-maleficence	**Nuksaan na karna** ("**do no harm**" – primum non nocere)	Side rails up, galat dawa na dena
Justice	**Fairness** – sabko barabar care	Ameer-gareeb me farak nahi
Veracity	**Truthfulness** – sach bolna	Patient ko sahi information
Fidelity	**Promise nibhana**, loyalty	"15 min me aaungi" – to aana
Confidentiality	Patient ki information **gupt** rakhna	HIV status kisi ko na batana
Accountability	Apne kaam ki **zimmedari** lena	Galti report karna
Respect for persons / dignity		Privacy maintain karna

"A Big Nurse Just Visited Friends Carefully"

Autonomy · **B**eneficence · **N**on-maleficence · **J**ustice · **V**eracity · **F**idelity · **C**onfidentiality

- **Ethical dilemma** – do equally (un)desirable options ke beech choose karna (e.g. terminally ill patient – euthanasia request).
- **Euthanasia** – **mercy killing**. **Active** (lethal injection – **illegal in India**) vs **Passive** (life support withdraw) – **Supreme Court (Aruna Shanbaug case, 2011)** – passive euthanasia allowed under guidelines; **Common Cause vs Union of India (2018)** – **living will / advance directive** valid; simplified procedure **2023**.
- **ICN Code of Ethics (2021)** – 4 elements: **Nurses & patients/people**, **Nurses & practice**, **Nurses & the profession**, **Nurses & global health**. Fundamental responsibilities of nurse: **promote health, prevent illness, restore health, alleviate suffering**.
- **INC Code of ethics / Code of professional conduct** – respect, confidentiality, competence, accountability.

1.6 Legal Aspects of Nursing

Term	Meaning
Negligence	Expected standard of care na dena (**omission or commission**) – unintentional; e.g. side rails up na karna → fall
Malpractice	**Professional negligence** – professional dwara negligence
Elements of malpractice (4 D's)	**D**uty, **D**ereliction (breach) of duty, **D**amage (injury), **D**irect cause (causation)
Tort	Civil wrong (not crime) – **intentional** (assault, battery, false imprisonment, defamation, invasion of privacy) / **unintentional** (negligence)
Assault	**Threat** to touch / harm (without touching) – "Dawa nahi li to injection lagaungi"
Battery	**Actual touching without consent** – consent ke bina procedure/ injection
False imprisonment	Bina legal authority ke patient ko rokna – **unnecessary restraints**, discharge rokna (bill ke liye)
Defamation	Kisi ki reputation ko nuksaan – **Libel** (written) / **Slander** (spoken)
Invasion of privacy	Bina permission photo, information share
Crime	Against society/state – e.g. intentional overdose, theft of narcotics
Res ipsa loquitur	"**The thing speaks for itself**" – e.g. sponge left in abdomen
Respondeat superior	"**Let the master answer**" – employer bhi responsible for employee's negligence (vicarious liability)
Good Samaritan law	Emergency me **free/ voluntary help** karne wale ko **legal protection** (India – **Good Samaritan guidelines 2015/ MV Amendment Act 2019 Section 134A**)
Statute of limitations	Case file karne ki time limit
Subpoena	Court me hazir hone ka order
Affidavit	Written statement under oath
Plaintiff / Defendant	Case karne wala / jis par case

Consent

- **Informed consent** – patient ko procedure ka **nature, purpose, risks, benefits, alternatives** samjha ke, **voluntary**, **written**, **signed**.

- Valid consent ke liye: **adult (≥ 18 yrs)**, **mentally competent**, **conscious, not under sedation/ drugs**, **voluntary (no force)**, **informed**, language samajh me aaye.
- **Minor (< 18)** – parent/ guardian; **emergency (life-threatening)** – **implied consent** (no time) – 2 doctors' opinion; unconscious – nearest relative.
- **Responsibility**: explain karna – **doctor/surgeon**; nurse – **witness** & check ki patient ne samjha hai. Doubts ho to nurse doctor ko inform kare.
- Consent **anytime withdraw** ho sakta hai.
- Types: **Express** (oral/ written) vs **Implied** (action se – arm forward karna injection ke liye).

! Legal protection – nurse kya kare?

- **Accurate, timely, legible documentation** ("not documented = not done")
- **Follow hospital policy**, doctor's written order (verbal order – emergency only, **read back**, sign within 24 hrs)
- Galat/ doubtful order → **clarify with doctor** (blindly follow nahi)
- **Incident report** – fill karo (patient record me nahi lagate, mention nahi karte)
- Know limits – **scope of practice**, continuing education, **professional liability insurance**
- Patient's belongings/ valuables – documented, witnessed handover

Patient's Bill of Rights (AHA 1973) / Charter of Patients' Rights (NHRC, India 2018)

Right to information, records & reports, **emergency medical care**, informed consent, confidentiality, **dignity & privacy**, second opinion, transparency in rates, non-discrimination, safety & quality, choose alternative treatment, choose source of medicines/tests, proper referral & transfer, protection in clinical trials, discharge (body cannot be detained for bills), patient education, grievance redressal.

1.7 Nursing Theories

Theorist	Theory / Model	Key concept
Florence Nightingale	**Environmental theory** (1859)	Environment ko modify karke healing
Hildegard Peplau	**Interpersonal relations theory** (1952)	Nurse-patient relationship – **4 phases: Orientation, Identification, Exploitation, Resolution**; "Mother of psychiatric nursing"
Virginia Henderson	**Need theory / Definition of nursing** (1955/66)	**14 basic human needs**; independence
Faye Abdellah	**21 Nursing problems** (1960)	Patient-centred approach, problem solving
Dorothy Johnson	**Behavioural system model**	7 subsystems
Ida Jean Orlando	Deliberative nursing process	Nurse-patient interaction
Myra Levine	**Conservation model**	4 conservation principles (energy, structural, personal, social integrity)
Martha Rogers	**Science of unitary human beings**	Human = energy field

Theorist	Theory / Model	Key concept
Dorothea Orem	**Self-care deficit theory** (1971)	**3 nursing systems: Wholly compensatory, Partly compensatory, Supportive-educative**
Imogene King	**Goal attainment theory**	Personal, interpersonal, social systems; **transaction**
Betty Neuman	**Systems model**	**Stressors**, lines of defence; primary/ secondary/ tertiary prevention
Sister Callista Roy	**Adaptation model** (1970)	**4 adaptive modes: Physiological, Self-concept, Role function, Interdependence**; stimuli – focal, contextual, residual
Jean Watson	**Theory of human caring**	**10 carative factors** (caritas processes)
Madeleine Leininger	**Transcultural nursing** / Culture care theory	**Sunrise model**; culturally congruent care
Patricia Benner	**From Novice to Expert** (1984)	**5 stages: Novice → Advanced beginner → Competent → Proficient → Expert** (Dreyfus model)
Nola Pender	**Health promotion model**	Health-promoting behaviours
Rosemarie Parse	Human becoming theory	
Lydia Hall	**Care, Core, Cure model** (1955); first used term "**nursing process**"	Loeb Centre
Ernestine Wiedenbach	Helping art of clinical nursing	
Joyce Travelbee	Human-to-human relationship model	
Kolcaba	**Comfort theory**	Relief, ease, transcendence
Ramona Mercer	Maternal role attainment	

♫ Theories yaad karne ki trick

- **O**rem = **O**wn care (**Self-care**) · **R**oy = **R**eact/adapt (**Adaptation**) · **K**ing = **K**ey goal (**Goal attainment**) · **N**euman = **N**ervous stress (**Stressors / systems**)
- **P**eplau = **P**eople relation (**Interpersonal**) · **L**eininger = **L**ocal culture (**Transcultural**) · **W**atson = **W**arm caring (**Caring**) · **B**enner = **B**eginner → expert
- **Metaparadigm of nursing (4 concepts)**: **Person, Environment, Health, Nursing** (Fawcett).

✎ PYQ style

- "Self-care deficit" theory – **Orem** · "Adaptation model" – **Roy** · "Interpersonal relations" – **Peplau** · "Transcultural nursing" – **Leininger** · "14 basic needs" – **Henderson** · "21 nursing problems" – **Abdellah** · "Novice to expert" – **Benner** · "Caring" – **Watson** · "Unitary human being" – **Rogers** · "Goal attainment" – **King** · "Environment" – **Nightingale**.
- Orientation phase of Peplau – **first meeting, trust building**; Resolution – **termination**.

2 Health, Illness, Hospital & Patient Safety

2.1 Health & Illness

- **WHO (1948)**: "Health is a state of complete **physical, mental and social well-being** and not merely the absence of disease or infirmity." (Spiritual dimension – later suggested.) **7 April = World Health Day**.
- **Dimensions of health**: physical, mental, social, spiritual, emotional, vocational, environmental, intellectual.
- **Wellness** – active process of achieving full potential (Halbert Dunn – "high-level wellness").
- **Health-illness continuum** (Dunn/ Travis) – high-level wellness ←→ premature death.

Health model	Key idea
Health Belief Model (Rosenstock, Becker)	Perceived susceptibility, seriousness, benefits, barriers → behaviour
Health Promotion Model (Pender)	Positive health behaviours
Agent-Host-Environment (Epidemiological triad) (Leavell)	Disease = interaction of agent + host + environment
Holistic model	Whole person – body, mind, spirit
Maslow's model	Hierarchy of needs

- **Stages of illness behaviour (Suchman)**: (1) Symptom experience → (2) Assumption of sick role → (3) Medical care contact → (4) Dependent patient role → (5) **Recovery/ rehabilitation**.
- **Levels of prevention** – Primordial, Primary (health promotion + specific protection), Secondary (early diagnosis & treatment), Tertiary (disability limitation & rehabilitation).
- **Risk factors**: genetic, age, lifestyle, environment.

2.2 Health Care Delivery & Hospital

- **Levels of health care (India)**: **Primary** – Sub-centre/ Health & Wellness Centre (Ayushman Arogya Mandir – pop. 5000 plain/ 3000 hilly), **PHC** (30,000/ 20,000), **Secondary** – **CHC** (1,20,000/ 80,000 – 30 beds, 4 specialists), Sub-district & District hospital; **Tertiary** – Medical college hospitals, AIIMS, super-speciality.
- **Hospital** – types by ownership (government, private, trust), by function (general, speciality, teaching), by bed (small < 100, medium 100–300, large > 300).
- **Ward** – unit of hospital; nurse-patient ratio (INC): general ward **1 : 6** (per shift), ICU **1 : 1**, step-down 1 : 2–3.
- **Patient unit** – space & equipment for one patient (bed, side locker, over-bed table, chair, call bell, light, oxygen & suction outlets). Distance between beds – **at least 1–1.5 m (3–6 feet)** to prevent cross-infection.
- **Ideal ward temperature** – **20–24 °C (68–75 °F)**; humidity **30–60 %**; noise level – WHO **< 35–40 dB** (day) in wards.
- **Lighting** – natural light best; night light for safety.

2.3 Admission, Transfer & Discharge

Admission

- Types: **Routine/ planned (elective)**, **Emergency**, **Day care**, **Transfer from other unit/hospital**.
- Steps: **Receive & welcome** (address by name), identify (**ID band – 2 identifiers: name + UHID/DOB**), orient to ward (call bell, toilet, routines, visiting hours), vitals, height & weight, **nursing admission assessment & history**, allergies (allergy band – **red**), valuables list (with witness), consent, inform doctor, **plan of care**.
- **Nurse's first responsibility** on admission – **make patient comfortable & safe**, orient, assess.
- **Admission bed** – prepared to receive patient (open bed / ether/ post-op bed as required).

Transfer

- **Intra-hospital** (ward to ward/ ICU) or **inter-hospital** (referral). Prepare patient & family, transfer summary/ notes, belongings, medications, **nurse-to-nurse handover (SBAR)**, accompany critical patient with O_2, monitor.

Discharge

- **Discharge planning starts on admission** (most important point!).
- Types: **Routine/ planned**, **LAMA/ DAMA** (Leave/ Discharge Against Medical Advice), **Absconding** (bina bataaye bhaag jaana – report immediately), **Transfer**, **Death**.
- Steps: doctor's written order, **discharge teaching (medicines, diet, activity, follow-up, warning signs)**, return valuables, discharge summary/ certificate, remove ID band (as per policy) at exit, escort to vehicle in **wheelchair**, after discharge – **terminal cleaning/ disinfection** of unit (closed bed).
- **LAMA** – patient ko risks samjhao, inform doctor, **signed LAMA form** (patient/ relative) with witness; refuse karne par document; can't be forced (autonomy) – but detaining = false imprisonment.
- **Medical/ dead body cannot be detained for unpaid bills** (Patients' Charter).

> ♫ **METHOD – Discharge teaching**
> **M**edications · **E**nvironment (home safety) · **T**reatment (dressing, exercises) · **H**ealth teaching (warning signs) · **O**utpatient follow-up · **D**iet

2.4 Patient Safety

- **WHO International Patient Safety Goals (IPSG – JCI)**:
  1. **Identify patients correctly** – **2 identifiers** (name + DOB/ UHID) – **room/ bed number is NOT an identifier**
  2. Improve effective communication (read-back of verbal orders, critical values)
  3. Improve safety of **high-alert medications** (KCl concentrated, insulin, heparin, opioids – double check)
  4. Ensure **correct site, correct procedure, correct patient surgery** (site marking, time-out)
  5. Reduce risk of **health care-associated infections** (**hand hygiene**)
  6. Reduce risk of **patient harm from falls**
- **17 September = World Patient Safety Day** (WHO, since 2019).
- **Root cause analysis (RCA)** – sentinel event ke baad; **Near miss** – error that didn't reach patient; **Sentinel event** – unexpected death/ serious injury.

Fall prevention

- High-risk: **elderly (> 65)**, history of falls, sedatives/ diuretics/ antihypertensives, confusion, weakness, postural hypotension, impaired vision, urgency/ nocturia, IV poles/ tubes.
- Assessment tools: **Morse Fall Scale** (0–125; **≥ 45 high risk**), Hendrich II.
- Interventions: **bed in lowest position, wheels locked**, side rails (as per policy – not all 4 – entrapment), **call bell within reach**, non-slip footwear, adequate lighting/ **night light**, clear clutter, frequent rounds & toileting schedule, **dangle legs before standing**, yellow fall-risk band/ sign.
- **Patient found on floor** → **first assess** (ABC, injury, LOC) **before moving**, call for help, inform doctor, incident report.

Restraints

- **Last resort** – only when other alternatives fail; to protect patient/ others; **doctor's written order** (not PRN), time-limited (adult **renew every 4 hrs**, children 2 hrs, < 9 yrs 1 hr – JC standard), **informed family**.
- Types: **physical** (wrist, ankle, vest/ jacket, mitten, belt, elbow restraint – infants), **chemical** (sedatives), environmental (side rails).
- **Check every 15–30 min**: circulation, skin, **2 fingers space** under restraint; **release every 2 hrs** for ROM, toileting, food, fluids.
- Tie to **bed frame (movable part that moves with head of bed) – NOT side rails**; **quick-release knot** (clove hitch for limb; **half-bow knot** to frame).
- Unnecessary restraint = **false imprisonment**. **Mummy restraint** – infants (procedures on head/neck); **elbow restraint** – after cleft lip repair (prevent touching).

Fire safety

- **RACE**: **R**escue (patient ko hatao) → **A**larm (fire alarm, call) → **C**ontain/ confine (doors & windows band) → **E**xtinguish/ Evacuate.
- **PASS** (extinguisher): **P**ull pin → **A**im at base → **S**queeze → **S**weep.
- Evacuation order: **ambulatory patients first → wheelchair → bed-ridden** (horizontal evacuation first – same floor, then vertical). Don't use lifts.
- Fire classes: **A** – wood, paper, cloth (water, foam) · **B** – flammable liquids, oil, petrol (foam, CO_2, dry chemical) · **C** – electrical (**CO_2, dry chemical – NOT water**) · D – metals · K – kitchen oils.
- **Fire triangle** – heat + fuel + oxygen. O_2 in use → no smoking, no oil/ grease on fittings.
- **Electrical safety** – 3-pin plug (grounded), no frayed cords, don't overload, report sparks, **never touch patient & electrical equipment with wet hands**.

Other hazards

- **Poisoning** (children – keep medicines locked), **burns** (hot water bottles – check temperature), **suffocation/ choking**, **radiation** (TDS principle), **needle stick injury**, **chemical (glutaraldehyde, cytotoxic)**, **back injury** of nurses (body mechanics).
- **Hospital codes (common)**: **Code Blue** – cardiac/ respiratory arrest · **Code Red** – fire · **Code Pink** – infant/ child abduction · **Code Black** – bomb threat · **Code Orange** – hazardous spill/ mass casualty · **Code Yellow** – disaster/ mass casualty (varies) · **Code Grey/ White** – violent person · **Code Silver** – person with weapon.

PYQ style

- Discharge planning starts – **at the time of admission**.
- Patient identification ke liye minimum – **2 identifiers**; bed number **not** an identifier.
- Restraint ko bandhte hain – **bed frame** se, side rail se nahi; check every **15–30 min**, release every **2 hrs**.
- Unnecessary restraint is – **false imprisonment**.
- Electrical fire – **CO_2** extinguisher; **water nahi**.
- Ideal distance between two beds – **3–6 feet (1–2 m)**.

3 Communication, Health Education & Documentation

3.1 Communication

D Communication

Information, ideas, feelings ka **do ya zyada logon ke beech exchange**. Latin "**communis**" = common. Components: **Sender → Message → Channel/ Medium → Receiver → Feedback** (+ context/ environment, noise).

- **Verbal** – spoken/ written words. **Non-verbal** – body language, facial expression, eye contact, touch, posture, gestures, tone (paralanguage), personal space (**proxemics**). **~65–93 % of communication is non-verbal**.
- **Proxemics (Edward Hall) – zones**: **Intimate 0–18 inch (0–45 cm)** – physical care; **Personal 18 inch–4 ft** – interview/ history; **Social 4–12 ft** – group; **Public > 12 ft** – lecture.
- **SOLER** (Egan) – active listening posture: **S**it squarely, **O**pen posture, **L**ean forward, **E**ye contact, **R**elaxed.
- Levels: intrapersonal (self-talk), interpersonal (2 persons), small group, public, transpersonal.
- **Therapeutic communication** – patient-centred, goal-directed, helps patient express feelings.

Therapeutic vs Non-therapeutic techniques

Therapeutic ✓	Example	Non-therapeutic ✘	Example
Active listening	Attention, nodding	**False reassurance**	"Sab theek ho jayega"
Open-ended questions	"Aap kaisa mehsoos kar rahe hain?"	**Closed questions (yes/no) excessively**	"Dard hai?"
Silence	Patient ko sochne ka time	**Giving advice**	"Aapko surgery karwani chahiye"
Restating / Paraphrasing	"Aap keh rahe hain ki neend nahi aati?"	**Asking "Why"**	"Aapne dawa kyun nahi li?" (defensive banata hai)
Reflecting (feelings wapas)	"Aap pareshan lag rahe hain"	**Changing subject**	
Clarifying	"Aapka matlab kya hai?"	**Approval/ disapproval, judging**	"Ye galat hai"
Focusing	Main issue pe laana	**Stereotyping / clichés**	"Har koi aisa feel karta hai"
Summarizing	End me recap	**Belittling, defending**	"Doctor bahut ache hain"
Offering self	"Main aapke saath baithti hoon"	**Probing/ excessive questioning**	
Sharing observation, Giving information, Exploring, Validating		**Literal responses, Disagreeing, Arguing**	



Exam trick – therapeutic answer kaise pehchane?

Jis option me nurse **patient ki feelings ko accept/ explore** kare (open-ended, reflection, "Tell me more…") wo **therapeutic** hai. "Don't worry", "Why…", advice, doctor par baat taal dena – **non-therapeutic**.

Barriers to communication

Physical (noise, pain, hearing loss), language/ cultural, psychological (anxiety, fear, anger), **semantic (medical jargon)**, environmental, status difference, information overload, poor listening.

- **Communication with special patients**: hearing impaired – face the patient, good light, speak slowly, don't shout, written/ sign; visually impaired – announce yourself before touching, explain surroundings (clock positions for food), don't move furniture; aphasia – simple yes/no, picture boards; unconscious – **assume patient can hear** (hearing last sense lost) – talk, explain before touching; children – age appropriate, play; elderly – patience, low-pitch.

3.2 Nurse–Patient Relationship (NPR)

- **Therapeutic relationship** – **patient-centred, goal-directed, time-limited**, professional (not social).
- **Peplau's phases**: **Orientation** (meet, trust, contract, assess – patient seeks help) → **Identification** (patient identifies helper) → **Exploitation (working)** (patient uses services) → **Resolution (termination)** (goals achieved, independence).
- Standard phases (Stuart): **Pre-interaction** (nurse self-preparation, read records) → **Orientation (introductory)** → **Working** → **Termination**.
- **Trust** – foundation; **empathy, genuineness, respect, unconditional positive regard (Carl Rogers)**.
- **Transference** (patient apni purani feelings nurse par daale), **Counter-transference** (nurse apni feelings patient par).
- **Boundaries** – no gifts, no personal relationships, no self-disclosure overuse.

3.3 Health Education

D **Health Education**

"Health education is a process that informs, motivates and helps people to **adopt and maintain healthy practices and lifestyles**" (WHO). Aim – **behaviour change**.

- **Principles**: interest, participation, known to unknown, **simple to complex**, motivation, comprehension, reinforcement, learning by doing, credibility, good human relations, **feedback**, setting an example.
- **Domains of learning (Bloom's taxonomy)**: **Cognitive** (knowledge – thinking), **Affective** (attitude/ feelings – values), **Psychomotor** (skills – e.g. insulin injection practice → **return demonstration**).
- Teaching methods: **lecture, demonstration & return demonstration (best for skills)**, discussion, role play, **group discussion, panel, symposium, workshop, seminar**, counselling (individual).
- **AV aids**: **auditory** (radio, tape), **visual** (charts, posters, flash cards, models, flannel graph, slides), **audio-visual** (TV, film, video). "**Cone of experience**" (Edgar Dale) – direct purposeful experience most effective; verbal symbols least.
- **Approaches**: regulatory, service, **educational**, primary health care.

- **Communication for change**: Knowledge → Attitude → Practice (**KAP**). **Stages of adoption**: Awareness → Interest → Evaluation → Trial → Adoption.
- Readiness to learn – first assess (pain, anxiety reduce karo pehle).

3.4 Records & Reports

D Record vs Report

Record – permanent **written** document of patient's care (legal document). **Report** – **oral or written** exchange of information among health team (e.g. shift handover, incident report).

Purposes of records

Communication, legal evidence, **quality assurance/ audit**, education, research, statistics, billing/ reimbursement, planning care.

Principles of good documentation (ABC + more)

- **Accurate, Brief, Complete**, **Current/ timely** (immediately after care), **Factual (objective)**, **Legible**, **organized/ chronological**, **confidential**.
- **Blue/ black permanent ink**; date, time, **signature with designation** after each entry.
- **Error**: draw **single line**, write "**error/ mistaken entry**", initial/ sign – **never erase, white-out, overwrite, or tear page**.
- **No blank lines** – draw line through empty space.
- Chart **only your own care** (never for others); **never chart in advance**.
- Use **approved abbreviations** only; avoid "**U**" (write units), "**QD**" (write daily), trailing zero (**1.0 mg ✗ → 1 mg**), leading zero needed (**.5 ✗ → 0.5 mg**).
- Quote patient's words for subjective data ("...").
- Avoid vague terms ("good", "normal", "adequate", "seems").

Methods of documentation

Method	Description
Narrative	Story-like, chronological (traditional)
SOAP / SOAPIE / SOAPIER	**S**ubjective, **O**bjective, **A**ssessment, **P**lan, **I**ntervention, **E**valuation, **R**evision – **POMR** (Problem-Oriented Medical Record – **Lawrence Weed**)
PIE	**P**roblem, **I**ntervention, **E**valuation
Focus charting / DAR	**D**ata, **A**ction, **R**esponse
Charting by exception (CBE)	Only abnormal findings documented
Kardex	Card file – quick summary of care plan (not permanent record)
Computerized/ EHR	Electronic; password confidentiality

Common records & reports

- **Temperature/ TPR chart**, **I/O chart**, nurses' notes, medication administration record (**MAR**), doctor's order sheet, **consent form**, **incident report** (**NOT kept in patient file**, not mentioned in notes), **diet**

sheet, **Kardex**, **24-hr report/ census** (admissions, discharges, deaths, transfers, births), **change-of-shift report (handover)** – bedside preferred, **SBAR**.

- **Anecdotal record** – teacher's record of student's behaviour; **Cumulative record** – student's overall record.
- **Retention of records** – inpatient records usually **3 years** (minimum, MCI regulation); medico-legal cases – longer/ permanent; birth & death – permanent.

SBAR – handover/ communication with doctor

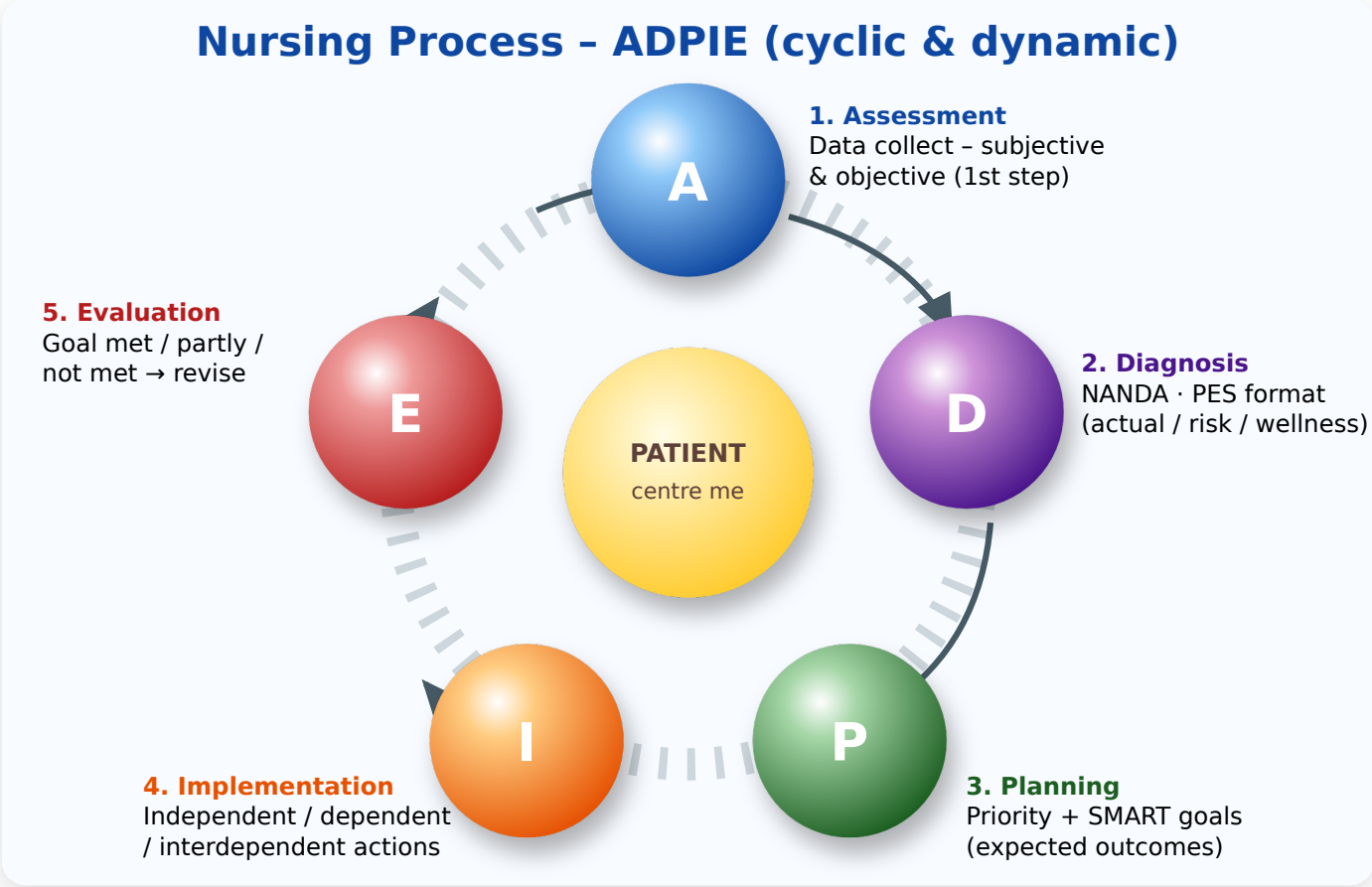
Situation (abhi kya ho raha hai) · **B**ackground (history) · **A**ssessment (aapka analysis) · **R**ecommendation (kya chahiye) – **ISBAR** me pehle **I**dentify yourself.

PYQ style

- Problem-oriented record (SOAP) introduced by – **Lawrence Weed (1969)**.
- Incident report kahan rakhte hain – **hospital administration/ risk management – patient chart me nahi**.
- Charting me galti hone par – **single line strike, "error", sign** (erase nahi).
- Best method to teach a skill – **demonstration & return demonstration**.
- Patient "mujhe surgery se darr lag raha hai" → best response – "**Aap kis baat se darr rahe hain? Mujhe batayiye.**" (exploring).
- Intimate distance – **0–18 inches**.

4 Nursing Process, Critical Thinking & Priority Setting

4.1 Nursing Process – Overview



Nursing process ke 5 steps – cyclic, patient-centred

Basic facts

- **Definition**: Systematic, rational, **problem-solving method** of planning & giving individualized nursing care.
- Term "nursing process" first used – **Lydia Hall (1955)**. **Yura & Walsh (1967)** – first 4 steps (APIE). Nursing diagnosis step added – **NANDA (1973; NANDA-I 2002)**.
- 5 steps: **A**ssessment → **D**iagnosis → **P**lanning (outcome identification + planning) → **I**mplementation → **E**valuation. (ANA – 6 steps: Assessment, Diagnosis, **Outcome identification**, Planning, Implementation, Evaluation.)
- Characteristics: **cyclic, dynamic, systematic, goal-directed, client-centred, interpersonal, universally applicable**, uses critical thinking.

4.2 Step 1 – Assessment

- **Purpose**: complete **database** banana. **First & most important step**; continuous throughout care.
- **Types of assessment**:

- **Initial/ comprehensive** – on admission (within 24 hrs as per policy)
- **Focused/ problem-oriented** – specific problem (e.g. pain)
- **Emergency** – life-threatening (ABC)
- **Time-lapsed/ ongoing** – later comparison

- **Types of data**:

Subjective (Symptoms)	Objective (Signs)
Patient **khud batata** hai – feelings, perceptions	Nurse **observe/ measure** karti hai
Pain, nausea, anxiety, itching, dizziness	BP, pulse, temp, wound redness, vomiting, lab values, weight
Quote in patient's words	Measurable, verifiable
"Covert" data	"Overt" data

- **Sources**: **Primary – patient (best/ most reliable)**; **Secondary** – family, other health team, records, lab reports, literature.
- **Methods**: **Interview (nursing history)**, **Observation**, **Physical examination (IPPA)**.
- **Validation** – data verify karna (double-check), esp. if inconsistent.
- **Data organization frameworks**: Gordon's **11 functional health patterns**, Maslow, body systems, Henderson's 14 needs.
- **Cues** (data) → **Inferences** (interpretation).

4.3 Step 2 – Nursing Diagnosis

D NANDA-I definition

"A **clinical judgment** concerning a human **response** to health conditions/ life processes, or vulnerability to that response, by an individual, family, group or community." Nursing diagnosis **nurse independently treat** kar sakti hai.

Type	Format	Example
Actual (problem-focused)	**3 parts – PES**: Problem + **related to** (Etiology) + **as evidenced by** (Signs/symptoms)	Ineffective airway clearance **r/t** thick secretions **AEB** crackles & ineffective cough
Risk	**2 parts – P + risk factors** ("**related to**/ as evidenced by risk factors") – **no S**	Risk for infection r/t surgical incision
Health promotion	1 or 2 parts; "**Readiness for enhanced** …"	Readiness for enhanced nutrition
Syndrome	1 part – cluster	Disuse syndrome, Rape-trauma syndrome
Possible (older)	Insufficient data	Possible social isolation

- **PES** = **Problem (diagnostic label)** + **Etiology** + **Signs & Symptoms (defining characteristics)**.
- **Components**: **Label** (name), **Definition**, **Defining characteristics**, **Related/ risk factors**.
- **NANDA-I** – current taxonomy II: **13 domains**, 47 classes, **267 diagnoses (2021–23 edition)**, 277 (2024–26 edition); NIC (Nursing Interventions Classification), NOC (Outcomes Classification).

Nursing vs Medical diagnosis vs Collaborative problem

Nursing diagnosis	Medical diagnosis	Collaborative problem
Patient's response to illness	**Disease/ pathology**	Potential complication (PC) needing doctor + nurse
Changes as response changes	Remains same till cured	e.g. "PC: Hypoglycaemia"
Nurse treats independently	Doctor treats	Nurse monitors + both treat
e.g. **Acute pain**, Impaired mobility	e.g. **Appendicitis**, Fracture femur	

Common errors in writing nursing diagnosis

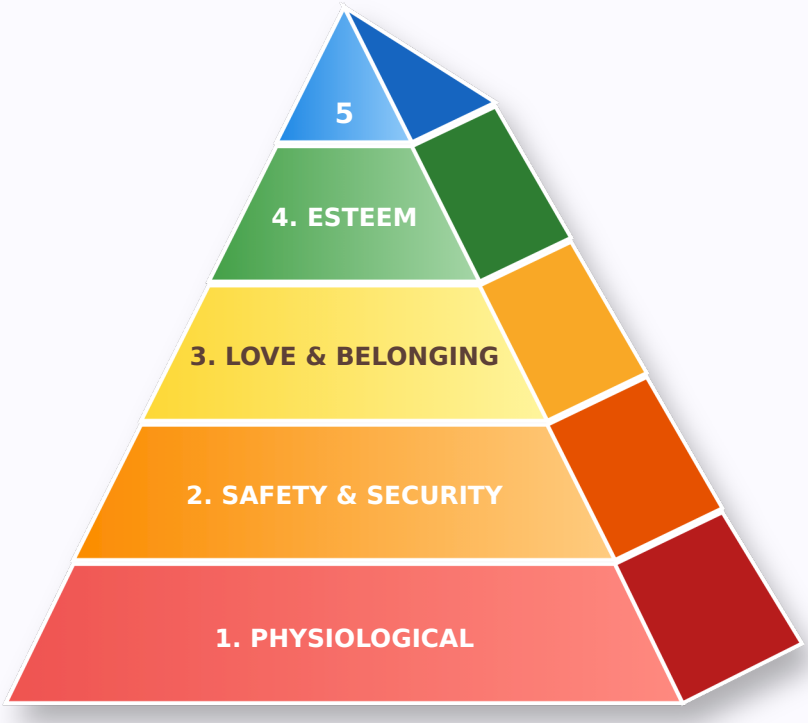
- Medical diagnosis ko etiology bana dena ✘ ("Acute pain r/t appendicitis") → ✔ "r/t inflammation & surgical incision"
- Problem & etiology same likhna ✘; legally inadvisable statements ✘ ("r/t nurse's negligence")
- Nursing intervention/ equipment ko diagnosis likhna ✘ ("Needs suctioning")
- Value judgements ✘

4.4 Step 3 – Planning

- **Priority setting** → **Goals/ outcomes** → **Interventions** → **Care plan documentation**.

Priority setting

Maslow's Hierarchy of Needs (1943) – Priority setting



5. Self-actualization
Apni full potential achieve karna

4. Esteem
Self-respect, recognition, dignity

3. Love & belonging
Family, friends, acceptance

2. Safety & security
Physical safety, fall/ infection prevention, freedom from fear

1. Physiological (sabse pehle)
O_2/airway, water, food, elimination, sleep, temperature, sex

Neeche wali need pehle puri karo → upar wali. Physiological me bhi order: **Airway → Breathing → Circulation**.

Maslow ka pyramid – neeche wali need pehle

- **High priority** – life-threatening (airway, breathing, circulation, severe bleeding, shock). **Medium** – non-life threatening but health threat (acute pain, unsafe mobility). **Low** – developmental/ future needs (teaching about diet).
- Priority rules: **ABC → Maslow → Acute before chronic → Actual before risk (usually) → Safety → Patient's own priority** (if not conflicting). **Assessment before intervention** (unless emergency).
- **Kalish** – modified Maslow (adds **stimulation needs** – sex, activity, exploration, novelty – between physiological & safety).

Goals & expected outcomes

- **Long-term** (weeks/ months) vs **Short-term** (hours/ days).
- **SMART**: **S**pecific, **M**easurable, **A**chievable/ attainable, **R**ealistic/ relevant, **T**ime-bound.
- Components of outcome statement: **Subject** (patient) + **Verb (measurable)** + **Condition** + **Criteria (performance)** + **Time**. e.g. "**Patient will walk** with walker **50 m** in corridor **by 3rd post-op day**."
- **Measurable verbs** ✔: walk, list, state, demonstrate, identify, drink, verbalize. **Non-measurable** ✘: know, understand, learn, appreciate, think.

Nursing interventions & care plan

- **Independent** (nurse-initiated) – turning, teaching, positioning, hygiene. **Dependent** (physician-initiated) – medications, IV. **Collaborative/ interdependent** – with physiotherapist, dietician.
- **Rationale** – scientific reason for each intervention.
- **Nursing care plan columns**: Assessment – Nursing diagnosis – Goal/ Objectives – Interventions – Rationale – Evaluation.
- Types: **Standardized** (protocols), **Individualized**, **Concept map**, Critical pathway (clinical pathway – interdisciplinary, timeline).

4.5 Step 4 – Implementation

- **Action phase** – plan ko execute karna; **reassess patient before implementing** (status change ho sakta hai).
- Skills needed: **cognitive** (intellectual), **interpersonal**, **technical (psychomotor)**.
- **Delegation** (5 rights): **Right task, Right circumstance, Right person, Right direction/ communication, Right supervision/ evaluation**. **Assessment, teaching, evaluation & unstable patients – delegate nahi** kar sakte (to ANM/ assistant).
- **Documentation** after implementation.

4.6 Step 5 – Evaluation

- Compare patient's **response with expected outcomes** → **Goal met / partially met / not met**.
- Not met → **reassess, revise plan** (diagnosis/ goals/ interventions). Continue/ modify/ terminate plan.
- Types: **Structure** (setting/ resources), **Process** (nurse's performance), **Outcome** (patient's change – most important), **Concurrent (formative)** & **Retrospective (summative)**.
- **Quality assurance (QA)** – nursing audit (Phaneuf's audit – 7 functions).

4.7 Critical Thinking & Problem Solving

- **Critical thinking** – purposeful, reflective reasoning for clinical judgment. Skills: **interpretation, analysis, inference, evaluation, explanation, self-regulation** (Facione).
- **Attitudes**: confidence, independence, fairness, responsibility, risk-taking, discipline, perseverance, creativity, curiosity, integrity, humility.
- **Problem-solving method (scientific)**: Identify problem → collect data → hypothesis → plan → implement → evaluate. **Trial & error** – least effective; **Intuition** – expert nurses (Benner).
- **Decision making** – choosing best alternative.
- **Evidence-based practice (EBP)** – best research evidence + clinical expertise + patient preferences. **PICO** – Population, Intervention, Comparison, Outcome.



Priority questions – step-by-step approach

1. **Unstable vs stable** – unstable pehle (e.g. post-op with falling BP > diabetic asking for breakfast)
2. **ABC** – airway sabse pehle (stridor > chest pain > fracture)
3. **Acute > chronic**, **actual > potential**, **safety**
4. Agar option me "**assess**" hai aur situation emergency nahi – **assessment pehle**
5. Delegation question me – **stable, routine, predictable** task hi delegate

PYQ style

- Nursing process ka first step – **Assessment**; last – **Evaluation**.
- "Patient states pain 8/10" – **subjective data**; "BP 150/90" – **objective**.
- Primary source of data – **patient**.
- NANDA ka full form – **North American Nursing Diagnosis Association** (2002 se NANDA-International).
- Nursing diagnosis me "**related to**" = **etiology**; "**as evidenced by**" = **signs/symptoms**.
- Risk diagnosis has – **2 parts** (no S).
- Goal statement me measurable verb – **demonstrate/ list/ walk** ("understand" nahi).
- Maslow ka highest need – **Self-actualization**; lowest – **Physiological**.

5 Vital Signs – Temperature, Pulse, Respiration, BP, SpO_2 & Pain

Vital signs – normal adult values (sabse important table)

Vital sign	Normal	Ranges / terms
Temperature (oral)	**37 °C (98.6 °F)**; range 36.5–37.5 °C	Rectal +0.5 °C, Axillary −0.5 °C, Tympanic ≈ core
Pulse	**60–100 /min**	Newborn 120–160, infant 100–140, child 80–120
Respiration	**12–20 /min**	Newborn 30–60, infant 25–40, child 20–30
BP	**< 120/80 mmHg**	Newborn ~65/40
SpO_2	**95–100 %**	< 90 % hypoxaemia
Pain	0 (no pain)	5th vital sign
Pulse : Respiration ratio ≈ 4 : 1 · Temperature 1 °F ↑ → pulse ~10 beats ↑ (1 °C → ~18 beats)		

- Vital signs kab lete hain: **admission**, as ordered/ routine (4 hourly), **before & after surgery/ invasive procedure**, **before & after drugs affecting cardio-respiratory system** (digoxin, antihypertensives, opioids), **blood transfusion**, **condition change**, before & after nursing interventions affecting vitals (ambulation).
- **Order of checking** (child): **R → P → T** (rona shuru hone se pehle respiration). Adult: T, P, R, BP.
- Abnormal value → **recheck** first, then report.

5.1 Body Temperature

- **Thermoregulation centre – Hypothalamus** ("body's thermostat"). Anterior hypothalamus – heat loss; posterior – heat production/ conservation.
- **Heat production**: basal metabolism (BMR), muscle activity (**shivering** – heat ↑ up to 4–5 times), thyroxine, adrenaline, fever.
- **Heat loss mechanisms**:

Mechanism	Meaning	Example
Radiation (most heat loss ~60 %)	Without contact – electromagnetic waves	Body heat to cold room
Conduction	**Direct contact** with cooler object	**Ice pack, cold bed**, cold stethoscope
Convection	**Air/ water current**	**Fan**, wind
Evaporation	Liquid → vapour (sweat, insensible)	**Tepid sponge**, sweating

- **Factors**: age (newborn – poor regulation; elderly – lower temp), **circadian rhythm (lowest 4–6 am, highest 4–8 pm)**, exercise, hormones (**ovulation ↑ 0.3–0.6 °C – progesterone**), stress, environment.

Terms

Term	Meaning
Afebrile	Normal temperature
Pyrexia/ Fever/ Febrile	Above normal (> 37.5 °C oral / > 38 °C)
Hyperpyrexia	**> 41 °C (105.8 °F)**
Hypothermia	**< 35 °C (95 °F)**
Heat stroke	> 40 °C with CNS dysfunction, hot dry skin

Types of fever

Type	Pattern	Example
Continuous/ sustained	Remains above normal, fluctuation **< 1 °C** in 24 hr	**Typhoid (2nd week)**, lobar pneumonia
Remittent	Above normal throughout, fluctuation **> 1 °C (2 °C)**, never normal	**Typhoid (early)**, infective endocarditis
Intermittent	Fever present for some hours, **returns to normal** daily	**Malaria**, septicaemia, TB (pyaemia)
Relapsing/ Recurrent	Fever days alternating with afebrile days	**Brucellosis (undulant)**, Pel-Ebstein (Hodgkin's), Borrelia
Quotidian	Every 24 hrs	P. knowlesi
Tertian	Every **48 hrs** (alternate day)	**P. vivax, P. ovale**, falciparum (malignant tertian)
Quartan	Every **72 hrs**	**P. malariae**
Step-ladder	Gradual daily rise	**Typhoid (1st week)**

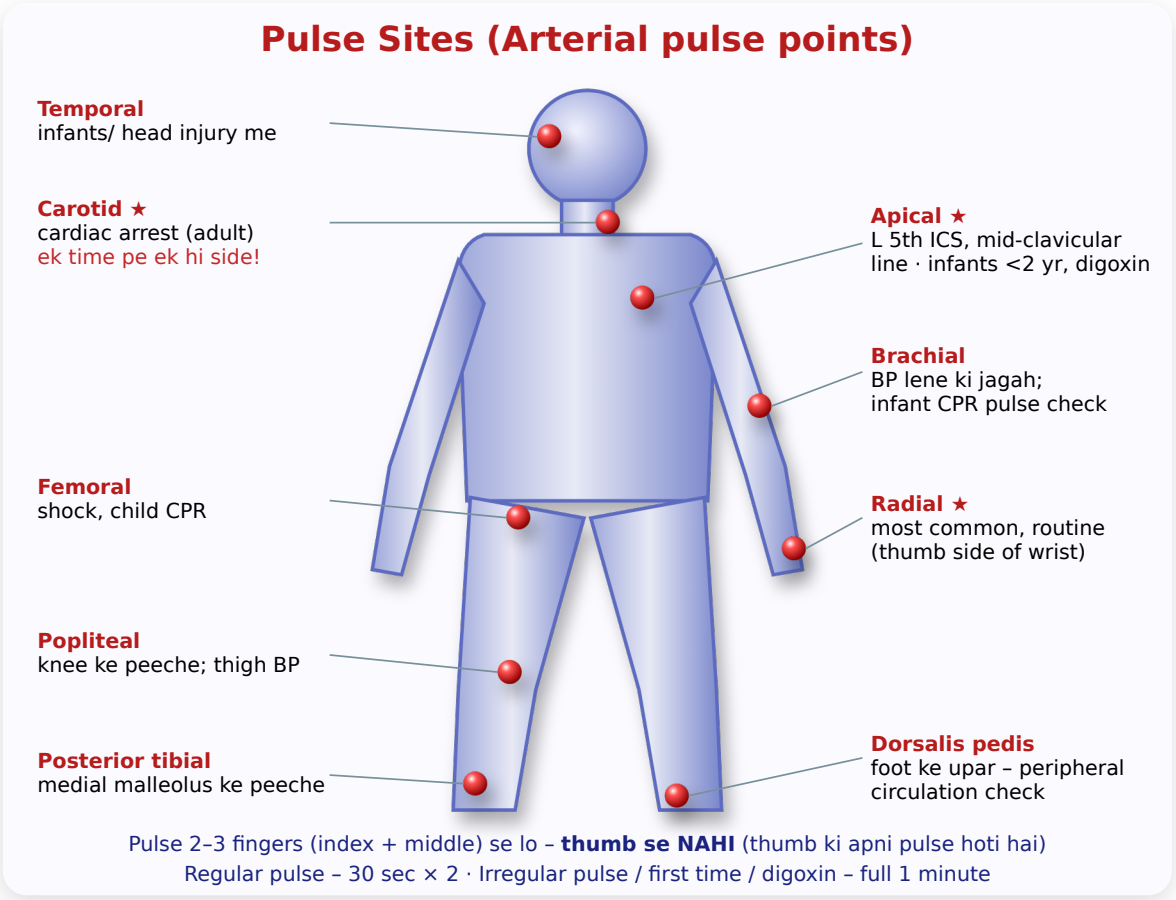
- **Stages of fever**: **Onset/ chill (cold) stage** – shivering, pale cold skin, **rigors** (heat production ↑ to reach new set-point) → **Course/ plateau (flush) stage** – hot dry flushed skin, thirst → **Defervescence/ termination** – **sweating**, temp falls. Fall: **Crisis** (sudden – within hours) vs **Lysis** (gradual – days).
- **Fever care**: rest, **fluids ↑ (2.5–3 L)**, light clothing, **tepid sponge (water 27–34 °C / ~1 °C below body)**, antipyretics (paracetamol), oral care, high-calorie diet; during chill stage – **cover with blanket (warmth)**; record 4-hourly.
- **Cold sponge** – avoid ice/ alcohol rub (shivering, vasoconstriction).

Temperature sites & thermometers

Site	Time (clinical glass)	Notes
Oral (sublingual pocket)	**3 min** (2–3)	Wait **15–30 min after eating, drinking hot/cold, smoking**; **contraindicated**: unconscious, infants/ children < 5–6 yrs, mouth breathers, oral surgery, seizures, confused, O_2 mask
Axillary	**5–10 min** (9 min)	**Safest**, least accurate; children; dry axilla
Rectal	**2–3 min**	**Most accurate (core)**; insert **adult 1.5 inch (3.5 cm)**, child 1 inch, infant 0.5 inch; **Sims' position**; lubricate; **contraindicated**: rectal surgery, diarrhoea, haemorrhoids, **MI/ cardiac patients (vagal stimulation → bradycardia)**, neutropenia, newborn (perforation risk)
Tympanic	1–3 sec	Infrared; pull pinna **up & back (adult)**, **down & back (child < 3 yr)**; close to core
Temporal artery	Seconds	Forehead scan
Groin		Rarely

- **Clinical thermometer (mercury)** – range **35–42 °C (95–110 °F)**; **constriction** prevents mercury fall-back; **shake down below 35 °C (95 °F)** before use. Mercury thermometers being **phased out** (toxicity) → **digital**.
- **Celsius ↔ Fahrenheit**: **°F = (°C × 9/5) + 32**; **°C = (°F − 32) × 5/9**. e.g. 38 °C = 100.4 °F; 40 °C = 104 °F; 104 °F = 40 °C.
- Thermometer **cleaning**: soap & **cold water**, wipe **from stem (clean area) towards bulb (dirty area)**, then 70 % alcohol; **hot water nahi** (mercury expand → thermometer toot sakta hai).
- **Mercury spill** – don't touch/ vacuum; use mercury spill kit, gloves, collect with card/ syringe, airtight container.

5.2 Pulse



Pulse sites – kaunsi site kab use karein

- **Pulse** = wave of blood created by **left ventricle contraction** felt over artery.
- Assess: **Rate, Rhythm, Volume (strength), Tension, Equality (bilateral), Character, Condition of arterial wall**.
- **Volume scale**: **0 = absent, 1+ = weak/ thready, 2+ = normal, 3+ = full/ bounding** (some 4-point scale: 3+ increased, 4+ bounding).
- Count: regular – 30 sec × 2; **irregular – full 1 minute**; apical always 1 min.

Term	Meaning
Tachycardia	> 100 /min
Bradycardia	< 60 /min
Thready/ weak pulse	Shock, haemorrhage, dehydration
Bounding pulse	Fever, fluid overload, anxiety, aortic regurgitation
Water-hammer (Corrigan's) pulse	**Aortic regurgitation**
Pulsus paradoxus	SBP falls > 10 mmHg on inspiration – **cardiac tamponade**, severe asthma
Pulsus alternans	Strong-weak alternate – **LV failure**
Pulsus bisferiens	Double peak – AR, HOCM
Pulse deficit	**Apical − Radial** (2 nurses, same time, 1 min) – **atrial fibrillation**
Sinus arrhythmia	Rate ↑ in inspiration, ↓ expiration – **normal in children**

Term	Meaning
Dicrotic pulse	Typhoid

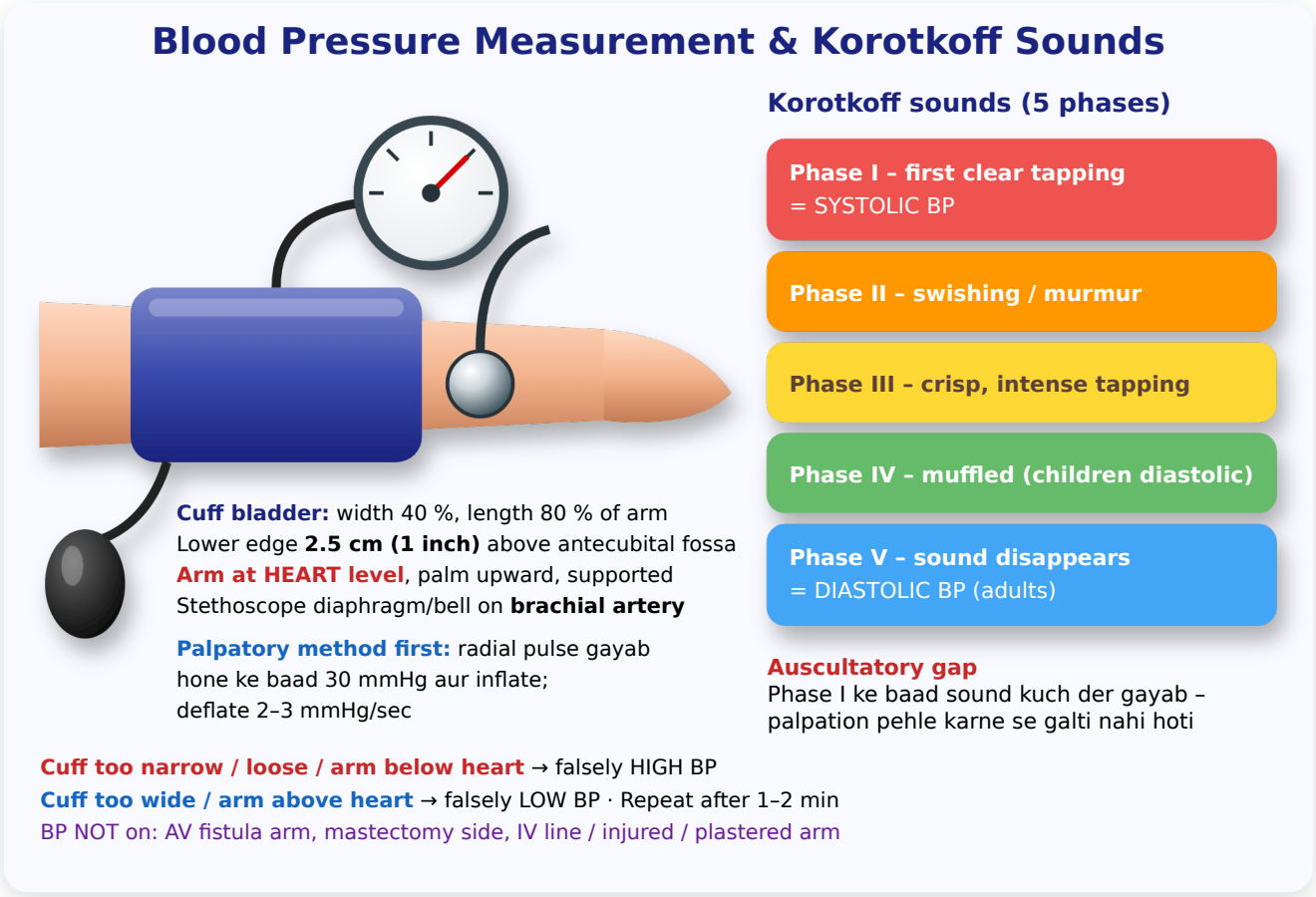
- **Apical pulse**: **5th ICS, left mid-clavicular line** (adult); **4th ICS, lateral to MCL in children < 7 yrs**. Use for **infants < 2 yrs**, irregular pulse, **before digoxin**.
- **Factors ↑ pulse**: exercise, fever, pain, anxiety, haemorrhage/ hypovolaemia, hypoxia, sympathomimetics (adrenaline, atropine), standing. **↓ pulse**: sleep, athletes, **digoxin, beta-blockers**, hypothermia, ↑ICP (late), vagal stimulation.

5.3 Respiration

- **External respiration** (lungs – gas exchange alveoli ↔ blood) & **internal respiration** (blood ↔ tissues). **Ventilation** = movement of air in/out.
- Count **without informing patient** (pretend pulse is being counted) – conscious control se pattern badal jata hai; **1 full minute** if irregular; one inspiration + expiration = 1 breath.
- **Inspiration – active** (diaphragm contracts), **expiration – passive** (normal).
- Respiratory centre: **medulla oblongata & pons**; main stimulus – **↑ CO_2 / H^+** (chemoreceptors); COPD – **hypoxic drive**.

Term	Meaning	Example
Eupnoea	Normal breathing	
Tachypnoea	> 20 /min, fast shallow	Fever, pneumonia, anxiety
Bradypnoea	< 12 /min	**Opioid overdose**, ↑ICP
Apnoea	Absence of breathing	Cardiac arrest, sleep apnoea
Dyspnoea	Difficult/ laboured breathing (subjective)	HF, asthma
Orthopnoea	Breathing difficulty **lying flat**, eased sitting up	**Left HF**
PND	Sudden night breathlessness	Left HF
Hyperventilation	↑ rate & depth → **↓ CO_2 (resp. alkalosis)**	Anxiety
Hypoventilation	↓ → ↑ CO_2	Sedation
Kussmaul	**Deep, rapid, regular**	**DKA / metabolic acidosis**
Cheyne-Stokes	**Waxing-waning depth + apnoea periods** (cyclic)	**Dying patient, HF, ↑ICP**, drug overdose
Biot's (ataxic)	Irregular, varying depth, sudden apnoea	**Meningitis**, brain damage (medulla)
Apneustic	Prolonged gasping inspiration	Pons lesion
Stertorous	**Snoring** sounds	Unconscious, airway obstruction
Stridor	High-pitched crowing **(inspiratory)**	**Upper airway obstruction** – laryngeal oedema, croup
Wheeze	Whistling **(expiratory)**	**Asthma**
Sighing	Deep inspiration + prolonged expiration	Anxiety, normal occasionally

5.4 Blood Pressure



BP lene ka sahi tareeka aur Korotkoff ki 5 sounds

- **BP = Cardiac output × Peripheral resistance**. **Systolic** – ventricular contraction; **Diastolic** – ventricular relaxation (minimum pressure).
- **Pulse pressure = SBP − DBP** (normal ~40); wide → aortic regurgitation, ↑ICP; narrow → shock, cardiac tamponade, aortic stenosis.
- **MAP = DBP + ⅓ pulse pressure = (SBP + 2 DBP) ÷ 3** (normal 70–100).
- Factors: age ↑, **exercise ↑**, stress ↑, **sex (women < men before menopause)**, **diurnal (lowest early morning, highest late afternoon)**, weight/ obesity ↑, **posture (lying > sitting > standing slightly)**, race, medications, smoking ↑, caffeine ↑.
- **Sphygmomanometer** – mercury (gold standard), aneroid, electronic. **Riva-Rocci** invented mercury sphygmomanometer (1896); **Korotkoff** described sounds (1905). Stephen Hales – first measured BP (horse, 1733).
- Cuff size: **bladder width = 40 % of arm circumference** (or 20 % wider than diameter); **length 80 %**. Adult standard 12–13 cm wide.
- Patient **seated 5 min rest**, **no caffeine/ smoking 30 min**, feet flat, legs uncrossed, **arm supported at heart level**, palm up; cuff **2.5 cm above antecubital fossa**; **palpatory method first** → auscultatory; deflate **2–3 mmHg/ sec**; wait **1–2 min** before repeat (venous congestion).
- **Thigh BP** – popliteal artery, **large cuff**, prone position; **SBP 10–40 mmHg higher** than arm.
- **Orthostatic (postural) hypotension**: SBP fall **≥ 20 mmHg** or DBP **≥ 10 mmHg** within 3 min of standing (lying → standing).

Error	Result
Cuff too narrow/ small	**Falsely HIGH**
Cuff too wide/ large	**Falsely LOW**
Cuff wrapped too loosely	**Falsely HIGH**
Arm below heart level	**Falsely HIGH**
Arm above heart level	**Falsely LOW**
Deflating too slowly	Falsely high diastolic (venous congestion)
Deflating too quickly	**Falsely low systolic** & high diastolic
Arm unsupported (muscle contraction)	Falsely high
Repeating too soon	Falsely high
Mercury not at eye level (reading from above/ below)	Parallax error (above eye → low; below eye → high)
Failure to identify auscultatory gap	**Falsely low systolic**

BP error trick

"**Small cuff, Loose cuff, Low arm → High BP**" · "**Big cuff, High arm → Low BP**"

- **BP classification** (JNC-7): Normal < 120/80; **Prehypertension** 120–139/80–89; **Stage 1** 140–159/90–99; **Stage 2 ≥ 160/≥ 100**. **Hypotension** < 90/60.
- **Avoid BP in arm with**: IV infusion, **AV fistula/ shunt**, **mastectomy (lymph node dissection) side**, injury/ burns, cast, PICC line.

5.5 Pulse Oximetry (SpO_2) & Pain

- **Pulse oximeter** – non-invasive, uses **red & infrared light** (Beer-Lambert law), sites: finger, toe, ear lobe, forehead, bridge of nose; infants – foot/ palm.
- Normal **95–100 %**; COPD target **88–92 %**; < 90 % = hypoxaemia (PaO_2 ~60).
- **False readings**: **nail polish (dark), acrylic nails**, cold extremities/ poor perfusion, movement, bright light, **CO poisoning (falsely HIGH)**, severe anaemia, jaundice (bilirubin), methylene blue dye, dark skin (slight overestimation). Change finger probe site every **2–4 hrs** (pressure).
- **Pain – 5th vital sign** (American Pain Society, 1996). Scales: NRS 0–10, **VAS 10 cm**, **Wong-Baker FACES (≥ 3 yrs)**, **FLACC** (2 months–7 yrs/ non-verbal), CRIES (neonates), PAINAD (dementia). Detail – Chapter 9.

PYQ style

- Thermoregulation centre – **hypothalamus**.
- Most accurate route of temperature – **rectal**; safest – **axillary**.
- Rectal temp contraindicated in – **MI, diarrhoea, rectal surgery**.
- Cheyne-Stokes respiration – **dying/ HF/ ↑ICP**; Kussmaul – **DKA**.
- Pulse deficit – **atrial fibrillation**; counted by **2 nurses** simultaneously.
- Apical pulse site – **left 5th ICS mid-clavicular line**.
- Cuff too narrow → **falsely high** BP.
- 1st Korotkoff sound – **systolic**; 5th – **diastolic** (adult).
- Malaria fever – **intermittent**; Typhoid – **step-ladder (1st wk) / continuous**.
- 39 °C = **102.2 °F**; 100 °F = **37.8 °C**.

6 Health Assessment & Physical Examination

6.1 Health History

- **Components**: Biographic data → **Chief complaint (CC – patient ke words me)** → **History of present illness (HPI)** → Past history → Family history (genogram) → Personal/ social history (diet, sleep, habits – smoking, alcohol) → Menstrual/ obstetric history → Immunization → Allergies → Review of systems.
- **HPI analysis – OLDCARTS / PQRST**: **O**nset, **L**ocation, **D**uration, **C**haracter, **A**ggravating, **R**elieving, **T**iming, **S**everity.
- Interview phases: **Introduction (orientation)** → **Working** → **Termination**. Privacy, comfortable seating, open-ended questions.

6.2 Techniques of Physical Examination

Technique	Meaning	Use / Notes
Inspection	**Dekhna** (+ smell, listen) – systematic observation	**First** technique; good light; compare both sides (symmetry)
Palpation	**Chhoona** – touch se feel	**Light** (1 cm – tenderness, texture) first, then **deep** (4 cm – organs, masses). Painful area **last**. **Fingertips** – fine discrimination (texture, pulse); **dorsum of hand** – **temperature**; **palm/ ulnar surface** – **vibration (fremitus)**; grasping – size, consistency
Percussion	**Thapthapana** – tap karke sound sunna	**Indirect/ mediate** (pleximeter finger on body, plexor finger strikes – most common), direct, blunt (fist – kidney/ CVA tenderness)
Auscultation	**Stethoscope se sunna**	**Diaphragm – high-pitched** (breath, bowel, normal heart S1, S2); **Bell – low-pitched** (murmurs, **S3, S4**, bruits) – placed **lightly**

! Order of examination

Normal order: I → P → P → A (Inspection, Palpation, Percussion, Auscultation).
Abdomen: I → A → P → P (Inspection, **Auscultation**, Percussion, Palpation) – kyunki palpation/ percussion bowel sounds badal deta hai.

Percussion notes

Note	Intensity / Pitch	Normal site	Abnormal example
Flat	Soft, high, short	**Muscle (thigh)**, bone	Large pleural effusion
Dull	Medium, thud-like	**Liver, heart, full bladder, spleen**	**Pneumonia (consolidation)**, pleural effusion, mass
Resonant	Loud, low, hollow	**Normal lung**	
Hyper-resonant	Very loud, booming	None in adult (child lung)	**Emphysema, pneumothorax**

Note	Intensity / Pitch	Normal site	Abnormal example
Tympanic	Loud, high, drum-like	**Stomach, gas-filled bowel**	Distended abdomen (gas)

Percussion – solid se hollow

Flat (bone/muscle) → Dull (liver) → Resonant (lung) → Hyper-resonant (emphysema) → Tympanic (stomach) = dense se khokla.

Equipment

Stethoscope, sphygmomanometer, **thermometer**, **ophthalmoscope** (retina/ fundus), **otoscope** (ear canal, TM), **tuning fork (512 Hz hearing; 128 Hz vibration)**, **percussion (reflex) hammer**, **Snellen chart (20 ft/ 6 m)**, tongue depressor, penlight/ torch, measuring tape, weighing scale, **nasal speculum**, **vaginal speculum** (Cusco's/ Sims'), gloves, **goniometer** (joint angle/ ROM), **skin-fold calliper** (fat), **Doppler** (weak pulses), cotton & pins (sensation).

Positions for examination

Position	Use
Sitting	Head, neck, chest (back & front), upper limbs, vitals
Supine (dorsal)	**Abdomen** (knees slightly flexed – relax muscles), breast, heart, extremities
Dorsal recumbent (supine, knees flexed, feet flat)	**Abdomen**, perineum, catheterization (female), vaginal exam
Lithotomy	**Vaginal/ pelvic exam, Pap smear**, delivery
Sims' (left lateral, right knee flexed)	**Rectal exam, enema**, vaginal (Sims' speculum)
Prone	Back, spine, hip joint movement
Knee-chest (genupectoral)	**Rectal/ prostate, sigmoidoscopy**
Standing	Gait, posture, spine, hernia, varicose veins, male genitalia
Fowler's/ semi-Fowler's	Dyspnoeic patient, head/ neck
Lateral recumbent	Heart (left lateral – mitral murmur better), **lumbar puncture**

6.3 General Survey & Measurements

- General survey: appearance, **level of consciousness**, posture, gait, build, hygiene, speech, mood, **signs of distress**, odours.
- **Height & weight**: same time (morning), same scale, same clothes, after voiding; infants – supine length (infantometer).
- **BMI = Weight (kg) ÷ Height (m)²** (Quetelet index).

BMI (WHO)	Category	Asian/ Indian cut-offs
< 18.5	**Underweight**	< 18.5
18.5–24.9	**Normal**	**18.5–22.9**

BMI (WHO)	Category	Asian/ Indian cut-offs
25–29.9	**Overweight**	**23–24.9**
≥ 30	**Obese**	**≥ 25**

- **Waist circumference** – risk: **men > 90 cm, women > 80 cm** (Asian); **Waist-hip ratio**: men > 0.9, women > 0.85 (abdominal obesity). **MUAC** (children 6–59 months) – **< 11.5 cm = SAM**.
- **Ideal body weight (Broca)** = Height (cm) − 100 (men); women − 105.
- **Mid-arm circumference, skin-fold thickness (triceps)** – nutrition.

6.4 Head-to-Toe – Important Findings

Skin

- **Colour**: **pallor** (anaemia, shock – see conjunctiva, palms, nail beds), **cyanosis** (central – lips/ tongue; peripheral – nail beds), **jaundice** (sclera first – bilirubin > 2–3 mg/dL), erythema, **mottling**.
- **Turgor** – pinch skin over **sternum/ clavicle (elderly)**, forearm/ abdomen; slow return (**tenting**) = **dehydration**.
- **Oedema** – **pitting**: **1+ (2 mm)**, **2+ (4 mm)**, **3+ (6 mm)**, **4+ (8 mm)**. Press over **bony surface (tibia, ankle, sacrum)** for 5 sec.
- **Capillary refill** – press nail bed → normal **< 2–3 sec**.
- **Clubbing** – **angle > 180°** (normal 160°); **Schamroth's window** lost – chronic hypoxia (COPD, bronchiectasis, CHD, lung cancer).
- **Koilonychia** (spoon nails – iron deficiency), **Beau's lines** (transverse grooves – acute illness), **splinter haemorrhage** (endocarditis), **paronychia**.
- Lesions – see MSN notes (macule, papule, vesicle...).

Head, Eyes, Ears, Nose, Throat

- Eyes: **PERRLA** (Pupils Equal, Round, Reactive to Light & Accommodation); normal pupil **2–6 mm** (3–5); **anisocoria** (unequal); **ptosis**; **nystagmus**; visual acuity (Snellen), **visual fields (confrontation test)**, **corneal reflex** (CN V & VII), conjunctiva (pallor), sclera (jaundice).
- Ears: otoscope – **adult pull pinna up & back; child < 3 yr – down & back**; **Rinne & Weber** tests; whisper test.
- Mouth: lips (cyanosis), tongue (coated – typhoid; beefy red – B12; **magenta – B2**), gums (bleeding – scurvy, gingival hyperplasia – phenytoin), teeth, tonsils (grading 1+ to 4+ – **4+ = touching midline "kissing tonsils"**), **gag reflex (CN IX, X)**, uvula (rises midline – "ah").
- Neck: lymph nodes, **thyroid (from behind)**, **trachea – midline**, **JVP (jugular venous pressure)** – patient at **45°**; normal **< 3–4 cm above sternal angle**.

Chest & Lungs

- Shape: **normal AP : lateral = 1 : 2**; **barrel chest 1 : 1** (emphysema); pigeon chest (pectus carinatum), funnel chest (pectus excavatum), kyphosis, scoliosis.
- **Tactile fremitus** ("99" / "ninety-nine" – palpate vibration): **↑ in consolidation (pneumonia)**; **↓ in pleural effusion, pneumothorax, emphysema**.
- **Vocal resonance/ bronchophony, egophony** ("ee" → "aa" over consolidation), **whispered pectoriloquy**.

- Breath sounds: **vesicular** (soft, low – most lung fields; inspiration > expiration), **bronchovesicular** (between scapulae, 1st–2nd ICS), **bronchial/ tubular** (loud, high – over trachea/ manubrium; abnormal elsewhere = consolidation).
- Adventitious: **crackles (rales)** – fine/ coarse – fluid (pneumonia, HF); **wheeze** – narrowed airway; **rhonchi** – secretions large airway (clear with cough); **pleural friction rub** – grating (pleurisy); **stridor**.

Heart

- Auscultation areas (**APETM** – "**All Physicians Enjoy Taking Money**"): **A**ortic (2nd ICS right sternal border) → **P**ulmonic (2nd ICS left SB) → **E**rb's point (3rd ICS left SB) → **T**ricuspid (4th/5th ICS left lower SB) → **M**itral/ apex (5th ICS left MCL – **PMI/ apex beat**).
- **S1 ("lub")** – closure of **AV (mitral & tricuspid)** valves – loudest at **apex**; start of systole. **S2 ("dub")** – closure of **semilunar (aortic & pulmonary)** – loudest at **base**. **S3** – "Ken-tuc-ky" – ventricular gallop (HF; normal in children/ athletes/ pregnancy). **S4** – "Ten-nes-see" – atrial gallop (stiff ventricle – HTN, MI).
- **Murmurs** graded **I–VI** (VI – heard with stethoscope off chest); **thrill** = palpable murmur.

Abdomen

- 4 quadrants / 9 regions; **bowel sounds 5–30 /min** (listen **5 min** before calling absent); **hyperactive (borborygmi)** – diarrhoea, early obstruction; **hypoactive/ absent** – paralytic ileus, peritonitis.
- **Liver span 6–12 cm (MCL)**; **spleen** – normally not palpable; **shifting dullness & fluid thrill** – ascites; **rebound tenderness (Blumberg)** – peritonitis; **Murphy's sign** – cholecystitis; **McBurney's point** – appendicitis.
- **CVA tenderness** (fist percussion at costovertebral angle) – **pyelonephritis/ kidney**.

Musculoskeletal & Neuro

- **Muscle strength grading (0–5)**: **0** – no contraction · **1** – flicker/ trace · **2** – movement with gravity eliminated · **3** – against gravity (no resistance) · **4** – against some resistance · **5** – normal full strength.
- **Deep tendon reflexes (0–4+)**: 0 absent · 1+ diminished · **2+ normal** · 3+ brisk · 4+ hyperactive with **clonus**. Biceps (C5–6), triceps (C7), brachioradialis, **patellar/ knee jerk (L2–L4)**, **Achilles/ ankle jerk (S1)**.
- **Babinski** – stroke sole → toes fan & great toe up = **positive (abnormal in adults; normal < 1–2 yrs)** → UMN lesion.
- **Romberg test** (balance – eyes closed, feet together), **finger-nose test** (cerebellum), **heel-shin test**, **gait**.
- **Level of consciousness**: Alert → Lethargic → Obtunded → Stuporous → Comatose. **AVPU** (Alert, Voice, Pain, Unresponsive). **GCS 3–15** (Eye 4, Verbal 5, Motor 6) – detail in MSN notes.
- **Mental status – orientation ×3 (time, place, person)**; **MMSE (30 points)**.

PYQ style

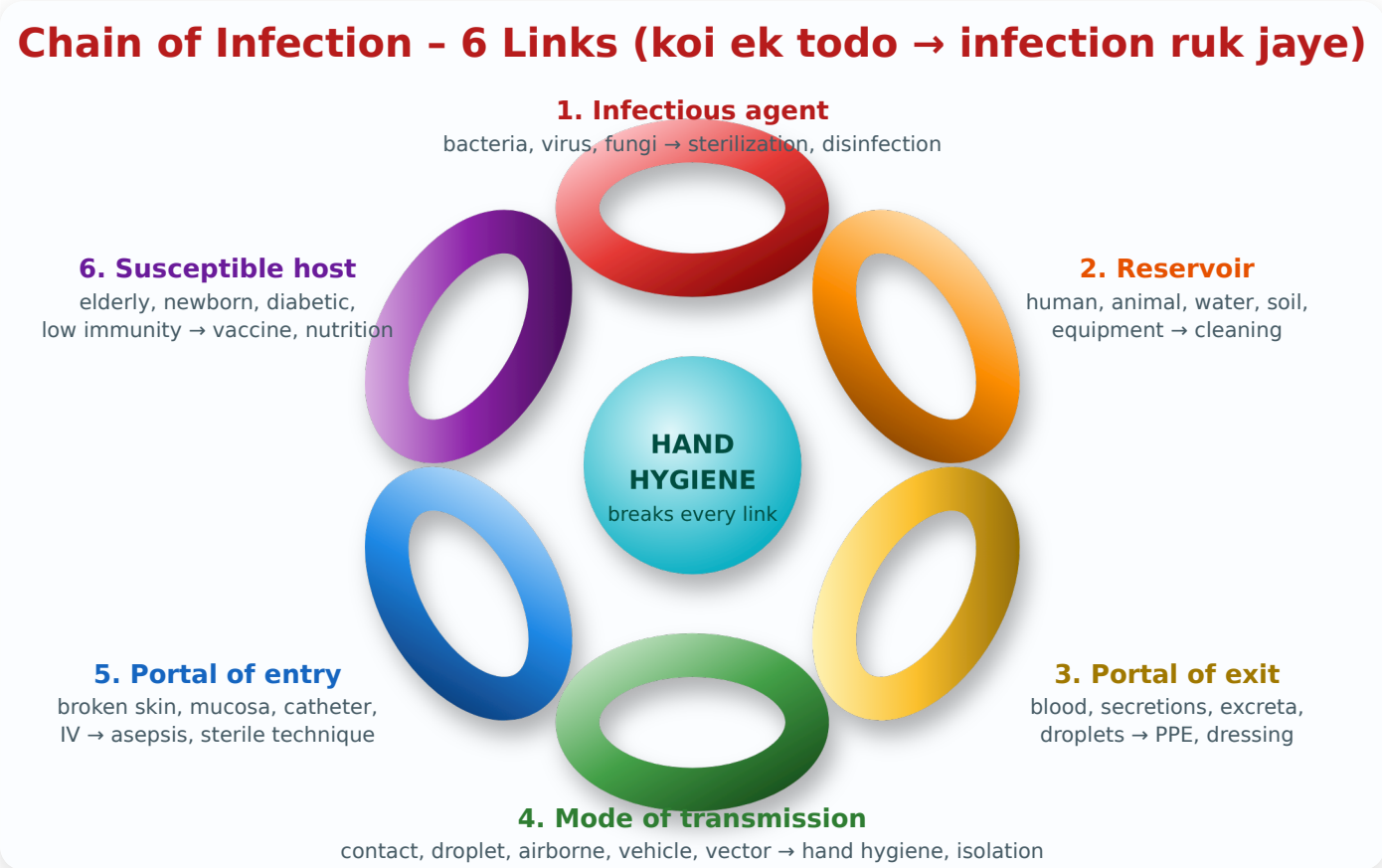
- Physical exam ka first technique – **inspection**.
- Abdomen me auscultation – **inspection ke baad, palpation se pehle**.
- Temperature palpate karte hain – **dorsum (back) of hand** se; vibration – **palm/ ulnar surface**.
- Percussion over liver – **dull**; over stomach – **tympanic**; emphysema – **hyper-resonant**.
- Bell of stethoscope – **low-pitched** sounds (murmurs, S3, S4).
- Skin turgor in elderly – **over sternum/ clavicle**.
- Normal capillary refill – **< 2–3 sec**.
- Muscle power against gravity but not resistance – **grade 3**.
- Clubbing angle – **> 180°**.
- BMI 27 (Indian) – **obese** (≥ 25).

7 Infection Control, Asepsis, Sterilization & BMW

7.1 Infection – Basics

- **Infection** – pathogen ka body me entry + multiplication → host response. **Colonization** – present but no disease. **Contamination** – object/ surface par microbes.
- **Stages of infection**: **Incubation** (entry → first symptoms) → **Prodromal** (non-specific symptoms – most infectious) → **Illness (acute)** → **Convalescence** (recovery).
- **Local** (redness, swelling, pain, heat, loss of function) vs **Systemic** (fever, malaise, ↑ WBC, ↑ ESR).
- Body defences: **1st line** – skin, mucosa, cilia, tears (lysozyme), stomach acid, normal flora; **2nd line** – inflammation, phagocytosis, fever, complement; **3rd line** – immune response (antibodies, T cells).
- **Normal WBC 4,000–11,000**; **neutrophils ↑ – bacterial**; **lymphocytes ↑ – viral/ TB**; **eosinophils ↑ – allergy/ parasites**.

Chain of infection



Chain of infection – 6 kadiyan aur unhe todne ke tareeke

- **Modes of transmission**: **Contact** (direct – touch; indirect – **fomites** = contaminated objects), **Droplet** (> 5 µm, travels **≤ 1 m/ 3 ft** – cough, sneeze), **Airborne** (< 5 µm droplet nuclei, long distance), **Vehicle** (water, food, blood), **Vector** (mosquito, fly – mechanical/ biological).

7.2 Health Care-Associated Infection (HAI / Nosocomial)

- Infection acquired in hospital – **not present/ incubating at admission**, appears **≥ 48 hrs after admission** (or within 3 days of discharge / 30 days of surgery).
- **Most common HAI – UTI (CAUTI)** (~30–40 %) → then **surgical site infection (SSI)**, **pneumonia (VAP)**, **bloodstream (CLABSI)**.
- Common organisms: **E. coli, Staph aureus (MRSA), Pseudomonas, Klebsiella, Enterococcus (VRE), Candida, C. difficile, Acinetobacter**.
- **Iatrogenic infection** – due to treatment/ procedure. **Exogenous** – from outside (other patient, staff); **Endogenous** – patient's own flora. **Cross infection** – patient to patient.
- **Single most important measure to prevent HAI = HAND HYGIENE** (Ignaz **Semmelweis** – 1847, Vienna – handwashing reduced puerperal fever – "**Father of hand hygiene/ infection control**"). **Joseph Lister** – antiseptic surgery (**carbolic acid/ phenol**, 1867) – "Father of antiseptic surgery".
- **Hospital Infection Control Committee (HICC)** – **Infection control nurse (ICN)** – surveillance, training, audit.

7.3 Hand Hygiene

Hand hygiene – WHO numbers

- **5 moments** (WHO 2009): (1) Before touching patient, (2) Before clean/ aseptic procedure, (3) After body fluid exposure risk, (4) After touching patient, (5) After touching patient surroundings.
- **Soap & water (handwash)**: **40–60 sec**; when hands **visibly soiled**, after **C. difficile/ spore-formers**, norovirus, before eating, after toilet.
- **Alcohol-based hand rub (ABHR, 60–80 % alcohol)**: **20–30 sec**; **preferred for routine** when not visibly soiled; **3–5 ml** (palmful).
- **Surgical hand scrub/ antisepsis**: **2–6 min** (traditional 5–10 min) with chlorhexidine/ povidone-iodine; **fingertips → elbows**; hands **held above elbow** (water flows from clean to less clean – hands to elbow); dry with **sterile towel**.
- **Medical handwash**: hands **lower than elbows** (water flows to fingertips/ sink).
- **WHO 6 steps** (7 with wrists): palm to palm → back of hands (right over left & vice versa) → palm to palm fingers interlaced → backs of fingers (interlocked) → **rotational rubbing of thumbs** → rotational rubbing of fingertips in palm (+ wrists).
- Nails short (< 0.5 cm/ ¼ inch), **no artificial nails, no rings/ watches**, cover cuts. Dry with **paper towel**; **turn off tap with towel** (elbow/ foot taps best).
- **5 May = World Hand Hygiene Day**; **15 October = Global Handwashing Day**.

7.4 Personal Protective Equipment (PPE)

- **Gloves**: **clean (examination)** – contact with body fluids; **sterile** – sterile procedures. Remove gloves – **glove-to-glove, skin-to-skin** technique; hand hygiene after removal. Gloves **don't replace hand hygiene**.
- **Gown** – fluid-resistant, back-tied; **Mask** – surgical (droplet, 3-ply); **N95 respirator** (filters ≥ **95 %** of particles ≥ **0.3 µm**; **fit-test**/ seal check; airborne – TB, measles, chickenpox, COVID aerosol); **goggles/ face shield**; cap, shoe cover.

- **Donning order (CDC)**: **Gown → Mask/ respirator → Goggles/ face shield → Gloves** ("**G-M-G-G**"; gloves last, over gown cuffs).
- **Doffing order (CDC)**: **Gloves → Goggles/ face shield → Gown → Mask/ respirator** (mask last – after leaving room) → hand hygiene. (Alternate: gown & gloves together.)
- Mask change – when **moist** (surgical mask effective ~4–6 hrs/ till damp); don't touch front.

7.5 Isolation Precautions (CDC 2007)

Precaution	Diseases	Key measures
Standard precautions (Universal 1985 + Body substance isolation)	**ALL patients – all blood, body fluids, secretions, excretions (except sweat), non-intact skin, mucosa**	Hand hygiene, PPE as needed, safe injection, respiratory hygiene/ cough etiquette, sharps safety
Contact	**MRSA, VRE, C. difficile, scabies, lice, impetigo, herpes simplex/ zoster (localized), wound infection, RSV**, diarrhoea (rotavirus)	Gloves + gown on entering, **dedicated equipment** (stethoscope, BP cuff), private/ cohort room
Droplet	**Influenza, pertussis, diphtheria (pharyngeal), mumps, rubella, meningococcal meningitis, Hib, streptococcal pharyngitis, adenovirus, mycoplasma pneumonia**	**Surgical mask** within 3–6 ft, private room/ cohort (beds > 3 ft apart), patient masked during transport
Airborne	**TB (pulmonary/ laryngeal), measles (rubeola), chickenpox (varicella), disseminated herpes zoster**, SARS, COVID (aerosol procedures)	**N95**, **AIIR – negative pressure room, 6–12 air changes/hr**, door closed, patient surgical mask during transport
Protective (reverse) isolation	**Neutropenic, burns, transplant, chemotherapy** patients	Protect **patient from outside**; positive pressure, HEPA, no fresh flowers/ plants, low-microbial diet

♫ **Airborne – "My Chicken Hez TB"**

Measles, **Chicken**pox, disseminated **He**rpes **z**oster, **TB** → Airborne + N95.
Droplet – "**SPIDERMAN**": **S**epsis/ **S**carlet fever/ **S**trep, **P**neumonia/ **P**ertussis, **I**nfluenza, **D**iphtheria, **E**pidemic parotitis (mumps), **R**ubella, **M**eningitis (meningococcal), **A**denovirus, **N** – Neisseria.

- **Cough etiquette** – cover mouth/ nose with tissue or **elbow (sleeve)**, dispose tissue, hand hygiene, mask, distance > 3 ft.

7.6 Asepsis

D **Medical vs Surgical asepsis**

Medical asepsis (clean technique) – microorganisms ki **sankhya kam karna & spread rokna** – handwashing, clean gloves, cleaning, disinfecting, isolation; e.g. **oral medication, bed bath, enema, NG feeding**.
Surgical asepsis (sterile technique) – **all microorganisms including spores ko destroy/ exclude** karna – **sterile field**; e.g. **catheterization, injections, IV cannulation, surgical dressing, OT, sterile specimen**.

Principles of sterile technique (sterile field rules)

- **Sterile touches sterile only** – sterile + unsterile = **unsterile**.
- Sterile objects **below waist level** or out of sight = **unsterile**.
- **1 inch (2.5 cm) border** of sterile field = **unsterile**.
- Sterile field **prolonged exposure to air** → contaminated; don't talk/ cough/ sneeze/ reach over sterile field.
- **Moisture → strike-through contamination** (wet sterile drape = contaminated).
- **Doubt = consider contaminated**.
- **Never turn back** to sterile field; face it always.
- **Sterile forceps (Cheattle forceps)** – tip always **down**; container with antiseptic **changed every 24 hrs** (replaced by individual sterile forceps now).
- **Pouring sterile solution**: label towards palm, **discard first few ml (lip cleaning)**, pour from **4–6 inches (10–15 cm)** height without splashing, don't touch bottle to container.
- Open sterile package: **first flap away from you**, then sides, **last flap towards you**.
- Sterile pack checks: **sterilization date/ expiry, indicator tape colour change**, intact, dry. **Event-related sterility** – sterile till package damaged/ wet (not fixed time) – old rule autoclaved packs ~**7 days (cloth wrap)**.
- **Sterile gloves**: **open gloving** (ward procedures) – **first glove picked by cuff (inside – skin to skin)**, second glove picked by **gloved fingers under the cuff (glove-to-glove)**. **Closed gloving** – OT with gown (hands inside sleeves).
- Hands **above waist & in front**.

7.7 Sterilization & Disinfection

Term	Meaning
Sterilization	**All microorganisms + spores** destroyed
Disinfection	Pathogens destroyed **except spores** (on inanimate objects)
Antisepsis	Disinfection **on living tissue (skin)** – antiseptic
Sanitization	Microbe count reduced to safe level (public health) – cleaning
Decontamination	Removal of pathogens to make items safe to handle (first step – e.g. 0.5 % hypochlorite soak 10 min)
Bacteriostatic	Growth inhibit (not kill)
Bactericidal	Kill bacteria
Fumigation	Room disinfection by gas (formaldehyde)

Spaulding classification (1957)

Category	Contact	Example	Processing
Critical	**Sterile tissue/ blood**	Surgical instruments, **catheters, needles, implants**	**Sterilization**
Semi-critical	**Mucous membrane/ non-intact skin**	**Endoscopes**, laryngoscope blade, **thermometer (rectal/ oral)**, respiratory equipment	**High-level disinfection (HLD)**

Category	Contact	Example	Processing
Non-critical	**Intact skin**	**BP cuff, stethoscope, bedpan, bed rails, crutches**	**Low/ intermediate-level disinfection**

Methods of sterilization

Method	Details	Used for
Autoclave (moist heat under pressure – steam) – most common, most reliable/ efficient	**121 °C, 15 psi (lb/inch²), 15–20 min** (wrapped 30 min); **134 °C, 30 psi, 3–4 min** (flash/ prevac)	**Surgical instruments, linen, dressings, glassware, culture media** – not for heat-sensitive/ oils/ powders
Hot air oven (dry heat)	**160 °C × 2 hrs** (or **170 °C × 1 hr**, **180 °C × 30 min**)	**Glassware, powders, oils, petroleum jelly, sharp instruments** (no rust/ blunt)
Boiling	**100 °C, 10–30 min** (min 10–20) – **NOT sterilization (spores survive)** – disinfection	Home use
Incineration	Burning to ashes (> 800–1000 °C)	**Anatomical waste**, dressings, infected material
Flaming	Direct flame	Inoculating loops, needle tips (lab)
Ethylene oxide (ETO) gas	Low temp (~37–55 °C), **aeration 8–12 hrs** (toxic, carcinogenic)	**Heat-sensitive – plastics, rubber, catheters, endoscopes, cardiac pacemakers**
Formaldehyde gas	Fumigation	Rooms, OT (now less)
Plasma sterilization (H_2O_2 gas plasma)	Low temp, short cycle	Heat & moisture sensitive
Radiation – Gamma (ionizing)	**Cold sterilization**	**Disposable syringes, needles, catheters, gloves, sutures** (industrial)
UV radiation (non-ionizing)	Surface disinfection only	OT/ lab air & surfaces
Filtration	Membrane filters (0.22 µm)	**Heat-sensitive liquids/ vaccines, sera**; HEPA (air)
Glutaraldehyde 2 % (Cidex)	**Sterilant: 10 hrs**; **HLD: 20 min**	**Endoscopes**, bronchoscopes

Sterilization indicators

- **Physical** – temperature, pressure, time gauges.
- **Chemical** – **autoclave indicator tape** (white lines → **black/ brown stripes**), **Bowie-Dick test** (daily air removal test, prevac autoclave).
- **Biological (most reliable)** – spores: **Geobacillus (Bacillus) stearothermophilus** – **autoclave**; **Bacillus atrophaeus (subtilis)** – **dry heat & ETO**.

Common disinfectants & antiseptics

Agent	Concentration / Use
Sodium hypochlorite (bleach)	**1 % – blood/ body fluid spills** (10 min contact); **0.5 %** – surface, decontamination of instruments/ gloves; 0.1 % – general surfaces

Agent	Concentration / Use
Alcohol (ethyl/ isopropyl)	**70 %** (most effective – water helps penetration) – skin antisepsis, thermometer, stethoscope; not sporicidal
Povidone-iodine (Betadine)	**5–10 %** – skin prep, wound; allergy (iodine)
Chlorhexidine gluconate	**2–4 %** skin prep, **0.12–0.2 % mouthwash** (VAP prevention), hand scrub – **longest residual action**
Hydrogen peroxide	**3 %** – wound cleaning (effervescence), 6 % disinfectant
Phenol (carbolic acid)	1–5 % – floors, excreta; **Lister** used; Lysol, cresol
Savlon	Chlorhexidine + cetrimide
Dettol	Chloroxylenol
Potassium permanganate ($KMnO_4$)	**1 : 1000 – 1 : 10,000** – wound, sitz bath, gastric lavage (some poisons)
Gentian violet	1 % – fungal, oral thrush
Formalin	40 % formaldehyde – specimen preservation (**10 % formalin**), fumigation
Lime (calcium oxide)	Faeces/ excreta disinfection
Bleaching powder	33 % available chlorine – water chlorination (**0.5 ppm residual chlorine** after 1 hr), latrines
Glutaraldehyde 2 %	Endoscopes (HLD)
Ortho-phthalaldehyde (OPA) 0.55 %	HLD – 12 min

- **Terminal disinfection** – **after patient discharge/ death/ transfer** – room & unit cleaned. **Concurrent disinfection** – **during illness** (daily, immediately) – excreta, linen.
- **Linen**: soiled linen in **red/ yellow bag** (as per hospital), **don't shake**, hold away from uniform, don't put on floor.

7.8 Biomedical Waste Management

Biomedical Waste Segregation – BMW Rules 2016 (amended 2018)



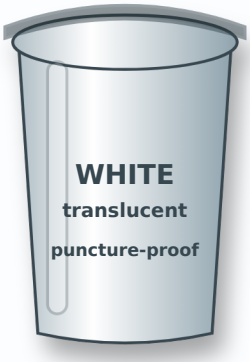
Infectious, anatomical
- Human/animal body parts
- Soiled dressing, cotton, plaster, blood bags
- Expired/ discarded drugs
- Chemical & lab waste

→ Incineration / deep burial



Contaminated recyclable
- IV sets, tubings, bottles
- Catheters, urine bags
- Syringes (without needle)
- Gloves, vacutainers

→ Autoclave + shred → recycle



Waste sharps
- Needles, syringes with fixed needles
- Scalpel, blades, lancets
- Any metal that can cut

→ Autoclave + shred / encapsulation



Glassware
- Broken/ discarded glass
- Vials, ampoules
- Metallic body implants

→ Disinfect → recycle

General (non-infectious) waste → BLACK bin (dry) / GREEN bin (wet)
BMW 48 hrs ke andar treat/ dispose · Bag 3/4 bharne par seal · Bar-code label

BMW segregation – 4 colour categories (BMW Rules 2016)

- **BMW Management Rules 2016** (Ministry of Environment, Forest & Climate Change), **amended 2018 & 2019** – replaced 1998 rules.
- **Segregation at source** (point of generation) – **most important step**.
- **4 categories**: **Yellow** (anatomical, soiled waste, expired medicines, chemical, lab), **Red** (contaminated **recyclable plastic** – tubing, catheters, IV sets, syringes without needles, gloves), **White translucent** (**sharps** – puncture-proof, leak-proof, tamper-proof), **Blue** (**glassware** – cardboard box with blue marking; metallic implants).
- **Untreated waste** storage **max 48 hrs**; **bar-code** labelling; **chlorinated plastic bags banned** (dioxins on burning); **pre-treatment of lab/ highly infectious waste** on-site (autoclave) before disposal.
- **Liquid waste** – disinfect before draining. **Cytotoxic drugs** – yellow (incineration at > 1200 °C).
- **Biohazard symbol**; incinerator: primary chamber **800 ± 50 °C**, secondary **1050 ± 50 °C**.

Needle-stick/ sharps injury

- Prevention: **NO recapping** (if must – **one-handed scoop technique**), dispose immediately in sharps container (**¾ full → seal**), **needle cutter/ destroyer**, safety-engineered devices, never bend/ break needles by hand, Hep B vaccination of staff.
- **Immediate action**: **wash with soap & running water** (do not scrub, **don't squeeze/ suck**, no bleach on skin), mucosa – flush with water, **report** to in-charge, test source & self (HIV, HBV, HCV), **PEP for HIV within 2 hrs (max 72 hrs) – 28 days**, HBV – HBIG + vaccine if non-immune.
- Risk of transmission after needle-stick: **HBV ~30 % (6–30 %)** > **HCV ~1.8 %** > **HIV ~0.3 %** ("**Rule of 3s: 30, 3, 0.3**").

PYQ style

- Most effective method to prevent cross infection – **hand washing**.
- Autoclave – **121 °C, 15 lb/inch², 15–20 min**.
- Biological indicator for autoclave – **Geobacillus stearothermophilus**.
- Hot air oven – **160 °C for 2 hrs** – glassware, oils, powders.
- Endoscope disinfection – **2 % glutaraldehyde (Cidex)**.
- Blood spill – **1 % sodium hypochlorite**.
- Disposable syringes sterilized by – **gamma radiation**.
- Most common HAI – **UTI**.
- Measles patient – **airborne** precautions; MRSA – **contact**; Influenza – **droplet**.
- PPE removal – **gloves first**; donning – **gloves last**.
- Ethylene oxide – **heat-sensitive (plastic, rubber) items**.
- First flap of sterile pack opened – **away from the nurse**.

8 Personal Hygiene, Bed Making & Comfort Devices

8.1 Personal Hygiene – Principles

- Purposes: cleanliness, **comfort**, circulation stimulation (massage), **prevent infection & pressure ulcers**, assessment opportunity (**skin inspection**), self-esteem, relaxation.
- General rules: **privacy** (screen, close door), explain, collect articles, **warm room (no draughts)**, **water temperature 43–46 °C (110–115 °F)** for bath (test with **elbow/ wrist/ thermometer**), clean → dirty, **top → bottom**, **distal → proximal** strokes on extremities (venous return), encourage self-care, **expose only the part being cleaned**.
- **Morning care (AM care)**: toileting, oral care, wash face & hands, bed straightening. **HS care (hour of sleep / evening care)**: toileting, oral care, back rub, bed straightening, comfort for sleep. **Early morning care** – before breakfast.

8.2 Bathing

Type	Description
Complete bed bath	Nurse bathes **dependent/ bed-ridden** patient fully
Partial bed bath	Only parts that cause odour/ discomfort – **face, hands, axillae, back, perineum**
Self-help bath / Assisted	Patient bathes, nurse assists
Tub bath, Shower	Ambulatory; check temp; non-slip mat
Sponge bath (tepid sponging)	**Fever reduction** – water 27–34 °C (~1 °C below body temp)
Therapeutic baths	**Sitz bath** (perineal/ rectal – 40–43 °C, **15–20 min**), **oatmeal/ starch bath** (pruritus), **medicated** ($KMnO_4$), **saline**, cold bath
Bag bath / chlorhexidine bath	Pre-moistened no-rinse cloths (ICU)

Bed bath sequence

Eyes (inner → outer canthus, plain water, no soap, separate corner of cloth each eye) → face → ears → neck → **arms (far arm first; distal → proximal, long firm strokes – venous return)** → hands (can soak) → chest (breast – circular) → abdomen → **legs** → feet → **change water** → back & buttocks (**back rub**) → **perineum last** (change water & gloves).
Change water – when cool, soapy or dirty, and **before perineal care**.

8.3 Oral Care (Mouth care)

- Purposes: remove plaque/ debris, prevent **halitosis, stomatitis, dental caries, parotitis, aspiration pneumonia (VAP)**, improve appetite.
- **Brushing**: soft brush, **45° angle to gum line**, **from gum to crown** (vibrating/ circular – Bass technique), **2 times/day, 2 min**; floss once daily; tongue cleaning.

- **Special mouth care** – for **unconscious, NPO, dehydrated, febrile, O_2 therapy, NG tube, mouth breathers, chemotherapy** patients – every **2–4 hrs**.

! Oral care of UNCONSCIOUS patient

- Position – **side-lying (lateral) with head turned to side / head of bed lowered** → fluid mouth se bahar nikle (**aspiration prevention**) – **most important point**.
- **Suction ready** at bedside.
- Use **padded tongue blade / bite block** to keep mouth open – **never use fingers**.
- **Small amount of solution**; use swab/ toothette; don't pour fluids.
- Lips – **petroleum jelly/ lip balm** (dryness, cracks).
- Solutions: **NS**, **chlorhexidine 0.12–0.2 %**, sodium bicarbonate (1 tsp in 500 ml), **glycothymoline**, H_2O_2 (diluted, 1:4 – short-term, can damage mucosa); **lemon-glycerine swabs ✗** (drying, erode enamel).

- **Dentures**: remove with gauze – **upper first** (break suction by pushing gently), lower next; clean over **basin of water/ towel (prevents breakage)**, **cool/ lukewarm water** (not hot – warping), store in **labelled cup with water**/ solution when out; reinsert – upper first. Unconscious patients – dentures removed & stored.
- **Oral conditions**: halitosis (bad breath), **stomatitis** (inflammation of mouth), **gingivitis** (gums), **glossitis** (tongue), **cheilosis** (cracking of lips/ corners – **B2 deficiency**), **parotitis** (salivary gland), **sordes** (crust on lips/ teeth – fever, dehydration), **dental caries**, **pyorrhoea**, **xerostomia** (dry mouth), **oral thrush/ candidiasis** (white patches – nystatin).

8.4 Hair & Scalp Care

- Brush/ comb daily (distribute sebum), from **ends upward to scalp** for tangles; braid long hair (**not tight** – alopecia).
- **Shampoo in bed** – patient **diagonal/ to edge of bed**, Kelly pad/ rubber sheet trough to bucket, cotton in ears, eyes covered; **contraindicated** – neck/ head injury (doctor's order), fresh surgery, very ill.
- **Pediculosis** – lice: **capitis (head)**, **corporis (body – lives in clothing seams)**, **pubis (crab louse – sexually transmitted)**. **Nits** – eggs glued to hair shaft (white, oval, can't be flicked off unlike dandruff).
- **Treatment**: **Permethrin 1 % (head lice – first line)** / 5 % for scabies, **Malathion 0.5 %**, gamma benzene hexachloride (**Lindane** – neurotoxic, not first line); apply as directed (permethrin 10 min, rinse), **fine-tooth comb (nit comb)**, **repeat after 7–10 days**; wash clothes/ linen in **hot water (≥ 60 °C)** or seal in plastic bag **2 weeks**; check all family members; **don't use kerosene**; nurse – gloves, gown, cap.
- **Alopecia** – hair loss; **hirsutism** – excess hair (female); **dandruff** – scaling.

8.5 Eye, Ear & Nose Care

- **Eye care**: **clean from inner canthus → outer canthus** (away from nasolacrimal duct), separate swab/ cotton for each eye & each stroke, **NS/ sterile water**. **Unconscious** – no blink reflex → **artificial tears/ lubricant, eye pads/ moist saline gauze, tape eyelids** (corneal abrasion/ ulcer prevention).

- **Contact lens** – remove for unconscious/ surgery; hard lens – slide off; soft lens – pinch off; store in labelled case with solution.
- **Artificial eye (prosthesis)** – remove by pulling lower lid down & pressing; clean with warm NS; store in water/ saline in labelled container.
- **Ear care**: clean external ear with washcloth; **no cotton buds/ hairpins inside canal** (wax impaction, TM perforation); **hearing aid** – turn off & remove battery when not in use (night), clean with dry cloth, **no water**, avoid hairspray.
- **Nose care**: clean nostrils with moist cotton applicator (not too deep); NG/ O_2 tube patients – check pressure areas.

8.6 Perineal Care & Foot/ Nail Care

- **Perineal care (pericare)**: Female – **front to back (pubis → anus)**, separate swab each stroke, **labia – outer then inner**, centre last (meatus → down). Male – retract foreskin (uncircumcised), clean **meatus outward in circular motion**, **replace foreskin** (paraphimosis prevention), then shaft, scrotum. Indications – catheter, post-delivery, incontinence, surgery.
- **Foot care**: soak in warm water (not for diabetics – avoid prolonged soaking), dry **between toes**, lotion **but not between toes**. **Nails – cut straight across** (toenails) – corners not cut (ingrown nail); **diabetics/ PVD – no cutting by nurse; podiatrist** (or file).
- Fingernails – **curved (oval)** cut. Nail biting, **paronychia**, **onychomycosis** (fungal).

8.7 Back Care / Back Rub (Back massage)

- Purposes: **stimulate circulation**, **prevent pressure ulcers**, relaxation & sleep, comfort, assess skin.
- Position: **prone or side-lying**. Lotion/ powder (not both); duration **3–5 min**; warm hands.
- **Massage strokes**: **Effleurage** (long, smooth, gliding strokes – most common, start & end), **Petrissage** (kneading – lifting/ squeezing muscles), **Tapotement** (percussion – tapping/ hacking), **Friction** (deep circular), **Vibration**.
- **Contraindications**: **reddened bony prominences (don't massage – tissue damage)**, **DVT/ thrombophlebitis** (embolism), rib/ vertebral fractures, burns, **cardiac surgery (some)**, skin lesions.
- Areas to watch: sacrum, scapulae, heels, elbows, occiput, trochanters.

8.8 Bed Making

- Purposes: comfort, safety, **clean wrinkle-free bed (pressure ulcer prevention)**, neat appearance, prevent cross-infection.
- **Principles**: **hand hygiene**; **mitred (hospital) corners** – square, neat (**45° angle**); linen **held away from uniform**; **don't shake/ fan linen** (spreads microbes); soiled linen **never on floor**; **bottom sheet tight & wrinkle-free**; **toe pleat** in top sheet (pressure on toes ↓ – footdrop prevention); open end of pillowcase **away from door**; complete **one side first** (save time & energy – body mechanics); bed at **working height (waist level)**, wheels locked; **economy of time, material & energy**.
- Linen order (bottom to top): **mattress → (mattress cover) → bottom sheet → mackintosh (rubber sheet) → draw sheet** (mackintosh & draw sheet placed **under buttocks – shoulder to knee/ middle third**) → top sheet → blanket → counterpane/ bedspread → pillows.
- **Draw sheet** – can be changed without changing whole bed; **mackintosh** – protect mattress (incontinence).

Type of bed	Purpose / Feature
Closed bed (unoccupied)	After discharge/ **terminal disinfection**, waiting for new patient; top linen pulled up to head
Open bed (unoccupied)	Ready for a patient who is up/ new admission – top linen **fan-folded to foot**
Occupied bed	Patient **can't get out of bed** – make with patient in bed (side to side – roll patient); **2 nurses** ideal
Admission bed	Ready for new admission – open bed
Post-operative / Surgical / Recovery / Ether bed	Receive patient from OT – top linen **fan-folded to side (far side/ opposite to door)**, mackintosh at head & middle, **hot water bottles (removed before patient placed)**, **kidney tray, tissue, suction, O_2, BP apparatus, IV stand** ready; pillow **removed / placed vertically at head (headboard protection)**
Cardiac bed	**Cardiac table/ backrest**, patient sits up (**orthopnoeic/ high Fowler's**), knee rest – **HF, dyspnoea**
Fracture bed	**Fracture boards (wooden) under mattress**, firm mattress, **Balkan beam/ overhead frame**, monkey bar (trapeze), cradle
Amputation bed	Top linen in 2 parts (stump exposed for observation), **sandbags, cradle, tourniquet at bedside**
Divided bed	Top linen in **2 halves** – leg/ lower limb dressing, fracture
Plaster bed	Fresh plaster cast – **firm mattress, fracture board, pillows with mackintosh**, **cast uncovered** to dry
Burn bed	Sterile linen, **bed cradle**, **air/ water bed**, warm room
Water bed / Air bed / Alternating pressure mattress	**Pressure ulcer prevention** (bed-ridden, paralysed)
Circo-electric bed / Stryker frame / Foster frame	Turning **spinal cord injury** patients (in alignment)
Clinitron (air-fluidized) bed	Burns, severe pressure ulcers
Rocking bed	Respiratory (poliomyelitis – historic)
Hammock (sling) bed	Pelvic fracture

8.9 Comfort Devices

Device	Use
Backrest (bed rest)	Fowler's position – dyspnoea, cardiac
Cardiac table (over-bed table)	Lean forward – **orthopnoea**
Bed cradle (arch)	Keep **top linen off** – burns, fracture, gout, wound, amputation, cast drying
Footboard / foot rest	**Prevents footdrop** (plantar flexion), maintains dorsiflexion
Sandbags	Immobilize/ support limb, prevent rotation

Device	Use
Trochanter roll	**Prevents external rotation of hip** (rolled blanket from iliac crest to mid-thigh)
Hand roll / hand splint	**Prevents finger flexion contracture**, functional position of hand
Trapeze bar (monkey bar)	Patient lifts self – **upper-body strengthening**, repositioning (contraindicated – some spinal/ abdominal surgery)
Side rails	Safety (fall prevention) – but entrapment risk; not a restraint substitute
Air ring / Donut cushion	**Old practice – now NOT recommended** (↓ circulation in centre → ulcer)
Knee rest/ knee roll	Prevents sliding down; but **prolonged knee gatch – DVT (popliteal compression)**
Heel protector / Elbow protector	Pressure relief
Pillows	Support, positioning
Hot water bottle	Warmth – **fill ⅔, expel air, temp 52 °C (adults), 40–46 °C (children/ elderly/ unconscious)**, cover, **never on unconscious/ paralysed/ diabetic** (burns)
Ice cap/ ice collar	Cold application (post-tonsillectomy, headache)

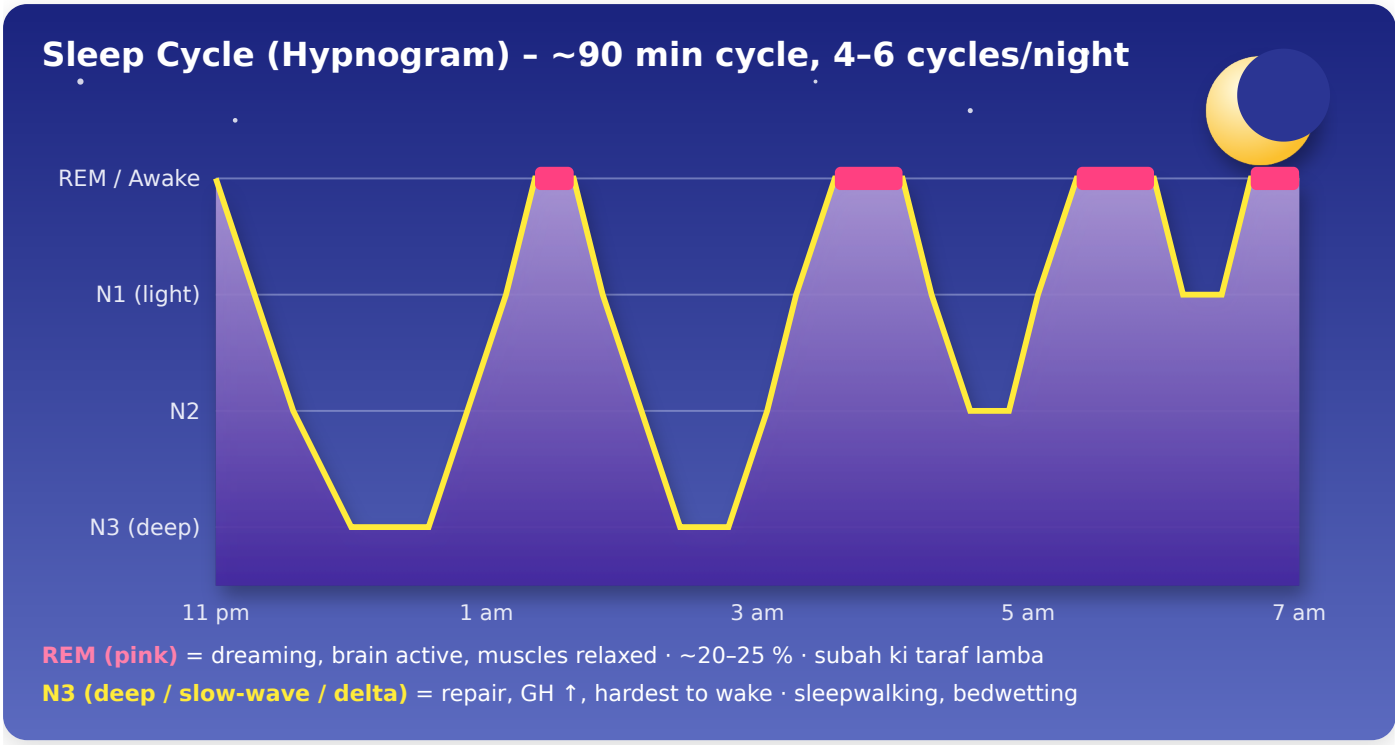
✎ PYQ style

- Eye cleaning – **inner to outer canthus**.
- Unconscious patient oral care – **side-lying position**, suction ready.
- Bath water temperature – **43–46 °C (110–115 °F)**.
- Perineal care female – **front to back**.
- Massage stroke – long gliding = **effleurage**; kneading = **petrissage**.
- Bed after discharge – **closed bed**; post-op bed – top linen **fan-folded to side**.
- Bed cradle used to – **keep weight of linen off the patient**.
- Footdrop prevention – **footboard**; external hip rotation prevention – **trochanter roll**.
- Head lice treatment – **permethrin 1 %**; repeat after **7–10 days**.
- Dentures stored in – **labelled cup with water**; remove **upper first**.
- Linen should not be – **shaken**; mitred corner angle – **45°**.

9 Rest, Sleep & Pain Management

9.1 Sleep – Physiology

- **Rest** – mental & physical calmness, relaxation (not necessarily sleep). **Sleep** – recurrent, altered state of consciousness; controlled by **RAS (reticular activating system)** & **BSR (bulbar synchronizing region)** in brainstem; **pineal gland – melatonin** (darkness ↑).
- **Circadian rhythm** – 24-hr biological clock (**suprachiasmatic nucleus of hypothalamus**); body temperature lowest early morning.



Raat bhar me 4–6 sleep cycles – NREM se REM tak

Stage	Features
NREM Stage 1 (N1)	**Lightest**, transition, few minutes, easily aroused; hypnic jerks
NREM Stage 2 (N2)	Light sleep, **~50 % of total sleep**, sleep spindles & K-complexes; vitals ↓
NREM Stage 3 (N3) – deep/ slow-wave/ delta sleep	**Deepest**, **hardest to arouse**, **tissue repair, growth hormone ↑**, muscles relaxed; **sleepwalking (somnambulism), night terrors, bedwetting (enuresis)** occur here
REM (Rapid Eye Movement) – paradoxical sleep	**Dreaming** (vivid), brain very active (EEG like awake), **muscle atonia**, irregular HR/ RR/ BP ↑, **memory consolidation & learning**; **~20–25 %** of adult sleep; first REM **~90 min** after sleep onset; REM periods **lengthen toward morning**

- **Sleep cycle ≈ 90–110 min**, **4–6 cycles/night**. Sequence: N1 → N2 → N3 → N2 → **REM** → N2…
- Newborn – **~50 % REM**; elderly – **↓ N3 (deep sleep)**, frequent awakening.

Sleep requirements (approx.)

Age	Hours/day
Newborn (0–3 months)	**14–17 (up to 16–18)**
Infant (4–11 months)	12–15
Toddler (1–2 yrs)	11–14
Preschool (3–5 yrs)	10–13
School age (6–13 yrs)	9–11
Adolescent (14–17 yrs)	8–10
Adult (18–64)	**7–9**
Elderly (65+)	**7–8** (but less deep sleep)

Sleep disorders

Disorder	Meaning
Insomnia	**Difficulty falling/ staying asleep** – most common sleep disorder
Hypersomnia	Excessive sleep
Narcolepsy	Sudden uncontrollable sleep attacks in daytime (goes directly into REM), **cataplexy** (sudden muscle weakness with emotion) – stimulants
Sleep apnoea	Breathing stops ≥ 10 sec during sleep; **obstructive (OSA – snoring, obesity) → CPAP**; central
Somnambulism	**Sleepwalking** (N3) – safety, don't wake forcefully
Somniloquy	Sleep talking
Bruxism	Teeth grinding (N2)
Enuresis	Bedwetting (children)
Night terrors (pavor nocturnus)	Child screams, can't be consoled, **no memory** (N3) vs **Nightmares** – frightening dreams (**REM**), remembers
Restless leg syndrome	Urge to move legs at night
Sleep deprivation	Irritability, fatigue, ↓ concentration, hallucinations; **ICU psychosis**

Promoting sleep – nursing measures

- **Sleep hygiene**: regular bedtime & wake time, **avoid caffeine (4–6 hrs before), alcohol, nicotine, heavy meals** near bedtime, **light snack/ warm milk (tryptophan)**, exercise (not 2–3 hrs before sleep), **avoid daytime naps** (if insomnia), bed only for sleep, dark-quiet-cool room, screens off.
- Hospital: **cluster nursing activities**, dim lights, reduce noise, back rub, **relieve pain**, toileting before sleep, comfortable position, bedtime rituals, **HS care**, relaxation.
- Sedatives/ hypnotics – last option (benzodiazepines – dependence; elderly – fall risk, confusion).

9.2 Pain

D Pain

IASP (2020): "An unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage."
Margo McCaffery (1968): "Pain is **whatever the experiencing person says it is**, existing **whenever he says it does**." → **Patient's self-report = most reliable indicator of pain**.

- **Nociception** – 4 processes: **Transduction** (stimulus → impulse) → **Transmission** (A-delta – fast, sharp, myelinated; C fibres – slow, dull, unmyelinated) → **Perception** (cortex – conscious awareness) → **Modulation** (endorphins, enkephalins – body's natural opioids).
- **Gate Control Theory – Melzack & Wall (1965)**: substantia gelatinosa in dorsal horn acts as gate; **large-fibre stimulation (rubbing, massage, TENS)** closes gate → pain ↓.
- **Specificity theory** (von Frey), **Pattern theory** (Goldschneider).

Types of pain

Type	Description
Acute	Sudden, < 3–6 months, protective, **sympathetic signs (↑ HR, BP, RR, sweating, dilated pupils)**
Chronic	> 3–6 months, persistent; **no sympathetic signs** (adaptation), depression, fatigue
Cutaneous (superficial)	Skin – sharp, burning, localized
Deep somatic	Bone, muscle, tendons – diffuse
Visceral	Organs – cramping, poorly localized, may be referred
Referred	Felt at site distant from origin – **MI → left arm/ jaw**; **gallbladder → right shoulder**; **diaphragm → shoulder tip**; **kidney → groin**; **appendix → periumbilical (early)**
Radiating	Spreads from origin (sciatica)
Phantom	In amputated limb – **real pain**
Neuropathic	Nerve damage – burning, shooting, tingling (diabetic neuropathy, post-herpetic) – **gabapentin, pregabalin, amitriptyline**
Intractable	Resistant to treatment
Psychogenic	No physical cause found (still real to patient)
Breakthrough pain	Flare above controlled chronic pain (cancer) → rescue dose

Pain assessment

- **PQRST / OLDCART**: **P**rovokes/ palliates, **Q**uality, **R**egion/ radiation, **S**everity, **T**iming.
- Scales:

Scale	Use
Numeric rating scale (NRS) 0–10	Adults (most common); **1–3 mild, 4–6 moderate, 7–10 severe**
Visual analogue scale (VAS)	**10 cm line** – no pain to worst pain

Scale	Use
Verbal descriptor scale	None – mild – moderate – severe
Wong-Baker FACES	**Children ≥ 3 yrs**, low literacy, language barrier
FLACC (Face, Legs, Activity, Cry, Consolability)	**2 months–7 yrs / non-verbal** (0–10)
CRIES / NIPS / PIPP	**Neonates**
Oucher scale	Children 3–12 (photos)
PAINAD	**Advanced dementia**
CPOT / BPS	**Ventilated/ ICU** non-verbal

- **Pain threshold** – minimum stimulus at which pain is felt (fairly same in all). **Pain tolerance** – maximum pain a person can bear (varies – culture, past experience, anxiety). **Anxiety ↑ → pain perception ↑**.
- **Reassess pain** after intervention: **IV opioid – 15–30 min**, **oral – 30–60 min**.
- Elderly/ cognitively impaired – pain **undertreated** – observe behaviour (grimacing, restlessness, guarding).

Pain management

Pharmacological:

- **Non-opioids**: **Paracetamol** (max **4 g/day adult**; hepatotoxic), **NSAIDs** (ibuprofen, diclofenac – GI bleeding, renal; **give with food**), aspirin (**not in children < 16 – Reye's syndrome**).
- **Opioids**: **Morphine** (gold standard), fentanyl (patch – 72 hrs), tramadol, codeine, pethidine (meperidine – avoid – toxic metabolite), buprenorphine. Side effects – **respiratory depression (most serious – check RR; < 12 hold)**, **constipation (most common – does not develop tolerance; give laxative prophylactically)**, nausea, sedation, **pinpoint pupils**, urinary retention, pruritus. Antidote – **Naloxone**.
- **Adjuvants (co-analgesics)**: antidepressants, anticonvulsants, corticosteroids, muscle relaxants.
- **WHO analgesic ladder (1986)**: Step 1 mild – non-opioid ± adjuvant → Step 2 moderate – **weak opioid (codeine, tramadol)** + non-opioid → Step 3 severe – **strong opioid (morphine)** + non-opioid. Principles: **by mouth, by the clock, by the ladder, for the individual, attention to detail**.
- **PCA (Patient-Controlled Analgesia)** – **only patient presses button** (not family/ nurse) – "PCA by proxy" dangerous; lockout interval.
- **Epidural analgesia**, nerve blocks, local anaesthetics (lignocaine).
- **Placebo** – not used to "test" pain (unethical without consent).
- **Tolerance** (need higher dose), **physical dependence** (withdrawal on stopping) ≠ **addiction** (psychological craving – rare in pain patients). **Equianalgesia** – dose conversion between opioids/ routes.

Non-pharmacological:

- **Physical**: **cutaneous stimulation – massage, heat & cold, TENS (Transcutaneous Electrical Nerve Stimulation), acupressure, acupuncture**, positioning, immobilization, exercise, contralateral stimulation.
- **Cognitive-behavioural**: **distraction** (music, TV, conversation – best for **short, acute** procedural pain), **relaxation/ deep breathing**, **guided imagery**, **biofeedback**, hypnosis, meditation, prayer, **humour**, **reiki/ therapeutic touch**.

- **Environmental**: quiet, dim light, comfortable.
- Patient education, **prevent pain** (premedicate **30 min before** painful procedure like dressing/ ambulation).

PYQ style

- Most reliable indicator of pain – **patient's self-report**.
- Gate control theory – **Melzack & Wall (1965)**.
- Deepest sleep stage – **NREM stage 3 (delta)**; dreaming – **REM**.
- Growth hormone release – **deep sleep (N3)**.
- Pain scale for 4-year child – **Wong-Baker FACES**; infant – **FLACC/ CRIES**.
- Opioid side effect that doesn't develop tolerance – **constipation**.
- Pain ki medicine dressing se kitni der pehle – **30 min**.
- Pain in left arm during MI – **referred pain**.
- Massage/ TENS work by – **gate control (closing gate)**.
- Warm milk promotes sleep due to – **tryptophan**.

10 Body Mechanics, Mobility, Positioning & Pressure Ulcers

10.1 Body Mechanics

D Body mechanics

Body ka **coordinated, efficient & safe use** (musculoskeletal + nervous system) during movement, lifting, turning – **nurse ki back injury (most common occupational injury)** aur patient injury rokne ke liye.

- **Principles**:
  - **Wide base of support** (feet apart, **shoulder width / ~30 cm**, one foot slightly forward) → **stability ↑**
  - **Lower centre of gravity** – **bend knees & hips, not the back** (squat)
  - Keep object **close to body** (centre of gravity) – lifting leverage
  - **Use large muscles of legs & thighs** (not back) – "**lift with legs**"
  - **Back straight**, avoid **twisting** (turn whole body – pivot on feet)
  - **Push, pull, slide or roll** rather than lift – **sliding/ rolling needs less energy than lifting**
  - Work at **waist height** (adjust bed)
  - Tighten **abdominal & gluteal muscles** before moving (internal girdle)
  - Get help/ **mechanical lift** for heavy patients; **friction ↓** (slide sheet)
  - Use body weight as counterbalance; move rhythmically
- **Centre of gravity** in standing adult – **2nd sacral vertebra (S2)**, centre of pelvis. **Line of gravity** must fall within base of support for balance.
- **Good posture (alignment)** – head erect, shoulders back, chest out, abdomen in, knees slightly flexed, feet parallel.

10.2 Range of Motion (ROM)

Movement	Meaning
Flexion	Angle ↓ (bending)
Extension	Angle ↑ (straightening to 180°)
Hyperextension	Beyond 180°
Abduction	**Away from midline**
Adduction	**Towards midline**
Internal/ external rotation	Turning towards/ away from midline
Circumduction	Circular movement (combination) – shoulder, hip
Pronation	**Palm down** (posterior)
Supination	**Palm up** (anterior) – "**soup in palm**"

Movement	Meaning
Inversion / Eversion	Sole **inward / outward**
Dorsiflexion	Toes **up** (towards shin)
Plantar flexion	Toes **down** (pointing) – **footdrop** = fixed plantar flexion
Opposition	Thumb touches fingertips
Radial/ ulnar deviation	Wrist sideways

- **Active ROM** – patient khud karta hai; **Active-assisted** – nurse partial help; **Passive ROM** – **nurse moves joints** (unconscious, paralysed) – **maintains joint mobility only, not muscle strength**. **Resistive** – against resistance (strength ↑).
- Each movement **3–5 times**, **2–3 times/day**, slowly, **support joint above & below**, **stop at point of pain/ resistance**; never force.
- **Contraindications to ROM**: acute inflammation/ arthritis flare, **fractures (unhealed)**, **DVT in that limb**, recent MI (passive caution), joint dislocation, torn ligaments.
- **Isotonic exercise** – muscle shortens, movement (walking, swimming, ROM) – **↑ muscle tone, mass, strength & circulation**. **Isometric** – **muscle contracts without joint movement** (quadriceps setting, gluteal setting) – **↑ strength**; **↑ BP & HR – avoid in cardiac patients**; useful for patients **in cast/ traction**. **Isokinetic** – constant-speed resistance (machines). **Aerobic** – O_2 use ↑ (cardiac endurance).
- **Quadriceps setting** – press knee into bed, hold 5–10 sec – **prepare for ambulation**, prevent DVT.

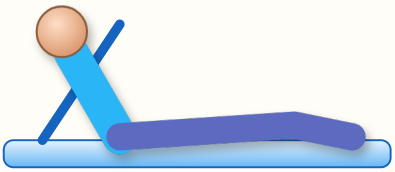
10.3 Hazards of Immobility

System	Complications
Musculoskeletal	**Disuse atrophy** (muscle mass ↓ ~3 %/day / 10–15 % per week), **contractures** (**footdrop**, hip flexion), **osteoporosis** (Ca loss from bones – disuse), **joint ankylosis**
Cardiovascular	**Orthostatic hypotension** (most common), **DVT/ thrombus** (**Virchow's triad – stasis, hypercoagulability, vessel injury**), ↑ cardiac workload, **dependent oedema**
Respiratory	**Hypostatic pneumonia**, **atelectasis** (secretions pool), ↓ chest expansion
Integumentary	**Pressure ulcers**
Urinary	**Urinary stasis**, **renal calculi** (hypercalciuria), **UTI**, retention/ incontinence
GI	**Constipation** (most common GI), anorexia, faecal impaction
Metabolic	↓ BMR, **negative nitrogen balance** (protein catabolism), **hypercalcaemia**, ↓ appetite
Psychosocial	Depression, boredom, **sensory deprivation**, isolation, altered sleep, dependence

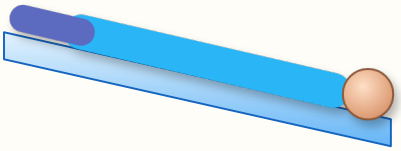
- Prevention: **turn every 2 hrs**, **early ambulation (most important)**, ROM, deep breathing & coughing, incentive spirometry, fluids 2–3 L, high-fibre & high-protein diet, **compression stockings/ SCD**, positioning, footboard, skin care.

10.4 Positioning

Important Patient Positions



Fowler's (45-60°) / High (90°)
Dyspnoea, HF, COPD, post-thyroid



Trendelenburg (head low)
Pelvic surgery; (shock me ab NOT preferred)



Sim's (left lateral, R knee bent)
Enema, rectal exam, unconscious



Supine / Dorsal recumbent
Spinal injury, post spinal anaes.



Prone (pet ke bal)
ARDS (proning), back surgery



Lithotomy (legs in stirrups)
Vaginal exam, delivery, TURP, cysto

More yaad rakho:
Knee-chest → sigmoidoscopy, cord prolapse · Orthopneic (table pe jhukna) → severe dyspnoea
Left lateral → enema, liver biopsy ke baad RIGHT side · Semi-Fowler (30°) → ↑ICP, NG feed, post-op
Modified Trendelenburg (sirf legs up) → shock · Pneumonectomy → operated side down / supine

Important patient positions

Position	Description	Uses
Supine/ Dorsal	Flat on back	Post spinal anaesthesia, spinal injury, rest
Dorsal recumbent	Supine + knees flexed, feet flat	Perineal care, female catheterization, vaginal exam
Prone	On abdomen, head to side	Back care, **after tonsillectomy (semi-prone)**, ARDS (proning), spine surgery, prevents hip flexion contracture
Lateral (side-lying)	On side, upper leg flexed	Rest, pressure relief, back care
Sims' (semi-prone)	**Left lateral**, **right knee & thigh flexed**, left arm behind	**Enema, rectal temperature/ exam**, suppository, **unconscious (drain secretions)**, perineal exam
Fowler's	HOB **45–60°** (semi-Fowler's **30–45°**, high Fowler's **60–90°**, low Fowler's 15–30°)	**Dyspnoea**, cardiac, eating, NG feeding, post-thyroidectomy, ↑ICP (30°)
Orthopnoeic	Sitting up, leaning forward over **over-bed table with pillow**	**Severe dyspnoea, asthma, COPD**, thoracentesis
Trendelenburg	Whole bed tilted – **head low, feet high** (~15–30°)	Pelvic surgery, **postural drainage (lower lobes)**; shock (old practice)
Reverse Trendelenburg	**Head high, feet low**	Head & neck surgery, GERD, hiatal hernia
Modified Trendelenburg	Supine, only **legs raised 20–30°**	**Shock**

Position	Description	Uses
Lithotomy	Supine, hips & knees flexed, **legs in stirrups**	**Vaginal exam, delivery, cystoscopy, TURP**, Pap smear
Knee-chest (genupectoral)	Kneeling, chest on bed, buttocks up	**Sigmoidoscopy, rectal exam**, cord prolapse, retroverted uterus
Jack-knife (Kraske)	Prone, hips flexed over break of table	**Proctological (rectal) surgery**, haemorrhoidectomy
Lateral kidney	Lateral with bridge under flank	Nephrectomy
Tripod	Sitting, leaning forward on arms	COPD (accessory muscles)

10.5 Moving, Lifting & Transfer

- **Moving up in bed**: lower HOB flat, patient bends knees & pushes with feet (on count of 3), **2 nurses + draw/ lift sheet** (avoid dragging – **shear**), trapeze.
- **Log rolling** – **spinal cord injury/ spinal surgery** – body moved as **one unit (like a log)**, **3–4 persons**, one person **controls head & neck** & gives commands, pillow between legs.
- **Dangling** – sit at bed edge with legs hanging **few minutes before standing** – **prevents orthostatic hypotension** (check dizziness, pulse).
- **Bed → wheelchair transfer**: wheelchair **at 45° angle to bed (parallel/ close), on patient's STRONGER (unaffected) side**, **wheels locked**, footrests up, patient wears non-skid shoes, **pivot on strong leg**, gait belt (transfer belt) at waist.
- **Bed → stretcher**: stretcher **parallel & touching bed**, both **locked**, same height, **3 persons (3-man lift)** or **slide board/ sheet**; stretcher par **side rails up & safety strap**; normal travel me **patient ke feet aage**, nurse **head end** par (airway observe).
- **Wheelchair down ramp** – **backwards** (nurse walks backwards); entering lift – **back in**.
- **Mechanical lift (Hoyer lift)** – heavy/ immobile patients.
- Hemiplegic – nurse stands on **affected (weak) side** while ambulating (support).

10.6 Ambulation Aids (Crutch, Cane, Walker)

Crutches

- **Axillary crutch length** = patient's height − **40 cm (16 inches)**; or **2–3 finger-widths (~5 cm) below axilla**; handgrip so **elbow flexed 20–30°**; crutch tips **15 cm (6 inch) in front & 15 cm to side** of feet (tripod position – standing stance).
- **Body weight on hands/ palms – NOT axilla** (**crutch palsy – radial/ brachial plexus nerve damage**).
- **Gaits**:
  - **4-point gait** – **most stable/ safest, slowest**; partial weight-bearing **both legs** (R crutch → L foot → L crutch → R foot) – arthritis, cerebral palsy
  - **2-point gait** – faster version of 4-point (R crutch + L foot together, then L crutch + R foot)
  - **3-point gait** – **non-weight bearing on one leg** (both crutches + affected leg forward together, then unaffected leg) – fracture/ amputation of one leg
  - **Swing-to / Swing-through** – **paraplegia** (both legs paralysed, with braces) – swing-through fastest
- **Stairs**: **UP – "Up with the GOOD"** (unaffected leg first → then crutches + affected leg). **DOWN – "Down with the BAD"** (crutches + affected leg first → then good leg). ("**Good goes to heaven, bad goes to hell**")
- Sitting: back to chair, both crutches in hand on **affected side**, grip armrest with other hand, lower.

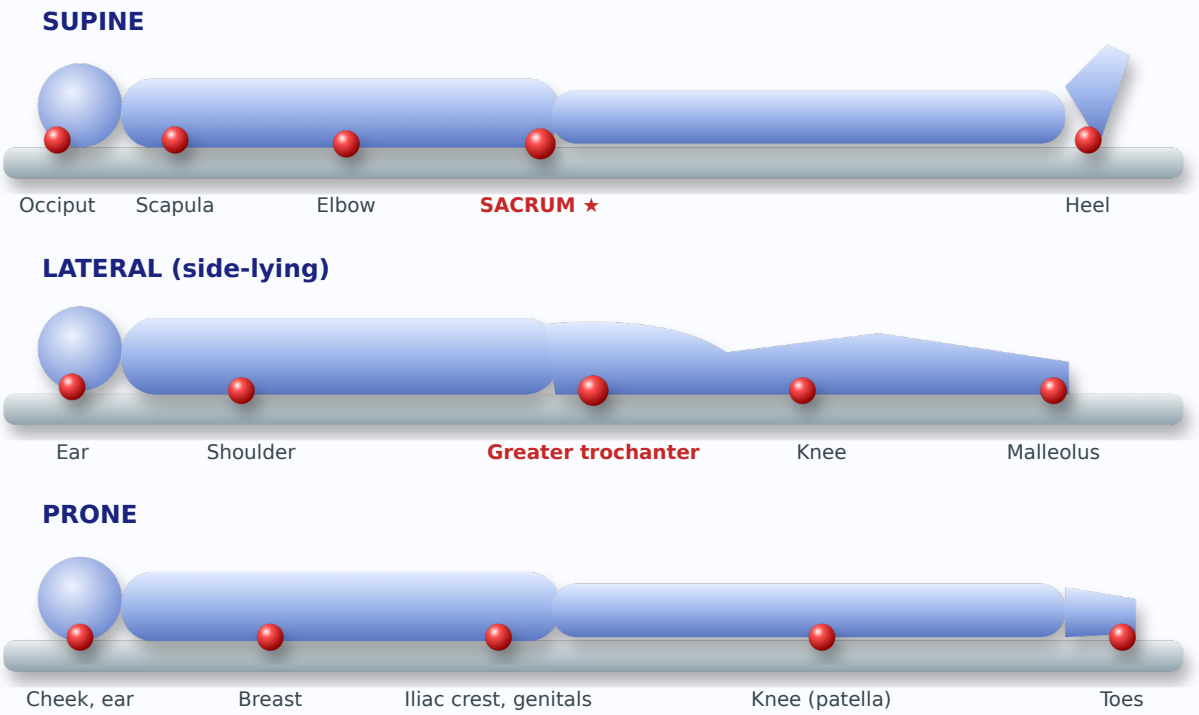
- **Cane**: held in hand **opposite (unaffected/ stronger side)** to affected leg; height – top at **greater trochanter / wrist crease** (elbow 15–30° flexed); move **cane + affected leg together** → then unaffected leg. Quad cane (4 legs) – more stable.
- **Walker**: most stable aid; height – wrist crease (elbow 15–30° flexed); **lift walker & move 15–20 cm (6–8 inch) forward → step with affected (weak) leg → then unaffected leg**; **never on stairs**; wheeled walker – rolled.
- **Gait belt** – while assisting ambulation; if patient starts to fall → **ease patient gently to floor** (protect head) – don't try to hold up.

10.7 Pressure Ulcer / Pressure Injury (Bed Sore / Decubitus Ulcer)

Pressure injury (NPIAP 2016)

Localized damage to skin and/or underlying soft tissue, usually over a bony prominence, as a result of **intense and/or prolonged pressure** or pressure in combination with **shear**. Capillary closing pressure ~**32 mmHg** – isse zyada pressure → ischaemia.

Pressure Points – Bed sore ke risky areas (position ke hisaab se)

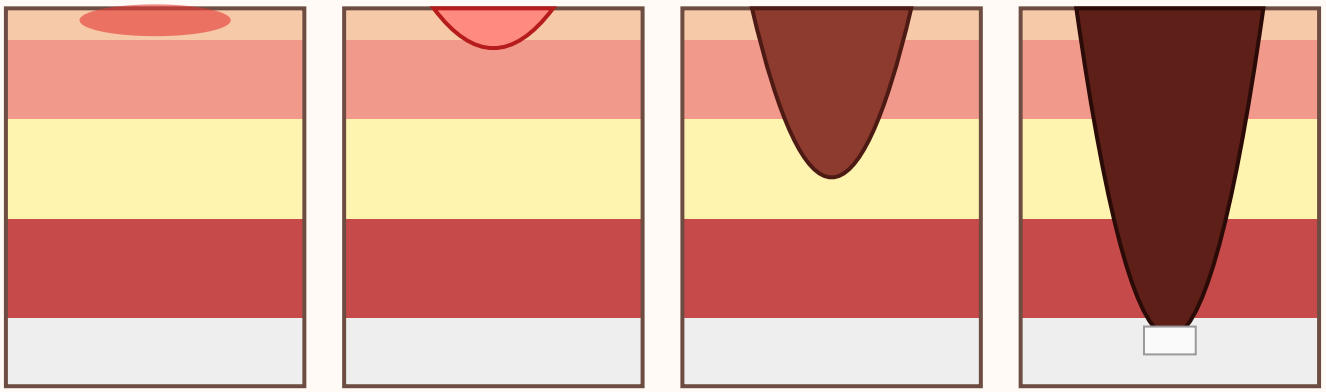


Sitting (Fowler's/ wheelchair) → ischial tuberosity, sacrum, scapula, heels · Most common site overall = SACRUM/ coccyx

Position ke hisaab se pressure points

- **Most common site – sacrum/ coccyx** (supine), then **heels**, ischial tuberosity (sitting), greater trochanter (lateral), occiput (children), elbows, scapula, ears.
- **Causes**: **pressure (main)**, **shear** (sliding down when **HOB > 30°** – skin stays, deep tissue moves → vessels stretch), **friction** (skin rubbing on sheet – abrasion), **moisture** (incontinence, sweat – maceration).
- **Risk factors**: **immobility (most important)**, ↓ sensation (paralysis, diabetes), malnutrition (**albumin < 3.5 g/dL**, protein deficiency), dehydration, anaemia, incontinence, elderly, ↓ LOC, obesity/ emaciation, sedation, fever, poor circulation, steroids.

Pressure Ulcer / Bed Sore – Stages (NPIAP)



Layers (upar se neeche): Epidermis · Dermis · Subcutaneous fat · Muscle · Bone

Unstageable = base slough/eschar se dhaka · **Prevention:** turn every 2 hours, Braden scale

Pressure ulcer stages – depth ke hisaab se

Stage (NPIAP)	Description
Stage 1	**Intact skin, non-blanchable erythema** (redness persists after pressure removed > 30 min); dark skin – purple/ blue, warm/ cool, firm
Stage 2	**Partial-thickness skin loss** with exposed dermis – **shallow open ulcer, red-pink wound bed** or **intact/ ruptured serum-filled blister**; no slough
Stage 3	**Full-thickness skin loss** – **subcutaneous fat visible**; slough may be present; **undermining & tunnelling** may occur; **no bone/ tendon/ muscle**
Stage 4	**Full-thickness skin & tissue loss** – **exposed bone, tendon, muscle, cartilage**; osteomyelitis risk
Unstageable	Base covered by **slough (yellow) or eschar (black)** – depth can't be seen (stable dry heel eschar – **don't remove**)
Deep tissue pressure injury (DTPI)	Intact/ non-intact skin with **persistent deep red, maroon or purple discolouration** or blood-filled blister
Medical device-related; Mucosal membrane pressure injury	From devices (NG, O_2 tubing, catheter)

- **Reverse staging is NOT done** – healing stage 4 ulcer is "healing stage 4", not stage 3.
- **Assessment tools**: **Braden scale** (most used) – **6 subscales**: **Sensory perception, Moisture, Activity, Mobility, Nutrition, Friction & shear**; score **6–23**; **lower score = higher risk**: **≤ 18 at risk**, 15–18 mild, 13–14 moderate, 10–12 high, **≤ 9 very high risk**. **Norton scale** (5 items – physical condition, mental, activity, mobility, incontinence; **5–20; ≤ 14 at risk**). **Waterlow** (UK). **Push tool** – healing assessment.

♫ Braden – "SMAMN-F"

Sensory · **M**oisture · **A**ctivity · **M**obility · **N**utrition · **F**riction & shear → max **23**, min **6**; kam score = zyada khatra.

Prevention & management

- **Reposition every 2 hrs** (bed), **every 1 hr in chair** (patient shifts weight every 15 min); turning clock; **30° lateral tilt** (avoid direct trochanter pressure).
- **HOB ≤ 30°** (shear ↓) unless contraindicated; **lift (don't drag)** – lift sheet/ trapeze.
- **Pressure-redistribution surfaces** – alternating pressure air mattress, water bed, foam, gel; **heels floated** (pillow under calves).
- **Don't massage reddened areas**; **no donut/ ring cushions**.
- Skin **clean & dry**, mild cleanser, moisturize dry skin, **barrier cream** (incontinence); check skin each shift.
- **Nutrition: high protein (1.25–1.5 g/kg), calories, Vit C, zinc, fluids**.
- Wound care: **moist wound healing**; cleanse with **NS**; **debridement** of necrotic tissue – **sharp/ surgical, mechanical (wet-to-dry), enzymatic (collagenase), autolytic (hydrocolloid/ hydrogel – body's own enzymes), biological (maggot/ larval therapy)**; dressings – **transparent film** (stage 1–2), **hydrocolloid** (stage 2–3, light exudate), **foam**, **alginate** (heavy exudate – absorbs, from seaweed), **hydrogel** (dry wound – adds moisture), **NPWT – negative pressure (VAC) therapy** (stage 3–4).

PYQ style

- Most common site of bed sore – **sacrum**.
- Turn bed-ridden patient every – **2 hours**.
- Stage with blister – **Stage 2**; bone exposed – **Stage 4**; only redness – **Stage 1**.
- Braden scale ka score kam = **risk zyada**; total parameters – **6**.
- Centre of gravity – **S2 (pelvis)**.
- Passive ROM ka purpose – **joint mobility maintain**, strength nahi badhata.
- Isometric exercise – **muscle contraction without movement** (quadriceps setting) – cardiac me avoid.
- Footdrop = **plantar flexion contracture** – footboard.
- Crutch length – **axilla se 2–3 finger (5 cm) neeche**; weight on – **hands**.
- Cane – **strong (unaffected) side** ke haath me.
- Stairs up – **good leg first**; down – **crutches + bad leg first**.
- Spinal injury moving – **log rolling**.
- Wheelchair placed on – **unaffected side**, brakes locked.

11 Nutrition & Feeding

11.1 Nutrients – Basics

Nutrient	Calories/g	Function	Sources
Carbohydrate	**4 kcal**	Main energy source (50–60 % of calories); fibre	Cereals, sugar, fruits
Protein	**4 kcal**	**Tissue building & repair**, enzymes, antibodies; adult **0.8–1 g/kg/day** (ICMR ~0.83 g/kg)	Egg, milk, pulses, meat, fish, soya
Fat	**9 kcal** (highest)	Energy store, fat-soluble vitamins (A, D, E, K) absorption, insulation	Oil, ghee, butter, nuts
Alcohol	**7 kcal**	–	
Vitamins, minerals, water	0	Regulatory	

- **Water** – ~60 % of body; intake **2–3 L/day**. **Fibre** – 25–40 g/day – constipation ↓.
- **Egg protein** – **reference protein** (biological value ~100). **Complete protein** – all essential amino acids (animal sources, soya). Cereal + pulse combination → complementary.
- **Essential amino acids** – 9 (body can't make). **Essential fatty acids** – linoleic, linolenic.
- **Calorie requirement (ICMR)**: sedentary man ~**2100–2300 kcal**, moderate ~2700–2800, heavy ~3500+; woman sedentary ~1700–1900; **pregnancy +350 kcal (2nd/3rd trimester)**, **lactation +500–600 kcal**.
- **BMR** – minimum energy for vital functions at rest (~1 kcal/kg/hr).

Vitamins

Vitamin	Deficiency	Sources
A (Retinol)	**Night blindness** (first), **Bitot's spots**, xerophthalmia, keratomalacia	Carrot, green leafy veg, liver, fish liver oil, mango, papaya
D (Calciferol)	**Rickets (children)**, **osteomalacia (adults)**	**Sunlight**, fish liver oil, egg yolk
E (Tocopherol)	Haemolytic anaemia, infertility (rats); antioxidant	Vegetable oils, nuts
K (Phylloquinone)	**Bleeding** (↓ prothrombin), haemorrhagic disease of newborn	Green leafy veg, **gut bacteria**; newborn – **1 mg IM at birth**
B1 (Thiamine)	**Beriberi** (wet – heart; dry – neuritis), **Wernicke-Korsakoff** (alcoholics)	Whole cereals, pulses (polishing rice removes)
B2 (Riboflavin)	**Angular stomatitis, cheilosis, glossitis (magenta tongue)**	Milk, eggs, green veg
B3 (Niacin)	**Pellagra – 3 D's: Dermatitis, Diarrhoea, Dementia** (+ Death) – **maize/ jowar eaters**	Meat, groundnut
B6 (Pyridoxine)	Peripheral neuritis (**INH therapy**), convulsions	Cereals, meat
B9 (Folic acid)	**Megaloblastic anaemia**, **neural tube defects** (pregnancy)	Green leafy veg, liver

Vitamin	Deficiency	Sources
B12 (Cyanocobalamin)	**Pernicious/ megaloblastic anaemia + neurological (subacute combined degeneration)** – vegans	**Only animal foods** – meat, milk, egg
C (Ascorbic acid)	**Scurvy** – **bleeding/ spongy gums**, delayed wound healing, perifollicular haemorrhage	**Amla (richest)**, guava, citrus, tomato; **destroyed by heat**

- **Fat-soluble**: **A, D, E, K** (stored – toxicity possible). **Water-soluble**: B complex, C (excreted – daily needed).

Minerals

Mineral	Deficiency	Notes
Iron	**Microcytic hypochromic anaemia** (most common nutritional deficiency)	Haem iron (meat) better absorbed; **Vit C ↑ absorption**; tea/ coffee, calcium, phytates ↓
Calcium	Rickets, osteoporosis, tetany	Milk (richest), ragi
Iodine	**Goitre**, cretinism	**Iodized salt (≥ 15 ppm at consumer)**
Zinc	Growth retardation, poor wound healing, diarrhoea (give zinc with ORS)	
Fluoride	Dental caries (deficiency); **fluorosis** (excess – > 1.5 ppm in water)	
Sodium	Hyponatraemia	Salt < 5 g/day (WHO)
Potassium	Hypokalaemia	Banana, orange, coconut water, potato

Malnutrition (PEM)

- **Kwashiorkor** – **protein deficiency** (calories adequate) – **oedema**, flag sign (hair), flaky-paint dermatosis, moon face, fatty liver, apathy; "**the disease of the displaced child**" (Cicely Williams).
- **Marasmus** – **protein + calorie deficiency** – **severe wasting (skin & bones), no oedema**, "**old man face**", alert & hungry.
- **SAM (Severe Acute Malnutrition)** – **WHZ < −3 SD** or **MUAC < 11.5 cm** or bilateral pitting oedema → **NRC (Nutrition Rehabilitation Centre)**; F-75 (stabilization) → F-100 (rehabilitation).

11.2 Therapeutic Diets

Diet	Description	Indication
Clear liquid	Transparent fluids – **water, tea, clear broth/ soup, apple juice, glucose water, ORS, jelly (gelatin)**; **no milk**; 1–2 days only	**First post-op diet**, before colonoscopy/ bowel prep, acute GI upset
Full liquid	Clear + **milk, custard, ice cream, strained cream soups, yogurt**	Transition, chewing/ swallowing difficulty
Soft diet	Easily chewed & digested, low fibre – khichdi, mashed potato, eggs, dal	Post-op progression, GI disorders, elderly
Mechanical soft/ pureed	Minced/ blended	Dental problems, dysphagia
Bland diet	Non-irritating – no spices, caffeine, alcohol, fried	Peptic ulcer (traditional), gastritis

Diet	Description	Indication
High-protein, high-calorie		**Burns, wound healing, pressure ulcers, TB, malnutrition, cancer, fever**
Low-protein		**Hepatic encephalopathy (acute), CKD (pre-dialysis)**
Low-sodium (salt-restricted)	Mild < 2–3 g Na; strict < 1 g; avoid pickles, papad, chutney, canned, baking soda	**Hypertension, heart failure, oedema, cirrhosis/ ascites, renal disease**
Low-fat		**Gallbladder disease, pancreatitis, obesity, CAD**
Low-cholesterol / low saturated fat		CAD, hyperlipidaemia
High-fibre (high-residue)	Whole grains, fruits, vegetables	**Constipation, diverticulosis, diabetes, IBS**
Low-residue (low-fibre)		**Diarrhoea, IBD flare, before bowel surgery/ colonoscopy**
Diabetic diet	Complex carbs, high fibre, low GI, regular timing, no simple sugars	Diabetes
Low-purine	Avoid organ meats, red meat, sea food, alcohol	**Gout**
Gluten-free	No wheat, barley, rye, oats (cross-contamination) – rice, maize OK	**Celiac disease**
Ketogenic	High fat, low carb	Refractory epilepsy (children)
Low-potassium, low-phosphorus		CKD
Acid-ash / Alkaline-ash	Urine acidify/ alkalinize	Renal stones
Tyramine-restricted	No aged cheese, wine, pickled/ smoked foods	**MAO inhibitors** (hypertensive crisis)
Lactose-free		Lactose intolerance
Sippy diet (historic)	Milk + cream hourly	Peptic ulcer (obsolete)

11.3 Feeding the Helpless Patient

- **Position – high Fowler's / sitting upright 90°**, head slightly flexed (**chin tuck**) – aspiration prevention; remain upright **30–60 min after** meal.
- Oral care before meals (appetite ↑), hand hygiene, clean pleasant environment, remove bedpan/ unpleasant sights.
- **Small bites**, food on **unaffected side** of mouth (stroke), allow time to chew & swallow, check for **pocketing**, alternate solids & liquids, **thickened liquids** for dysphagia; **never rush**; sit at eye level.
- **Visually impaired** – describe food position using **clock method** (e.g. rice at 12 o'clock, dal at 3).
- **Dysphagia screening** before first oral intake (stroke) – water swallow test; signs of aspiration – **coughing, choking, wet/ gurgling voice, drooling, ↓ SpO_2**.

- Record intake (% of meal), I/O.

11.4 Enteral Feeding (Tube feeding)

- **Indications**: GI tract working but can't eat – **unconscious, dysphagia (stroke), head & neck surgery/ cancer, oesophageal obstruction, critical illness, anorexia nervosa, burns**. "**If the gut works, use it**" – enteral preferred over parenteral (maintains gut mucosa, ↓ infection, cheaper).
- **Routes**: **Nasogastric (NG/ Ryle's)** – short-term (< 4–6 weeks); **Nasoduodenal/ nasojejunal** – aspiration risk/ gastroparesis; **Gastrostomy (PEG)/ Jejunostomy (PEJ)** – long-term (> 4–6 weeks); orogastric (neonates, basilar skull fracture).

NG (Ryle's) tube insertion – key points

- **Tube size**: adult **14–18 Fr** (feeding 8–12 Fr fine-bore), child 8–10 Fr; Ryle's tube length ~**105 cm**; markings at **40, 50, 60, 70 cm**.
- **Measure – NEX method**: **N**ose tip → **E**arlobe → **X**iphoid process (+ ~**5–10 cm** for stomach); mark with tape. Adult usual **~50–55 cm** (45–55).
- Position – **high Fowler's (sitting upright)**, **head tilted back for insertion through nostril**, then **flex head forward (chin to chest)** when tube reaches nasopharynx (closes trachea, opens oesophagus).
- **Lubricate with water-soluble jelly (lignocaine/ KY)** – **never oil-based (lipid aspiration pneumonia)**.
- Ask patient to **swallow (sips of water if allowed)** as tube advances.
- **Coughing, choking, cyanosis, inability to speak** → tube in trachea → **withdraw immediately**.
- **Confirm placement**: **X-ray – gold standard (most accurate, before first feed)**; bedside – **aspirate gastric content: pH ≤ 5 (1–5.5)**, appearance (grassy green/ clear/ brown); auscultation of air (**whoosh test**) – **not reliable**, not recommended alone; **check placement before every feed/ medication & every 4–8 hrs**.
- Secure with tape to nose (no pressure on nares), mark exit point.

Feeding procedure

- **Check placement + gastric residual volume (GRV)** before intermittent feed (or 4-hrly in continuous). **GRV > 250–500 ml** (policy) → hold & reassess; **return aspirate** (electrolytes) unless excessive/ abnormal.
- Position – **semi-Fowler's/ HOB 30–45° during feed and for 30–60 min after** (aspiration prevention – **most important**).
- **Flush with 30 ml water (15–30 ml)** before & after each feed & medication, **between medicines**, every 4 hrs in continuous feed (tube patency).
- Feed at **room temperature**, by **gravity** – syringe barrel held **~15–30 cm (6–12 inch) above stomach/ insertion point**; **slowly over 15–30 min** (~200–400 ml per bolus); **never force with plunger**.
- **Medicines**: liquid form preferred; crush tablets individually, dissolve in water; **don't crush enteric-coated/ sustained-release/ sublingual**; don't mix with feed; flush between.
- **Continuous feeding** – pump; open system hang time **≤ 4–8 hrs**; change bag & tubing **every 24 hrs**.
- Oral & nasal care every 4 hrs; lip moisturizer; nostril care.
- **Complications**: **aspiration (most serious)**, **diarrhoea (most common** – hyperosmolar feed, rapid rate, contamination, antibiotics), tube displacement/ blockage (flush with **warm water**; **don't use**

cola/ juices routinely), nausea/ vomiting, distension, **dumping**, refeeding syndrome (↓ PO_4, K, Mg), sinusitis, nasal erosion, dehydration/ hyperglycaemia.

- **Removal**: flush with air/ water, **ask patient to hold breath** (closes epiglottis), withdraw quickly & steadily.
- **Gastrostomy (PEG)** care: clean site daily with NS, dry, rotate external bumper, check for leakage/ infection; feed same as NG.

11.5 Parenteral Nutrition (TPN)

- **Total Parenteral Nutrition** – IV nutrition – **dextrose (high concentration 20–50 %), amino acids, lipids (fat emulsion), electrolytes, vitamins, trace elements** – when **GI tract non-functional** (bowel obstruction, short bowel, severe pancreatitis, fistula, prolonged ileus).
- **Central line** (subclavian/ PICC) for TPN – **hypertonic (> 900 mOsm/L)**; **PPN (peripheral)** – < 900 mOsm, short-term.
- **Nursing**: **strict asepsis** (sepsis risk), **dedicated lumen – no other drugs/ blood through TPN line**, **infusion pump**, **don't speed up to "catch up"** (hyperglycaemia), **change bag & tubing every 24 hrs** (lipids – 12 hrs), **filter (1.2 µm for lipids, 0.22 µm for non-lipid)**, refrigerate bags (take out **30–60 min before** use), **monitor blood sugar every 4–6 hrs** (hyperglycaemia – most common metabolic complication), daily weight, I/O, electrolytes.
- **Never stop TPN abruptly** → **rebound hypoglycaemia** → if bag finishes & new not available, hang **10 % dextrose (D10W)** at same rate; **taper off** gradually.
- Complications: **infection/ sepsis (catheter-related)**, **hyperglycaemia**, hypoglycaemia (abrupt stop), **air embolism** (left side, head down), pneumothorax (insertion), fluid overload, refeeding syndrome, liver dysfunction.
- **Lipid emulsion (Intralipid)** – **check egg/ soy allergy**; milky; don't use if oil separates/ cracked.

11.6 Intake & Output (I/O) Chart

- **Intake**: oral fluids, IV fluids, tube feeds, **ice chips (½ volume)**, foods liquid at room temp (**ice cream, jelly, soup**), irrigations not removed.
- **Output**: urine, vomitus, diarrhoea, drainage (wound, chest tube, NG suction), blood loss, **insensible loss** (~800–900 ml/day – estimated, not measured).
- Record every shift, 24-hr total; **normal urine ≥ 30 ml/hr (0.5 ml/kg/hr)**. 1 kg weight = 1 L fluid.
- Measure urine in **graduated container at eye level**; weigh diapers (**1 g = 1 ml**).

PYQ style

- Calories – **carbohydrate 4, protein 4, fat 9** kcal/g.
- Richest source of Vit C – **amla**; Vit A first sign – **night blindness**; pellagra – **3 D's (niacin)**.
- First diet after surgery – **clear liquid**.
- Burns patient diet – **high protein, high calorie**.
- Hepatic encephalopathy – **low protein** (acute).
- NG tube length measured – **nose → earlobe → xiphoid (NEX)**.
- NG placement – **X-ray (gold standard)**; aspirate pH – **≤ 5**.
- Feeding position – **semi-Fowler's/ high Fowler's**, upright **30–60 min** after.
- Flush NG tube with – **30 ml water**.
- Most common complication of tube feeding – **diarrhoea**; most serious – **aspiration**.
- TPN stopped suddenly → **hypoglycaemia** → hang D10W.
- Lubricant for NG – **water-soluble**.

12 Urinary Elimination & Catheterization

12.1 Micturition – Basics

- Urinary system: kidneys → ureters (**25–30 cm**) → bladder → urethra (female **3–4 cm**, male **18–20 cm**).
- **Bladder capacity 300–600 ml** (avg 500); **urge to void at ~150–300 ml** (stretch receptors); can hold up to ~600–1000 ml (overdistension).
- **Micturition reflex** – **sacral spinal cord S2–S4** (parasympathetic – **pelvic nerve**) → detrusor contracts & internal sphincter relaxes; **external sphincter – voluntary (pudendal nerve)**. Micturition centre – **pons**; cortex – voluntary control.
- Normal adult output **1–2 L/day (1500 ml)**; **≥ 30 ml/hr (0.5 ml/kg/hr)**; child 1 ml/kg/hr; infant 2 ml/kg/hr.
- Normal voiding **5–6 times/day**, not at night.

Normal urine

Character	Normal	Abnormal
Colour	**Pale yellow / straw/ amber** (urochrome)	Dark amber (dehydration), **red/ pink (haematuria, beetroot, rifampicin-orange)**, **tea/ cola (bilirubin, AGN)**, **blue-green** (methylene blue, Pseudomonas), **milky (chyluria, pyuria)**, **black (alkaptonuria, melanoma)**
Clarity	**Clear/ transparent**	Cloudy (UTI, pus, phosphates)
Odour	**Faintly aromatic**; ammonia on standing	**Fruity/ acetone (ketoacidosis)**, foul (infection), **mousy (PKU)**, maple syrup (MSUD)
pH	**4.5–8 (avg 6, slightly acidic)**	
Specific gravity	**1.005–1.030 (1.010–1.025)**	↑ dehydration, SIADH, glycosuria; ↓ (fixed 1.010) – renal failure, DI
Constituents	Water 95 %, urea, creatinine, uric acid, electrolytes	**Glucose, protein, ketones, blood, bilirubin, casts** = abnormal

Terms

Term	Meaning
Polyuria	> **3 L/day** (DM, DI, diuretics)
Oliguria	< **400 ml/day** (< 0.5 ml/kg/hr)
Anuria	< **100 ml/day** (some < 50)
Nocturia	Waking at night to void
Dysuria	**Painful/ burning** urination
Frequency	Voiding often (small amounts)
Urgency	Sudden strong desire to void

Term	Meaning
Hesitancy	Difficulty in **starting** stream (BPH)
Haematuria	Blood in urine
Pyuria	Pus in urine
Glycosuria	Glucose (> renal threshold **180 mg/dL**)
Proteinuria/ Albuminuria	Protein
Ketonuria	Ketones (DKA, starvation)
Enuresis	Involuntary urination – **bedwetting (nocturnal) after age 5**
Retention	Unable to empty bladder (urine accumulates)
Residual urine	Urine left after voiding – **> 100 ml (some > 50–150) abnormal** – **PVR**
Suppression	Kidneys **not producing** urine (vs retention – producing but not voiding)
Incontinence	Involuntary loss of urine

12.2 Urine Specimen Collection

Specimen	Method
Routine (random)	Clean container, **first morning (most concentrated)** preferred, 10–50 ml; within 1 hr to lab (or refrigerate)
Clean-catch midstream (MSU) – for culture	Clean meatus (female – front to back; male – circular from meatus), **begin voiding into toilet → collect midstream in sterile container** → finish in toilet; don't touch inside of container; **send within 30 min / refrigerate (4 °C) max 24 hrs**
From catheter (CSU)	**Clamp tubing below port 15–30 min**, clean **sampling port with alcohol**, **aspirate with sterile syringe**; **never from drainage bag**; never disconnect catheter
24-hour urine	**Discard first morning void (note time)** → collect all urine for 24 hrs → **include final void at same time next day**; keep **on ice/ with preservative**; **if one sample missed/ discarded → restart** (creatinine clearance, protein, VMA, catecholamines)
Timed (2 hr, 12 hr)	Same principle
Infant	Adhesive **urine collection bag (U-bag)**
Suprapubic aspiration	Sterile – infants (most reliable culture)

- **Urine for sugar (Benedict's test)** – blue → green → yellow → orange → **brick red (+ + + +)**. **Rothera's test** – ketones (purple ring). **Heat & acetic acid test / Sulfosalicylic acid** – protein (albumin). **Dipstick (reagent strip)** – read at exact time.
- Specific gravity – **urinometer** at room temp; refractometer.

12.3 Bedpan & Urinal

- **Fracture bedpan (slipper pan)** – flat end goes **under buttocks** – **hip fracture, spinal injury, casts, elderly, can't raise hips**.
- Warm the bedpan (metal – rinse with warm water); powder rim (not on skin – friction ↓); patient in **semi-Fowler's (normal voiding/ defecation position)** if allowed; privacy; **toilet tissue & call bell in**

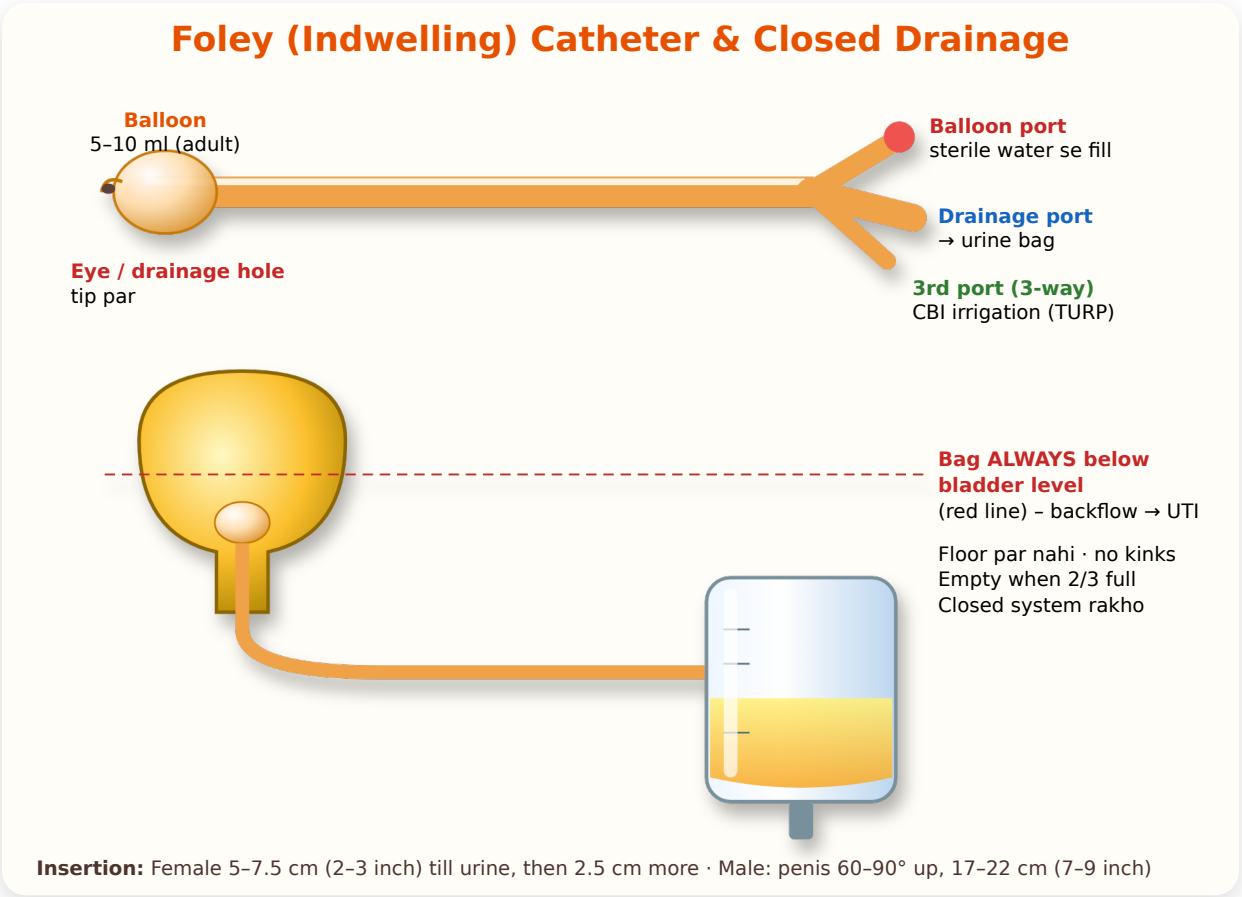
reach; don't leave patient on pan long (pressure).

- Male – **urinal**; **standing** position (if allowed) natural.
- **Commode** – bedside chair-toilet.

12.4 Urinary Catheterization

D Catheterization

Introduction of a catheter through the **urethra into the bladder** under **strict aseptic (sterile) technique** to drain urine. Types: **Intermittent (straight/ in-and-out – Nelaton/ Robinson)** & **Indwelling/ retention (Foley – self-retaining balloon)**; **Suprapubic** (through abdominal wall – long-term, urethral injury); **Condom (external – Texas) catheter** (male incontinence – change daily, leave **2.5–5 cm space** at tip, not too tight).



Foley catheter – balloon, ports aur closed drainage

Indications

Urinary retention (acute), **accurate hourly urine output** (critically ill/ shock), **pre/ intra-op** (pelvic surgery, long surgery), **bladder irrigation**, urinary obstruction (BPH), sterile specimen, incontinence with **stage 3–4 sacral ulcer**, end-of-life comfort, instillation of drugs, neurogenic bladder (intermittent).

Catheter facts

- Size in **French (Fr/ Ch)** – **1 Fr = ⅓ mm** (outer diameter). **Adult female 12–14 Fr, adult male 14–16 Fr**, children 8–10 Fr; haematuria/ clots – **18–24 Fr (3-way)**. **Smallest size that drains well** (urethral trauma ↓).

- **Balloon – 5 ml (10 ml) – fill with sterile water** (not saline – crystallizes & balloon may not deflate; not air – floats/ deflates); 30 ml balloon – post-TURP (haemostasis/ traction).
- Material: **latex** (short-term ≤ 2 weeks; latex allergy check), **silicone** (long-term – up to **12 weeks**), **silver-alloy/ hydrogel-coated** (↓ infection), PVC (intermittent).
- **2-way** (balloon + drainage), **3-way** (+ irrigation – **CBI after TURP**).

Female catheterization – steps

- **Dorsal recumbent position** (or Sims' – lateral if can't flex) – knees flexed, feet apart; good light.
- Perineal care first; open sterile tray; **sterile gloves**; drape.
- **Non-dominant hand separates labia (becomes contaminated – stays in place till catheter inserted)**; dominant (sterile) hand cleans.
- Cleaning: **front to back** – far labia minora → near labia minora → **meatus (centre) last**; one swab one stroke.
- Lubricated catheter inserted **5–7.5 cm (2–3 inch)** until urine flows, then advance **2.5–5 cm (1–2 inch) more** → inflate balloon → **gently pull back until resistance** felt.
- **Catheter in vagina by mistake → leave it there** (landmark) & insert **new sterile catheter** into meatus above it; then remove the wrong one.

Male catheterization – steps

- **Supine**, legs slightly apart. Retract foreskin (uncircumcised).
- Clean **meatus in circular motion outward** (glans).
- **Instil lignocaine gel (2 %) into urethra** – wait 3–5 min (anaesthesia + lubrication).
- Hold penis **perpendicular (60–90°) to body**, slight traction (straightens urethra) → insert **17–22.5 cm (7–9 inch)** till urine flows, then advance to **bifurcation (Y-port)** → inflate balloon → pull back gently → lower penis.
- **Replace foreskin** (prevent **paraphimosis**).
- Resistance at **prostate** – ask patient to take deep breath/ bear down (relaxes sphincter); **never force** (false passage).

- **Inflate balloon only after urine flows** (urethral trauma if inflated in urethra – pain → deflate, advance).
- **Drain max ~750–1000 ml at once** in acute retention (as per policy) → then clamp – **haematuria/ hypotension (decompression)** risk (evidence mixed; follow policy).
- **Secure catheter** – **female: inner thigh**; **male: upper thigh or lower abdomen (penis upward – prevents peno-scrotal fistula/ pressure)**; leave slack.

Catheter care (CAUTI prevention bundle)

- **Insert only when necessary; remove as soon as possible (daily review)** – **most important**.
- **Closed drainage system** – never disconnect catheter from tubing.
- **Bag always below bladder level** – not on floor; hang on bed frame (not side rail); no loops/ kinks (dependent loops ↓).
- **Empty bag when ⅔ full** (or every 8 hrs/ before transfer) using **separate clean container**, outlet not touching container.
- **Meatal care daily with soap & water** (during bath) – antiseptics not recommended routinely.

- **Fluids 2–3 L/day** (if not restricted) – flushing; **routine catheter change not needed** (change if blocked/ infected/ as per manufacturer).
- Strict hand hygiene; no routine bladder irrigation; no antibiotic prophylaxis routinely.
- Signs of CAUTI – fever, cloudy foul urine, suprapubic/ flank pain, confusion (elderly).

Catheter removal

- Check order; empty bag; **deflate balloon completely with syringe** (don't cut balloon port); ask patient to **breathe out/ relax**, withdraw gently; clean perineum.
- After removal: **first void expected within 6–8 hrs**; measure first 3 voids; encourage fluids; mild burning/ frequency normal initially; **can't void in 8 hrs/ distension → inform** (bladder scan).

Bladder irrigation

- **Intermittent** (closed – via port with sterile syringe, 30–60 ml NS) or **Continuous bladder irrigation (CBI) – 3-way catheter** – post-TURP/ bladder surgery – NS (room temp) – adjust rate to keep **output light pink (rosé)**; clots/ bright red → ↑ rate; **actual urine output = total drainage − irrigation volume**.
- **Bladder training** – timed voiding (every 2 hrs → gradually increase), Kegel exercises, fluid schedule.
- **Clean intermittent self-catheterization (CISC)** – neurogenic bladder (spinal injury, MS) – **clean (not sterile) technique at home**, every 4–6 hrs.

12.5 Urinary Retention & Incontinence

- **Retention** causes: **post-op/ post-anaesthesia (spinal)**, **BPH**, urethral stricture, drugs (**anticholinergics – atropine, antihistamines, opioids**), anxiety, lack of privacy, spinal cord injury, faecal impaction.
- Signs: **bladder distension (palpable, dull above symphysis)**, discomfort, dribbling (**overflow**), small frequent voids, restlessness.
- **Nursing measures to induce voiding** (before catheter): **privacy**, **normal position** (sit/ stand), **running tap water sound**, **pour warm water over perineum**, **warm sitz bath**, place hands in warm water, **stroke inner thigh**, **Credé's manoeuvre** (suprapubic pressure – neurogenic bladder; not if obstruction), adequate fluids, analgesia, ambulation. If not voided **8 hrs post-op** (or distended) → catheterize (as ordered); **bladder scan** for PVR.

Incontinence type	Cause / Feature	Management
Stress	Weak pelvic floor – leak on **coughing, sneezing, laughing, lifting** (multiparous women, post-menopause)	**Kegel exercises**, weight loss, surgery (sling)
Urge (overactive bladder)	Sudden urgency, can't reach toilet (UTI, stroke)	Bladder training, **anticholinergics (oxybutynin)**
Overflow	Bladder over-distended → dribbling (BPH, neurogenic, drugs)	Catheter, treat cause
Functional	Normal bladder, but **can't reach toilet** – mobility/ cognitive (dementia)	Toileting schedule, commode near bed, easy clothing
Reflex	Spinal cord lesion above S2 – no sensation	CISC
Total	Continuous unpredictable leak (fistula)	Surgery, pads

Incontinence type	Cause / Feature	Management
Mixed	Stress + urge	

- Care: skin care (barrier cream), **incontinence pads (not diapers routinely – dignity)**, timed voiding, adequate fluids (don't restrict – concentrated urine irritates bladder), avoid caffeine/ alcohol/ carbonated drinks, **external catheter (male)**, Kegel.

PYQ style

- Normal adult urine output – **1500 ml/day**; minimum **30 ml/hr**.
- Oliguria – **< 400 ml/day**; anuria – **< 100 ml/day**.
- Urine specific gravity – **1.005–1.030**.
- 24-hr urine – **first morning sample discarded**.
- Culture specimen from catheter – **sampling port with sterile syringe**, bag se nahi.
- Foley balloon – **sterile water**; female catheter insertion – **5–7.5 cm**; male – **17–22.5 cm**.
- Female catheterization position – **dorsal recumbent**.
- Catheter accidentally in vagina – **leave it, insert new one**.
- Urine bag – **below bladder level**.
- First nursing action for post-op retention – **non-invasive measures** (privacy, running water).
- Kegel exercise – **stress incontinence**.
- Most common HAI – **CAUTI**.

13 Bowel Elimination – Enema, Suppository & Stool Care

13.1 Defecation – Basics

- **Defecation reflex** – rectum stretch → **parasympathetic (S2–S4)** → urge; **internal anal sphincter (involuntary)**, **external (voluntary)**. **Gastrocolic reflex** – strongest **after breakfast/ meals** (best time for bowel training).
- **Valsalva manoeuvre** – forced expiration against closed glottis (straining) → **↑ intrathoracic pressure → ↓ venous return → bradycardia** (vagal) → **dangerous in cardiac, ↑ICP, post-eye/ abdominal surgery patients** → stool softeners.
- Normal frequency – **3 times/day to 3 times/week**; normal stool ~100–200 g/day.

Stool characteristics

Feature	Normal	Abnormal & cause
Colour	**Brown** (stercobilin); infant breast-fed – **yellow**, formula-fed – pale yellow	**Black tarry (melena)** – upper GI bleed, **iron supplements (black, not tarry)**, bismuth; **bright red (haematochezia)** – lower GI bleed, haemorrhoids, beetroot; **clay/ pale/ white (acholic)** – **obstructive jaundice (no bile)**, barium; **green** – rapid transit, spinach, infection
Consistency	Soft, formed (Bristol type 3–4)	**Hard (constipation)**, liquid (diarrhoea), **ribbon/ pencil-like (rectal obstruction/ cancer)**
Odour	Pungent (bacterial)	Very foul – infection, malabsorption, blood
Constituents	Water 75 %, undigested fibre, bacteria, bile	**Blood, mucus, pus, fat (steatorrhoea – frothy, floating, greasy – pancreatic/ malabsorption)**, worms, **rice-water (cholera)**, **red-currant jelly (intussusception)**

- **Bristol Stool Chart** – **7 types**: type 1–2 constipation, **3–4 normal (ideal 4 – smooth snake)**, 5–7 diarrhoea.
- **Meconium** – first stool of newborn – **greenish-black, tarry, sticky**, passed **within 24–48 hrs**; then transitional (green-yellow).

13.2 Common Bowel Problems

- **Constipation** – < 3 BMs/week, hard dry stool, straining. Causes: **low fibre, low fluids, immobility, ignoring urge**, opioids/ iron/ antacids (Al, Ca)/ anticholinergics, pregnancy, elderly, hypothyroidism, hypokalaemia.
  - Management: **fluids 2–3 L/day**, **high-fibre diet (fruits, vegetables, whole grains, bran – 25–30 g)**, **exercise/ ambulation**, **regular toilet time (after breakfast – gastrocolic reflex)**, privacy, sitting/ squatting position (knees above hips), warm drinks in morning, prune juice; laxatives (last).
- **Faecal impaction** – hardened stool in rectum can't be passed; sign – **oozing of liquid stool (paradoxical/ overflow diarrhoea)** around mass; Rx – **oil retention enema → digital removal**

(gloved, lubricated finger) (vagal stimulation risk – **stop if bradycardia/ dizziness**; doctor's order; contraindicated in cardiac patients).

- **Diarrhoea** – ≥ 3 loose stools/day; risk – **dehydration, electrolyte loss (K^+, HCO_3^- → metabolic acidosis)**, skin excoriation; **ORS**, fluids, BRAT diet (old), zinc (children), hand hygiene, perianal skin care; **C. difficile** – contact precautions + soap & water handwash.
- **Faecal incontinence** – skin care, bowel training, pads, rectal tube/ faecal management system (short-term).
- **Flatulence** – gas accumulation (abdominal distension); ambulation, **flatus tube**, avoid gas-forming foods (beans, cabbage, carbonated drinks), **carminatives**; **tympanites** – distension with gas.
- **Haemorrhoids** – dilated rectal veins; sitz bath, high fibre.

Laxatives

Type	Example	Note
Bulk-forming	**Isabgol (psyllium)**, methylcellulose, bran	**Safest, most natural** – take with **full glass of water** (else obstruction)
Stool softener (emollient)	**Docusate sodium**	Post-MI, post-surgery, haemorrhoids – avoid straining
Osmotic	**Lactulose**, PEG, **magnesium hydroxide (milk of magnesia)**, magnesium sulphate	Lactulose – hepatic encephalopathy; Mg – avoid in renal failure
Stimulant	**Bisacodyl (Dulcolax)**, senna	**Bisacodyl – don't crush, don't take with milk/ antacid** (enteric-coated)
Lubricant	**Liquid paraffin (mineral oil)**	↓ fat-soluble vitamin absorption; lipid aspiration pneumonia (elderly) – give at bedtime upright

13.3 Enema

D Enema

Introduction of **fluid/ solution into the rectum & sigmoid colon** through the anus. Greek – "**to send in / inject**".

Types

Type	Purpose / Solution	Volume (adult)
Cleansing – Soap suds enema (SSE)	Evacuate bowel – **castile/ mild soap (5 ml in 1000 ml water)** – irritant → peristalsis	**500–1000 ml** (750–1000)
Tap water (hypotonic)	Cleansing – **don't repeat** (water intoxication, electrolyte loss) – esp. infants	500–1000 ml
Normal saline (isotonic)	**Safest** – infants, children, elderly, repeated enemas	500–1000 ml
Hypertonic – Fleet/ phosphate enema	**Draws water into colon** – small volume, quick; **avoid in renal failure, dehydration, children < 2** (phosphate toxicity)	**120–250 ml (120 ml)** – retained **5–10 min**

Type	Purpose / Solution	Volume (adult)
Oil retention enema	**Softens hard stool** – olive oil, mineral oil, cottonseed oil – **constipation, impaction**	**90–150 ml (100–200)** – retain **1–3 hrs (≥ 30 min)**
Carminative enema	**Expel flatus** – e.g. **MGW enema: Magnesium sulphate 30 g + Glycerine 60 ml + Water 90 ml** (1-2-3) / turpentine	60–180 ml
Medicated/ therapeutic	**Kayexalate (hyperkalaemia)**, **neomycin/ lactulose (hepatic encephalopathy)**, steroid (IBD), anthelmintic	
Nutrient enema	Historic – glucose/ fluids	
Barium enema	**Diagnostic – X-ray of colon**	
Return-flow (Harris flush/ colonic irrigation)	**Flatus removal** – 100–200 ml in & out repeatedly	
Sedative enema	Paraldehyde/ chloral hydrate (historic)	
Anthelmintic	Worms	
Astringent	Bleeding	

- **Retention enemas** (oil, medicated, nutrient) – **retained for prolonged time**; **evacuant/ non-retention** (cleansing) – expelled quickly (retain **5–15 min**).

Enema procedure – golden numbers

- **Position – LEFT lateral (Sims') with right knee flexed** (follows natural anatomy of **sigmoid colon & descending colon on left side** – gravity flow). Children – dorsal recumbent; patient unable – dorsal recumbent on bedpan (less effective).
- **Solution temperature – 40.5–43 °C (105–110 °F)** for adults (cleansing); **37.7 °C (100 °F) for children**; oil retention – **body temperature (37 °C)**. Too hot → mucosal burn; too cold → cramps.
- **Rectal tube size**: **adult 22–30 Fr**, child 12–18 Fr, infant 10–12 Fr.
- **Insertion length**: **adult 7.5–10 cm (3–4 inch)**, **child 5–7.5 cm (2–3 inch)**, **infant 2.5–3.75 cm (1–1.5 inch)**.
- Lubricate tip **5 cm**; ask patient to **take deep breath / bear down (relaxes external sphincter)**; insert **towards umbilicus** (direction of rectum).
- **Height of can/ bag: 30–45 cm (12–18 inch) above anus** (adult) – **max 45–60 cm**; child ~7.5 cm (3 inch) above hips. Higher → faster flow → cramps, injury.
- Flow – **slow** (5–10 min for 1000 ml); **cramps → stop/ lower bag/ clamp**, deep breaths, resume.
- **Expel air** from tube before insertion (prime) – air → cramps/ distension.
- Ask patient to **retain 5–15 min** (cleansing); bedpan/ commode/ bathroom ready.
- Observe **result (returns)** – amount, colour, consistency; **"Enema till clear"** – **max 3 enemas** (fluid/ electrolyte imbalance).
- **Contraindications**: **acute abdomen/ appendicitis, intestinal obstruction, recent rectal/ colon/ prostate surgery, ↑ICP (Valsalva), severe haemorrhoids, rectal bleeding, perforation**, cardiac arrhythmia (vagal).

Age	Volume (cleansing)
Infant (< 18 months)	**50–150 ml (≤ 250)**

Age	Volume (cleansing)
Toddler/ preschool (18 months–5 yrs)	**250–350 ml**
School-age (5–12 yrs)	**300–500 ml**
Adolescent	500–750 ml
Adult	**750–1000 ml**

- **Bowel wash** – repeated enemas till clear return (pre-op, before colonoscopy/ barium).
- **Complications**: **vagal stimulation (bradycardia, fainting)**, fluid & electrolyte imbalance (water intoxication – tap water), **rectal mucosal injury/ perforation**, cramps.

13.4 Suppository & Flatus Tube

- **Suppository** – cone-shaped solid medication (cocoa butter/ glycerine base) melting at body temp. **Glycerine suppository** – laxative (softens, lubricates); **bisacodyl (Dulcolax)** – stimulant (works **15–60 min**).
- Procedure: **left lateral (Sims')**, gloves, lubricate, ask to breathe deep/ bear down, insert **beyond internal sphincter** along **rectal wall** (**not into faecal mass** – won't absorb), **adult 7.5–10 cm (3–4 inch; index finger length)**, **child/ infant 5 cm (2 inch) – little finger**; **pointed (tapered) end first** (Indian textbooks ki standard teaching); retain **15–20 min** (laxative) / till absorbed; refrigerate store (melting).
- **Contraindications**: rectal surgery/ bleeding, cardiac (vagal), ↓ platelets/ neutropenia (rectal route avoid).
- **Flatus tube (rectal tube)** – relieve flatulence; size **22–30 Fr** adult; insert **10 cm (4 inch)**; free end in **water container/ bag (bubbling shows gas)**; leave **max 20–30 min** (sphincter damage if longer); repeat after 2–3 hrs if needed.

13.5 Stool Specimen & Ostomy Basics

- **Stool specimen**: **1 inch (2.5 cm) formed / 15–30 ml liquid** in clean container with spatula; **not mixed with urine or toilet paper**; **ova & parasites – warm, send within 30 min (fresh)**; culture – sterile container; **occult blood (guaiac/ FOBT)** – avoid **red meat, iron, vitamin C, NSAIDs, aspirin 3 days** before (false +/−); 3 samples on 3 days; label & send promptly.
- **Ostomy** – surgical opening on abdomen: **colostomy** (colon – formed stool, sigmoid), **ileostomy** (ileum – **liquid, continuous, enzyme-rich – skin excoriation, dehydration**). Healthy stoma – **beefy red, moist**; **pale – anaemia; dusky/ purple/ black – ischaemia → report**. Pouch emptied when **⅓–½ full**; opening cut **2–3 mm (⅛ inch) larger** than stoma. Detail – MSN notes.

PYQ style

- Enema position – **left lateral (Sims')**.
- Rectal tube insertion adult – **7.5–10 cm (3–4 inch)**.
- Enema can height – **30–45 cm (12–18 inch)** above anus.
- Enema solution temperature – **40.5–43 °C (105–110 °F)**.
- Oil retention enema – retain **30 min–3 hrs**; purpose – **soften stool**.
- Soap suds enema – **5 ml castile soap in 1000 ml water**.
- Hypertonic (Fleet) enema volume – **120 ml**.
- Patient complains of cramps during enema – **stop/ lower the can, deep breaths**.
- Melena – **upper GI bleed** (black tarry); clay stool – **obstructive jaundice**.
- Valsalva is dangerous in – **cardiac & ↑ICP** patients.
- Faecal impaction sign – **liquid stool oozing**.
- Flatus tube kept for max – **20–30 min**.

14 Oxygenation & Respiratory Care

14.1 Basics

- **Hypoxaemia** – ↓ O_2 in **blood** (PaO_2 < 80; < 60 = respiratory failure). **Hypoxia** – ↓ O_2 at **tissue** level.
- **Types of hypoxia**: **Hypoxic (hypoxaemic)** – ↓ PaO_2 (altitude, lung disease); **Anaemic** – ↓ Hb (anaemia, **CO poisoning**); **Stagnant (circulatory/ ischaemic)** – ↓ blood flow (shock, HF); **Histotoxic** – cells can't use O_2 (**cyanide**).
- **Early signs of hypoxia**: **restlessness, anxiety, confusion (earliest)**, **tachycardia, tachypnoea**, ↑ BP, dyspnoea, use of accessory muscles, nasal flaring. **Late signs**: **cyanosis**, bradycardia, hypotension, arrhythmias, lethargy, coma.
- **Cyanosis** – bluish colour when **reduced (deoxygenated) Hb > 5 g/dL** – **late sign**; **central** (lips, tongue, mucosa – lung/ heart) vs **peripheral** (nail beds, extremities – cold, poor circulation). **Anaemic patients may not show cyanosis**; polycythaemia shows early.
- **Hypercapnia** – ↑ CO_2 (> 45) – headache, drowsiness, flushed warm skin, confusion.

14.2 Oxygen Therapy

Oxygen – basic facts

- O_2 is a **drug** – needs **prescription** (except emergency) – dose (flow L/min or FiO_2), device, duration, target SpO_2.
- Room air FiO_2 = **21 %**. Each **1 L/min via nasal cannula ↑ FiO_2 by ~4 %**.
- **Target SpO_2**: most patients **94–98 %**; **COPD/ CO_2 retainers 88–92 %**.
- **O_2 cylinder colour (India – IS 3933)**: **Black body with white shoulder (neck)**; **Medical air** – grey body, black & white shoulder; **N_2O – French blue**; **CO_2 – grey**; **Entonox (N_2O + O_2) – blue body, blue-white shoulder**; Cyclopropane – orange; **Helium – brown**; Nitrogen – grey with black shoulder.
- Portable cylinders: **D (~340–400 L)**, **E (~680 L)**; full cylinder pressure ~**2000 psi (~137 bar)**.
- **Duration (min) = (Cylinder pressure psi − 200 safe residual) × cylinder factor ÷ flow (L/min)** (D = 0.16, E = 0.28).
- **Liquid O_2** – central supply (−183 °C); **O_2 concentrator** – home (extracts O_2 from air, 90–95 %).

O_2 delivery devices

Device	Flow (L/min)	FiO_2	Key points
Nasal cannula (prongs)	**1–6** (> 6 – drying, no added benefit)	**24–44 %**	**Most common, comfortable**, can eat & talk; **humidify if > 4 L/min**; mouth breathing still gets some O_2
Simple face mask	**5–10** (**minimum 5** – else CO_2 rebreathing)	**35–55/60 %**	Short-term
Partial rebreather mask	6–11 (8–12)	**60–75 %**	Reservoir bag **⅓–½ full on inspiration**

Device	Flow (L/min)	FiO_2	Key points
Non-rebreather mask (NRBM)	**10–15**	**80–95 % (up to ~100 %)**	One-way valves; **bag must not collapse** (fill before applying); **highest concentration without intubation** – emergencies, CO poisoning
Venturi mask (air-entrainment)	4–12 (as per colour adaptor)	**Precise 24–60 %**	**Most accurate/ fixed FiO_2 – COPD**
Face tent	8–12	30–55 %	Facial trauma/ burns, claustrophobia
Tracheostomy collar/ T-piece (Briggs)	8–10	24–100 %	Tracheostomy/ ET tube (humidified)
Oxygen hood/ head box	7–12	30–90 %	**Infants**
Oxygen tent	10–15	~30–50 %	Children (croup) – historic; humidified, cool
High-flow nasal cannula (HFNC)	**up to 60**	21–100 %	Heated humidified – ICU
CPAP/ BiPAP (NIV)			Sleep apnoea, COPD exacerbation, pulmonary oedema
Bag-valve-mask (Ambu bag)	10–15 with reservoir	~100 %	Resuscitation

! Oxygen safety

- **No smoking / no open flame** – "**No Smoking – Oxygen in Use**" sign; O_2 **supports combustion** (not flammable itself).
- **No oil/ grease** on cylinder valves/ fittings (spontaneous combustion); no petroleum jelly on lips/ nose → use **water-soluble** lubricant.
- **Cotton** clothing & linen (static electricity – avoid wool/ synthetic).
- Electrical equipment grounded; cylinder **secured upright/ chained** on stand/ trolley (falls → valve breaks → "rocket"); store in cool place, away from heat.
- **Open cylinder valve slightly first ("cracking") to blow out dust – away from patient**.
- **Check flow meter** (ball/ float – read at **centre of ball**), humidifier water level (**sterile/ distilled water**, change every 24 hrs).
- **O_2 toxicity** – FiO_2 > **50–60 % for > 24–48 hrs** → substernal pain, dry cough, dyspnoea, **atelectasis**, ARDS-like; **Retinopathy of prematurity (ROP – retrolental fibroplasia)** in preterm infants → monitor SpO_2 (90–95 %).
- **COPD (CO_2 retainers)** – high O_2 → **↓ hypoxic drive → hypoventilation, CO_2 narcosis** → low flow, Venturi; but **never withhold O_2 from hypoxic patient**.
- Skin care – **behind ears, nares, bridge of nose** (pressure from tubing); oral care (drying).

- **Humidification** – O_2 is **dry** → dries mucosa → humidify with **sterile/ distilled water** (bubble humidifier) for > 4 L/min; heated humidifier for tracheostomy/ ventilator (bypass upper airway).

14.3 Airway Clearance & Lung Expansion

Deep breathing, coughing & incentive spirometry

- **Diaphragmatic (abdominal) breathing** – hand on abdomen rises on inhalation through nose, exhale through pursed lips.
- **Pursed-lip breathing** (COPD) – **inhale through nose (count 2), exhale slowly through pursed lips (count 4)** → keeps airways open (back pressure), **CO_2 out**, ↓ air trapping.
- **Controlled/ huff cough** – deep breath, hold 2–3 sec, 2–3 forceful coughs; **splint incision with pillow** (post-op).
- **Incentive spirometer** – **atelectasis & pneumonia prevention** (post-op): **sit upright**, exhale normally, **seal lips around mouthpiece**, **inhale slowly & deeply** (raise ball/ piston to target), **hold breath 3–5 sec**, exhale slowly; **10 times every 1–2 hrs while awake**; then cough. (**Volume-oriented** – piston; flow-oriented – balls.)

Steam inhalation

- Purposes: **loosen secretions, soothe mucosa, relieve congestion** (common cold, sinusitis, bronchitis, croup). Medications – **tincture benzoin (Friar's balsam)**, menthol, eucalyptus.
- **Nelson's inhaler** – jug with spout; water **~ 60–70 °C (just below boiling, 140–160 °F)**; mouthpiece/ spout **wrapped with gauze**; distance **~ 23–30 cm (9–12 inch)** from face; cover head with towel (tent); **15–20 min**; after – avoid going out in cold air for **30 min**; **eyes closed**.
- Precautions: **scalds/ burns** (children, elderly – supervise), don't fill above level of spout.

Nebulization

- Converts liquid medicine into **fine mist/ aerosol** (particles **1–5 µm** reach lower airways) – **bronchodilators (salbutamol, ipratropium), steroids (budesonide), NS, mucolytics (acetylcysteine), antibiotics**.
- Types: **jet (compressor)**, **ultrasonic**, mesh.
- Procedure: **sitting upright (Fowler's)**, medicine in cup (diluted with **NS 2–4 ml**), flow **6–8 L/min** (O_2 or air), breathe **slowly & deeply through mouth**, until mist stops (**10–15 min**), cough after; check pulse (salbutamol – **tachycardia, tremors**); **rinse mouth after steroids** (oral thrush); clean & dry equipment.
- **MDI (metered dose inhaler)** – shake, exhale, inhale slowly with actuation, **hold breath 10 sec**, wait **1 min** between puffs; **spacer** improves delivery; **bronchodilator first, steroid after (5 min gap)**.
- **DPI (dry powder inhaler)** – **rapid, forceful** inhalation; **don't shake**, don't exhale into device.

Chest physiotherapy (CPT)

- **Postural drainage** – positioning so **gravity drains secretions from specific lung segments** into larger airways; done **before meals (1 hr) or 2–3 hrs after meals** (vomiting/ aspiration ↓), **early morning (secretions accumulate overnight) & bedtime**; each position **5–15 min** (total 20–30 min); **after bronchodilator nebulization** (open airways first).
  - **Upper lobes** – **sitting upright / semi-Fowler's**
  - **Middle lobe (R)/ lingula (L)** – **side-lying, head down ~15°** (quarter turn back)
  - **Lower lobes** – **Trendelenburg (head down 30–45°)**, prone/ lateral as per segment

- **Percussion (clapping)** – **cupped hands**, rhythmic over affected segment on **thin cloth (towel)**, **1–2 min (3–5 min)**; **avoid spine, sternum, breasts, kidneys, liver, spleen, below ribs**.
- **Vibration** – flat hand, tensed arm, **during exhalation**.
- **Contraindications to CPT**: **haemoptysis, pulmonary embolism, ↑ICP, unstable cardiac, rib fractures, osteoporosis, recent spinal/ eye/ head surgery, bleeding disorders, pneumothorax (untreated), empyema**, immediately after meals; head-down contraindicated in ↑ICP, GERD, late pregnancy, severe HTN.

14.4 Suctioning

Type	Route	Technique
Oropharyngeal (Yankauer – tonsil tip)	Mouth	**Clean (medical asepsis)**
Nasopharyngeal/ Nasotracheal	Nose	**Sterile**
Endotracheal/ Tracheostomy	ET tube/ trach	**Sterile (open) or closed in-line system**

Suctioning – must-remember rules

- **Indications (as needed – not routine)**: audible/ visible secretions, **gurgling**, crackles/ rhonchi, ↓ SpO_2, ↑ peak airway pressure (ventilator), restlessness.
- **Position**: conscious – **semi-Fowler's**; unconscious – **lateral (facing nurse)**.
- **Hyperoxygenate with 100 % O_2 for 30–60 sec before & after** (hypoxia prevention).
- **Suction pressure**: **adult 100–150 mmHg (wall)** (some texts 80–120); **child 80–100 (100–120)**; **infant 60–80 (50–100)**; portable – adult 10–15 cmHg.
- **Catheter size**: **≤ half the internal diameter of artificial airway**; adult **12–16 Fr** (12–14 for trach), child 8–10, infant 5–8.
- **Insertion length**: nasopharyngeal – **tip of nose to earlobe (~13–15 cm adult)**; trach – until resistance felt (carina), then **withdraw 1 cm** before suction.
- **Insert WITHOUT suction** (thumb off port) → **apply intermittent suction while withdrawing with rotating motion**.
- **Each suction pass max 10–15 seconds** (time = 1 breath hold for nurse); **max 3 passes**; **rest 20–30 sec (1 min)** between passes.
- **Suction order**: **trachea first, then nose/ mouth** (same catheter: sterile → clean, never reverse; clean area to dirty).
- **Saline instillation – not routinely recommended** (pushes bacteria, ↓ SpO_2).
- Flush catheter with sterile NS between passes.
- **Complications**: **hypoxia (most common)**, **bradycardia/ arrhythmias (vagal)**, trauma/ bleeding, infection, atelectasis, laryngospasm, ↑ICP. **Stop if SpO_2 falls / HR drops** → reoxygenate.

14.5 Tracheostomy Care

- **Tracheostomy** – surgical opening in trachea (**2nd–3rd/ 3rd–4th tracheal ring**); tube – **outer cannula, inner cannula, obturator** (+ cuff in cuffed tubes).
- **Bedside always**: **obturator, spare tube (same size & one size smaller)**, tracheal dilator, suction, O_2, Ambu bag, sterile gloves.

- **Care** (every 8 hrs/ PRN): sterile technique (fresh trach), **clean inner cannula** (disposable or with **H_2O_2 then rinse with NS**), stoma cleaned with **NS** (H_2O_2 half-strength if crusted, rinse), **sterile drain/ split gauze (pre-cut keyhole – never cut gauze – fibres aspirated)**, change ties – **2 persons / new tie before removing old**; **one finger (1–2 finger) space** under ties.
- **Cuff pressure 20–25 mmHg (25–30 cmH_2O)** – check every 8 hrs (manometer); high → **tracheal necrosis/ stenosis/ fistula**; low → aspiration, air leak.
- **Humidification** essential (upper airway bypassed); fluids.
- **Accidental decannulation** – fresh (< 72 hrs–7 days, tract not formed): **call help, keep stoma open with tracheal dilator/ hemostat**, O_2, bag-mask over mouth & cover stoma; mature tract – **insert spare tube with obturator, then remove obturator**.
- First tube change – by **surgeon/ ENT after 5–7 days**.
- Communication – **writing board, picture cards, speaking valve (Passy-Muir – cuff must be deflated)**.
- **Fenestrated tube** – allows speaking/ weaning. **Decannulation** – after capping trials (24 hrs).

PYQ style

- Earliest sign of hypoxia – **restlessness/ confusion**; late – **cyanosis**.
- O_2 cylinder colour (India) – **black body, white shoulder**.
- Nasal cannula flow – **1–6 L/min**; FiO_2 – **24–44 %**.
- Highest O_2 without intubation – **non-rebreather mask**.
- Precise FiO_2/ COPD – **Venturi mask**.
- Simple mask minimum flow – **5 L/min**.
- Humidifier water – **sterile/ distilled**.
- Suction each pass – **10–15 sec**; adult pressure – **100–150 mmHg**.
- Suction applied – **while withdrawing**, not inserting.
- Incentive spirometer – **inhale slowly & deeply, hold 3–5 sec**.
- Postural drainage timing – **before meals / early morning**.
- Steam inhalation distance – **~23–30 cm (9–12 inch)**.
- Tracheostomy cuff pressure – **20–25 mmHg**.
- Tracheostomy bedside – **obturator & spare tube**.

15 Fluids, Electrolytes & IV Therapy

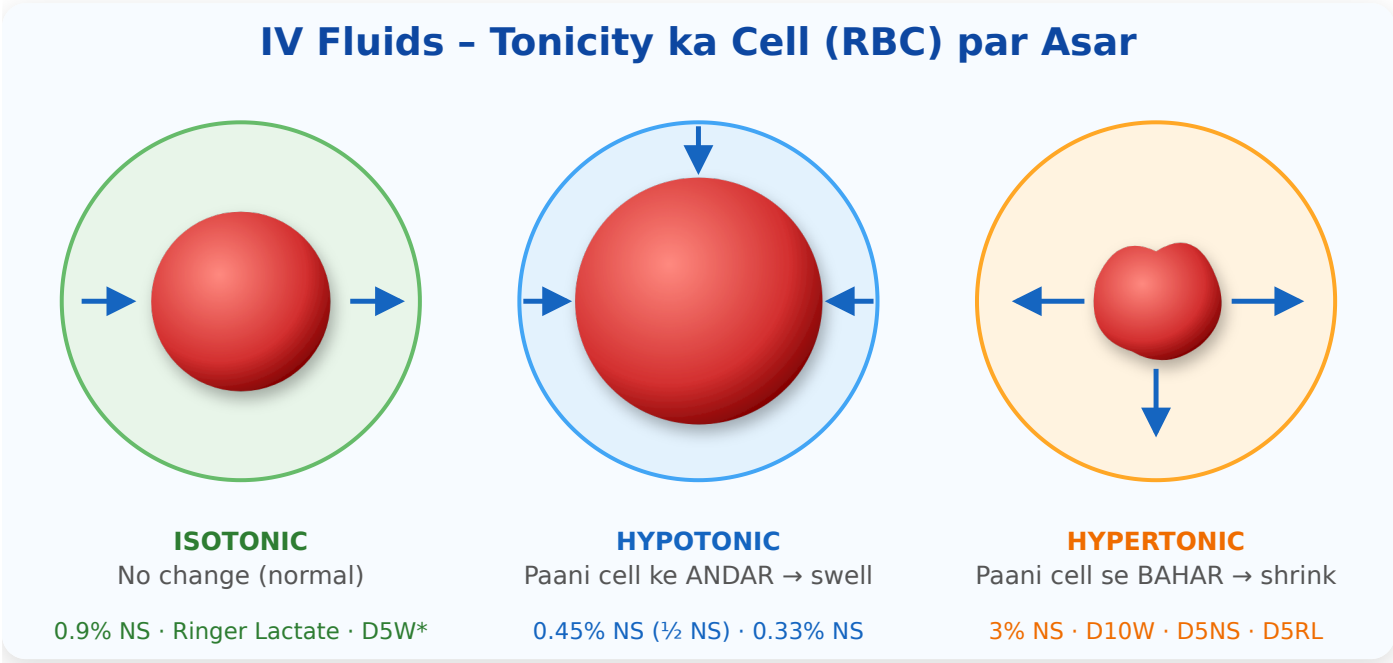
15.1 Body Fluids – Quick Facts

- Body water **~60 % adult** (male 60 %, female 50–55 %), **newborn 75–80 %**, elderly 45–50 %. **ICF ⅔ (40 %) – K^+ main**; **ECF ⅓ (20 %) – Na^+ main** (interstitial 15 % + plasma 5 %).
- **Daily intake ≈ output ≈ 2500 ml** (water 1500 + food 800 + metabolism 200 | urine 1500 + insensible skin 500 + lungs 300–400 + faeces 100–200).
- **Thirst centre – hypothalamus**. Hormones: **ADH** (water retain), **aldosterone** (Na retain, K excrete), **ANP** (Na & water excrete).
- Electrolytes normal: **Na 135–145**, **K 3.5–5.0**, **Ca 8.5–10.5 mg/dL**, **Cl 96–106**, **HCO_3 22–26**, **Mg 1.5–2.5**.

Fluid volume deficit (Dehydration)	Fluid volume excess (Overload)
Thirst, **dry mucosa**, **poor skin turgor**, sunken eyes, **sunken fontanelle (infant)**	**Oedema**, puffy eyelids, **weight gain**
Tachycardia, weak thready pulse, hypotension, orthostatic	**Bounding pulse, hypertension**, **JVD**
↓ urine, concentrated (SG > 1.030)	**Crackles**, dyspnoea, cough (pulmonary oedema)
↑ Hct, ↑ BUN, weight loss	↓ Hct (dilution)
Fever, confusion, flat neck veins	S3 sound

- **Best indicator of fluid balance – daily weight** (**1 kg = 1 L**), same time, scale, clothes.
- **ORS (WHO low osmolarity – 245 mOsm/L)**: per litre – **NaCl 2.6 g, glucose (anhydrous) 13.5 g, KCl 1.5 g, trisodium citrate 2.9 g**. Home-made – **6 level tsp sugar + ½ tsp salt in 1 L clean water**. **29 July = ORS Day**.

15.2 IV Fluids



IV fluids ki tonicity – RBC par asar

Fluid	Tonicity	Use
0.9 % NaCl (Normal saline)	**Isotonic** (308 mOsm)	Shock, hypovolaemia, **blood transfusion (only compatible fluid)**, hyponatraemia, DKA (first)
Ringer's Lactate (Hartmann's)	Isotonic	**Burns (Parkland)**, surgery, trauma; contains K & Ca – not with blood; avoid in liver failure (lactate), lactic acidosis
5 % Dextrose (D5W)	Isotonic in bag, **physiologically hypotonic**	Free water, hypernatraemia; **avoid in ↑ICP/ head injury**
0.45 % NaCl (½ NS)	**Hypotonic**	Cellular dehydration, hypernatraemia, DKA (later)
DNS (5 % dextrose in 0.9 % NaCl)	**Hypertonic**	Maintenance, fluid + calories
10 % Dextrose	Hypertonic	**Hypoglycaemia**, TPN stop/ transition
25 %/ 50 % Dextrose (D25/ D50)	Hypertonic	**Severe hypoglycaemia** (bolus)
3 % NaCl	Hypertonic	**Severe hyponatraemia, cerebral oedema** – slowly, ICU
Isolyte-P / Isolyte-M		Paediatric / maintenance
Mannitol 20 %	Hypertonic (osmotic diuretic)	**↑ICP**, cerebral oedema – **use filter**
Colloids – albumin 5 %/ 25 %, Haemaccel (gelatin), dextran, hetastarch	Plasma expanders	Hypovolaemia, hypoalbuminaemia

15.3 IV Cannulation

Gauge (G)	Colour	Flow (approx)	Use
14 G	**Orange**	~270–300 ml/min	**Trauma, massive blood loss** – largest
16 G	**Grey**	~200 ml/min	Major surgery, rapid fluids, trauma
18 G	**Green**	~90–100 ml/min	**Blood transfusion**, surgery, CT contrast
20 G	**Pink**	~60 ml/min	**Most adult IV therapy**, blood (acceptable)
22 G	**Blue**	~35 ml/min	Children, elderly, small veins
24 G	**Yellow**	~20 ml/min	**Neonates**, infants, fragile veins
26 G	Violet	~13 ml/min	Neonates

- **Smaller gauge number = larger diameter**.
- **Site selection**: **distal first (hand – metacarpal/ dorsal veins), then move proximal** (forearm – **cephalic, basilic, median antebrachial**); **non-dominant arm**; **avoid**: joints/ flexion areas (antecubital – except emergency/ short-term), **arm with AV fistula, mastectomy side, injured/ infected/ oedematous/ burned limb, phlebitis, legs in adults (DVT)**; infants – **scalp veins**, hand, foot.
- Tourniquet **10–15 cm (4–6 inch) above** site, **< 1–2 min**; skin antisepsis **chlorhexidine 2 % in 70 % alcohol** (or povidone-iodine) – **let dry 30 sec**; insert **bevel up at 10–30° (15–30°)**; look for **flashback**; advance cannula, withdraw stylet; **remove tourniquet before connecting**; secure with transparent dressing; label (date, time, gauge, initials).
- **Change peripheral cannula every 72–96 hrs** (or when clinically indicated, policy); **IV administration set** – continuous **every 72–96 hrs**; **blood set – after each unit / every 4 hrs**; **TPN/ lipid set – 24 hrs (lipids 12 hrs)**; IV fluid bag – **hang max 24 hrs**.
- **Phlebitis scale (VIP – Visual Infusion Phlebitis, Jackson)** – 0–5; score ≥ 2 → remove cannula.

15.4 IV Flow Rate Calculation

Formulas (exam favourite)

- **Drops/min (gtt/min) = Total volume (ml) × Drop factor (gtt/ml) ÷ Time (minutes)**
- **ml/hr = Total volume (ml) ÷ Time (hours)**
- **Drop factor**: **Macro-drip (adult) set = 10, 15 or 20 gtt/ml** (Indian sets commonly **15 or 20**); **Micro-drip (paediatric/ burette) = 60 gtt/ml → micro-drops/min = ml/hr**.
- **Blood set = 15 gtt/ml** (10–15).
- **Infusion time (hrs) = Volume ÷ ml/hr**.

Solved calculations

1. 1000 ml NS in 8 hrs, drop factor 15 → (1000 × 15) ÷ 480 = **31.25 ≈ 31 drops/min**.
2. 500 ml in 4 hrs, drop factor 20 → (500 × 20) ÷ 240 = **41.6 ≈ 42 drops/min**.
3. 1500 ml in 12 hrs → ml/hr = 1500 ÷ 12 = **125 ml/hr**.
4. Micro-drip 60 gtt/ml, 50 ml/hr → **50 microdrops/min**.
5. 1000 ml at 20 drops/min, DF 20 → 1 ml/min → **1000 min ≈ 16 hrs 40 min**.
6. 2 pints (1 pint ≈ 500 ml) blood – 1000 ml.

15.5 IV Complications

Complication	Signs	Nursing action
Infiltration (non-vesicant fluid into tissue)	**Swelling, coolness, pallor, taut skin**, ↓/ no flow, discomfort	**Stop, remove cannula, elevate limb**, warm/ cold compress (as per fluid), restart other limb
Extravasation (vesicant – chemo, dopamine, noradrenaline, calcium, KCl conc, dextrose 50 %)	Pain, burning, blistering → necrosis	**Stop, don't flush, aspirate**, antidote (**phentolamine** for vasopressors; hyaluronidase), mark area, notify
Phlebitis (vein inflammation)	**Redness, warmth, pain, red streak/ cord along vein**	Remove cannula, **warm moist compress**, elevate, re-site
Thrombophlebitis	Phlebitis + clot (hard cord)	Same; no massage
Local infection/ cellulitis	Pus, redness	Remove, culture tip, antibiotics
Septicaemia	Fever, chills, hypotension	Remove, blood culture, antibiotics
Fluid overload (circulatory)	**Dyspnoea, crackles, JVD, ↑BP, bounding pulse, cough (frothy)**	**Slow IV to KVO, high Fowler's, O_2**, inform, diuretic
Air embolism	**Sudden dyspnoea, chest pain, hypotension, cyanosis, churning murmur**, loss of consciousness	**Clamp line, turn patient to LEFT side in Trendelenburg (head down)** – air trapped in right atrium, O_2, inform
Speed shock (drug given too fast)	Flushing, headache, chest tightness, syncope, shock	Stop, inform; prevent – infusion pump
Haematoma	Bruising	Pressure, remove
Catheter embolism	Piece of catheter breaks	Tourniquet above site, X-ray
Nerve injury	Tingling	Remove

- **KVO (Keep Vein Open)/ TKO** – ~10–20 ml/hr (or 5–20 drops/min micro).
- Prevention of air embolism – **prime tubing fully**, all connections secure, never let bag run dry, **Luer-locks**.

15.6 Blood Transfusion – Nursing Basics

- **Pre-transfusion**: **written order & informed consent**; **blood group & cross-match**; check **2 nurses at bedside – patient ID, blood group, Rh, unit number, expiry, compatibility label**; **baseline vitals**; **18–20 G cannula**; **blood set with filter (170–200 µm)**; **prime with 0.9 % NS only**.
- Collect blood from bank **just before use**; start **within 30 min**; **complete within 4 hrs** (whole blood/ PRBC); **don't warm in hot water/ microwave** (blood warmer only); **no drugs in blood line**.
- **Rate**: **first 15 min slow (~2 ml/min / 15–20 drops/min) & nurse stays with patient** (most reactions occur in **first 15 min/ first 50 ml**); then increase.
- **Vitals: before, 15 min after start, every 30 min–1 hr, at end** (and 1 hr after).
- **Storage**: whole blood/ PRBC **2–6 °C** (CPDA-1: **35 days**; SAGM: 42 days); **platelets 20–24 °C with agitation – 5 days**; **FFP −30 °C / ≤ −18 °C – 1 year**.
- **Reaction → STOP transfusion immediately → keep vein open with NS (new tubing) → inform doctor & blood bank → vitals → send blood bag + set + patient's blood & urine sample → document**.

- Reactions: **febrile non-haemolytic (most common)**, **acute haemolytic (most dangerous – ABO mismatch – clerical error; fever, chills, flank/ back pain, haemoglobinuria, hypotension)**, allergic (urticaria – antihistamine), anaphylaxis, **TACO (circulatory overload)**, TRALI, bacterial, hypocalcaemia (citrate – massive), hyperkalaemia, hypothermia.
- **Universal donor O –ve**, **universal recipient AB +ve**. **1 unit PRBC ↑ Hb ~1 g/dL**.

PYQ style

- Fluid compatible with blood – **0.9 % NaCl**.
- Largest IV cannula – **14 G (orange)**; blood transfusion – **18 G (green)**; neonates – **24 G (yellow)**.
- Drops/min = **volume × drop factor ÷ time (min)**.
- Micro-drip drop factor – **60**.
- Infiltration – **cool, pale, swollen** site; phlebitis – **red, warm, tender** along vein.
- Air embolism position – **left lateral + Trendelenburg**.
- Fluid overload – **high Fowler's, slow IV**.
- Blood transfusion complete within – **4 hrs**; first **15 min** stay with patient.
- Transfusion reaction – **first stop transfusion**.
- Tourniquet placed – **10–15 cm above** site.
- IV site selection – **distal to proximal**.

16 Medication Administration & Drug Calculations

16.1 Basics & Rights

- **Pharmacology** – study of drugs. **Pharmacokinetics** – what body does to drug: **ADME – Absorption, Distribution, Metabolism (liver), Excretion (kidney)**. **Pharmacodynamics** – what drug does to body.
- **Half-life (t½)** – time for drug concentration to fall by 50 %. **Steady state** – after ~4–5 half-lives. **Therapeutic index** – safety margin (narrow – digoxin, lithium, warfarin, phenytoin, theophylline, aminoglycosides – **monitor levels**).
- **Peak level** – highest concentration (draw ~30–60 min after IV dose); **Trough level** – lowest (**draw just before next dose**).
- Drug names: **chemical**, **generic (non-proprietary – official)**, **trade/ brand (proprietary)**. Prescribe & use **generic names**.
- **Side effect** (predictable, mild, unintended), **adverse effect** (harmful), **toxic effect**, **allergic reaction**, **idiosyncrasy** (unexpected individual response), **tolerance**, **dependence**, **cumulative effect**, **synergism** (combined effect > sum), **antagonism**, **potentiation**.
- **Teratogenic** (fetal malformation), **iatrogenic** (treatment caused).

Rights of medication administration

Classic 5 rights: **Right Patient, Right Drug, Right Dose, Right Route, Right Time**.
Extended (10 rights): + **Right Documentation**, **Right Reason/ indication**, **Right Response (evaluation)**, **Right to Refuse**, **Right Education/ information**, (+ Right assessment, Right expiry, Right form/ approach).
3 checks: (1) when **taking from shelf/ drawer**, (2) when **preparing/ pouring**, (3) **before administering (at bedside)** / returning container.
2 patient identifiers (name + DOB/ UHID) – ask patient to state name; check ID band; **never use room/ bed number**.

- **Medication errors** – most common: wrong dose, wrong time, omission; **report immediately** – **first assess patient**, notify doctor, document, incident report.
- **Never give medicine prepared by someone else**; **document only after giving**; **don't leave medicines at bedside**; patient refuses → document & inform; **stay till patient swallows**.
- **Verbal/ telephone orders** – emergency only; **write down & read back**, doctor co-sign within 24 hrs.
- **Order types**: **Stat** (immediately, once), **PRN** (as needed), **Standing/ routine** (until discontinued), **Single/ one-time**, **Now** (within 90 min).

Common abbreviations

Abbreviation	Meaning	Abbreviation	Meaning
OD	Once daily	**a.c.**	**Before meals** (ante cibum)
BD/ BID	Twice daily	**p.c.**	**After meals** (post cibum)

Abbreviation	Meaning	Abbreviation	Meaning
TDS/ TID	Three times daily	**h.s.**	**At bedtime** (hora somni)
QID/ QDS	Four times daily	**stat**	Immediately
q4h	Every 4 hrs	**SOS**	If necessary (once)
PRN	When required	**NPO/ NBM**	Nothing by mouth
p.o.	By mouth	**SL**	Sublingual
IM, IV, SC, ID	Routes	**gtt**	Drops
Rx	Recipe/ take	**Tab, Cap, Syr, Inj**	Forms
qod (avoid)	Alternate day	**mg, mcg (µg), g, ml**	Units
ad lib	As desired	**OS/ OD/ OU** (avoid)	Left eye/ right eye/ both eyes
AS/ AD/ AU (avoid)	Left ear/ right ear/ both ears	**BBF / AF**	Before/ after food

- **ISMP "Do Not Use"**: **U/ IU** (write "units"), **QD, QOD**, trailing zero (**5.0 ✘**), missing leading zero (**.5 ✘ → 0.5**), MS/ MSO_4 (morphine vs magnesium).

16.2 Routes of Administration

- **Enteral**: oral, sublingual, buccal, NG/ PEG, rectal. **Parenteral**: ID, SC, IM, IV, intra-arterial, intrathecal, intraosseous, epidural, intracardiac, intra-articular. **Topical/ local**: skin, eye, ear, nose, vaginal, inhalation, transdermal.
- **Fastest action**: **IV > inhalation > sublingual > IM > SC > oral > rectal > transdermal** (approx). **IV – 100 % bioavailability**.
- **First-pass metabolism** (liver) – oral drugs; avoided by **sublingual, IV, rectal (partial), transdermal**.

Oral medications

- Position – **sitting upright** (aspiration); give with **full glass of water (60–100 ml)**; check **swallowing/ gag reflex**.
- **Don't crush**: **enteric-coated (EC), sustained/ extended-release (SR, ER, XL, CR, LA), sublingual, buccal, capsules with beads** – breaking destroys action/ causes irritation.
- **Liquid medicine** – shake; pour with **label towards palm**; read at **bottom of meniscus at eye level**; use oral syringe for accuracy.
- **Iron syrup** – **through straw** (teeth staining); **cough syrup – last, no water after** (soothing effect).
- **Sublingual (NTG)** – under tongue, don't swallow/ chew/ drink till dissolved; **Buccal** – between cheek & gum.
- **Antacids** – 1–3 hrs after meals & HS; separate from other drugs by 1–2 hrs.
- **Pulse/ BP check before**: **digoxin (apical 1 min – hold if < 60 adult)**, beta-blockers, antihypertensives; **RR before opioids**; **blood sugar before insulin/ OHA**; **K^+ before diuretics/ digoxin**.

16.3 Injections – Parenteral

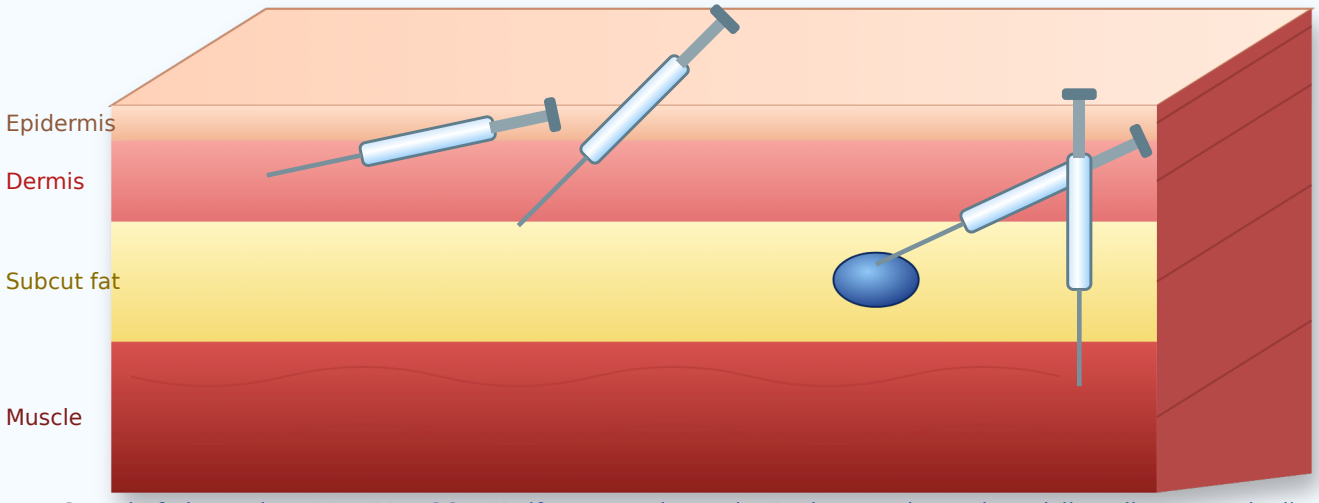
Injection Routes – Angle & Depth (3D skin block)

INTRADERMAL (ID)
10–15° · bevel UP
Dermis me · Mantoux, BCG, test dose
Wheal (bleb) bane

SUBCUTANEOUS (SC)
45° (90° obese)
Insulin, Heparin
≤ 1 ml (max 2 ml)

INTRAVENOUS (IV)
15–30° (≈ 25°)
Fastest action
bevel UP in vein

INTRAMUSCULAR
(IM) 90° · muscle
Deltoid ≤ 1 ml
Gluteal/thigh ≤ 3–5 ml



Speed of absorption: IV > IM > SC > ID (fastest → slowest) · ID slowest absorption – isliye allergy test ke liye

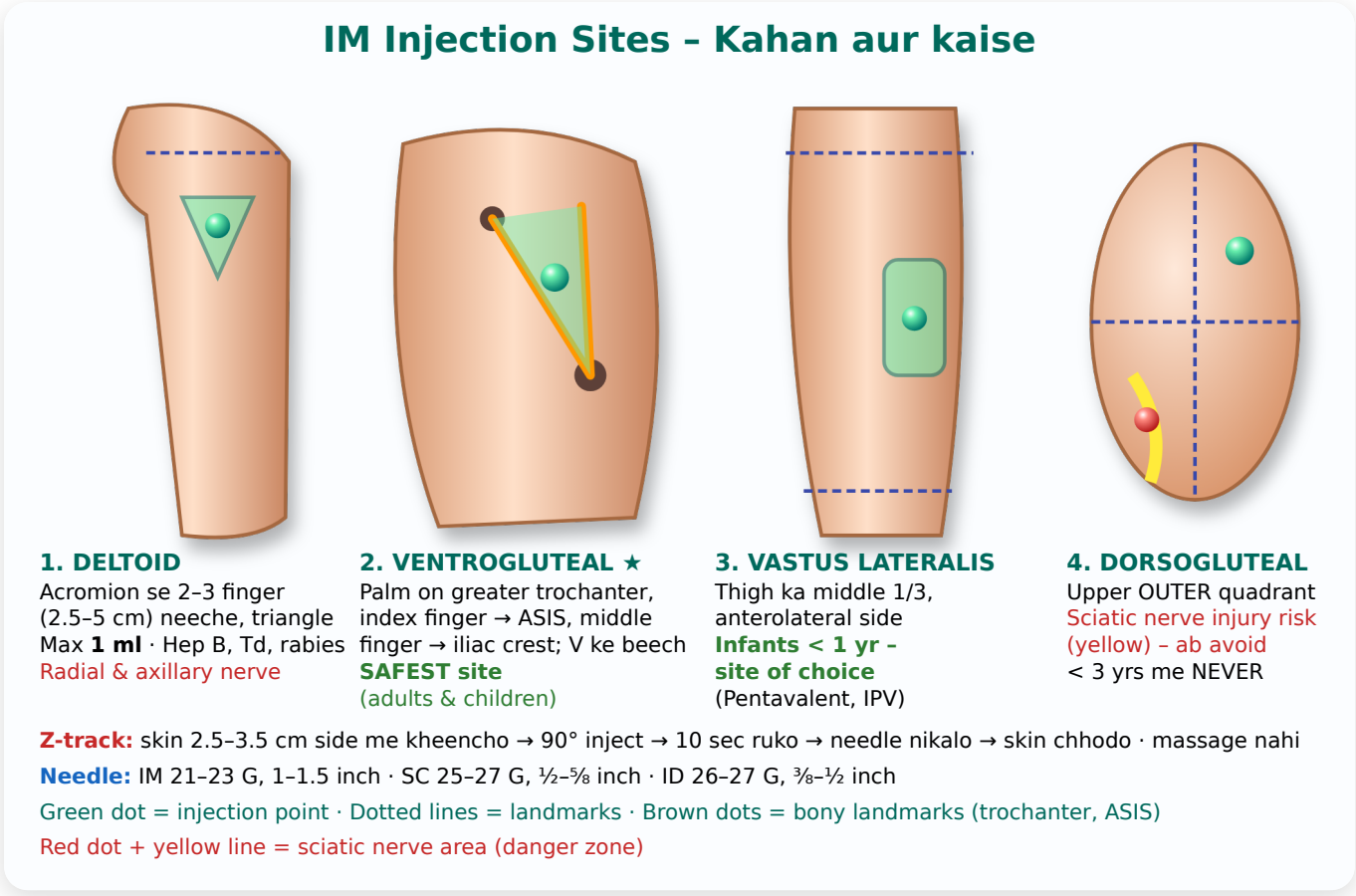
Injection routes – angle aur depth

Route	Angle	Needle gauge & length	Max volume	Sites
Intradermal (ID)	**10–15°, bevel UP**	**26–27 G, ⅜–½ inch** (tuberculin syringe 1 ml)	**0.01–0.1 ml**	**Inner (ventral) forearm** (Mantoux, sensitivity tests), **upper arm (BCG – left deltoid insertion)**, upper back
Subcutaneous (SC)	**45°** (thin) – **90°** (normal/ obese; pinch 2.5 cm/ 1 inch skin)	**25–27 G, ½–⅝ inch**	**≤ 1 ml (max 1.5–2 ml)**	**Abdomen** (2 inch/ 5 cm away from umbilicus – **fastest SC absorption**), **outer upper arm**, anterior thigh, scapular area, upper buttock
Intramuscular (IM)	**90°**	**21–23 G (19–25), 1–1.5 inch** (child ⅝–1 inch)	**Deltoid ≤ 1 ml (0.5–1)**; **ventrogluteal/ vastus lateralis 2–3 ml (up to 5 ml adult)**; infant vastus ≤ 0.5–1 ml	Deltoid, **ventrogluteal**, vastus lateralis, dorsogluteal, rectus femoris
Intravenous (IV)	**15–30° (≈ 25°)**, bevel up	18–22 G	Any (as ordered)	Peripheral veins

- **Mantoux/ sensitivity test** – **don't massage**, mark circle; read **48–72 hrs**; **bleb/ wheal** formation confirms correct ID injection.
- **Heparin & insulin (SC)**: **don't aspirate, don't massage** (haematoma / ↑ absorption); heparin – abdomen; **enoxaparin** – don't expel air bubble (pre-filled), inject in **"love handles"**, don't rub.

- **Aspiration** before IM – **not recommended for vaccines** (WHO/ CDC) & SC; for other IM drugs – per policy (dorsogluteal – yes).

IM injection sites



IM sites – deltoid, ventrogluteal, vastus lateralis, dorsogluteal

Site	Landmarks	Notes
Deltoid	**2.5–5 cm (2–3 finger-widths) below acromion process**, inverted triangle (acromion base – axillary line apex)	Small volume (**≤ 1 ml**); vaccines (Hep B, Td, COVID, rabies adult); **radial & axillary nerve, brachial artery** risk
Ventrogluteal	**Palm on greater trochanter**, **index finger on ASIS**, **middle finger along iliac crest** → inject in **V/ triangle** between fingers (left hand for right hip)	**Safest site** for adults & children (> 7 months) – **no major nerves/ vessels**, thick muscle, less fat
Vastus lateralis	**Middle third of anterolateral thigh** (hand's breadth below greater trochanter & above knee)	**Site of choice for infants (< 1 yr/ 12 months)** – vaccines (Pentavalent, PCV, IPV)
Dorsogluteal	**Upper outer quadrant** of buttock (imaginary line from **PSIS to greater trochanter** – inject above & lateral)	**Sciatic nerve injury & superior gluteal artery** risk – **avoid / last option**; **never < 3 yrs** (muscle undeveloped)
Rectus femoris	Anterior middle third of thigh	Self-injection

Z-track technique

- Purpose – **prevent leakage (tracking) of irritating/ staining drugs into subcutaneous tissue** – **iron dextran (Imferon)**, hydroxyzine, some antipsychotics (haloperidol decanoate).
- **Change needle after drawing up** (drug on outside of needle stains); **air-lock 0.2–0.5 ml** (optional).
- **Pull skin & SC tissue laterally 2.5–3.5 cm (1–1.5 inch)** → inject at 90° slowly (**10 sec/ml**) → **wait 10 sec** → withdraw needle → **release skin** (track sealed) → **don't massage**; preferred site – **ventrogluteal**/ dorsogluteal; walk to ↑ absorption.

Needles & syringes

- **Gauge** – diameter (**bigger number = thinner needle**); **length** – choose by site & patient size.
- Parts – **syringe: tip, barrel, plunger, flange**; **needle: hub, shaft, bevel** (**bevel up** for ID/ SC/ IV).
- **Insulin syringe – U-100 (100 units/ml)**; **tuberculin syringe – 1 ml (0.01 ml markings)**.
- **Ampoule – glass**: flick to bring drug down, **break away from you with gauze**, use **filter needle** (glass particles), no air injection. **Vial** – rubber stopper; **swab with alcohol**, **inject equal volume of air** (positive pressure) before withdrawing; **multi-dose vial – label date/ time opened, discard after 28 days** (or as per manufacturer).
- **Reconstitution** – diluent (sterile water/ NS) as per label; roll (don't shake proteins).

Insulin administration

- **Mixing two insulins – "Clear before Cloudy"** (Regular → NPH): inject air into **NPH (cloudy)** first → air into **Regular (clear)** → withdraw **Regular** → withdraw **NPH**. ("**RN**" – Regular, NPH). **Glargine/ detemir – never mix**.
- **Only Regular insulin (clear) can be given IV**.
- Roll cloudy insulin gently between palms (**don't shake**).
- **Abdomen – fastest absorption**; rotate sites **within same region** (lipohypertrophy prevention); **2 inch (5 cm) from umbilicus**; don't inject into exercising limb.
- Storage: unopened – **refrigerate 2–8 °C**; in-use vial/ pen – **room temp ≤ 28–30 days**; never freeze/ direct sunlight.

16.4 Other Routes

- **Eye drops**: patient **head tilted back, looks up**; **pull lower lid down** → drop into **lower conjunctival sac (centre/ outer third)** – **never on cornea**; dropper **1–2 cm** above (no touch); **close eye gently (not squeeze) & press inner canthus (nasolacrimal duct/ punctum) 1–2 min** (systemic absorption ↓); **multiple drops – wait 3–5 min**; **drops before ointment** (wait ~10 min); **ointment – thin ribbon from inner to outer canthus**.
- **Ear drops**: warm to **body temperature** (cold → **vertigo**, nausea); side-lying affected ear up; **adult – pinna UP & BACK**; **child < 3 yrs – DOWN & BACK**; drops on canal wall (not directly on TM); **remain lying 5–10 min**; cotton loosely (optional).
- **Nasal drops**: **head tilted back** (posterior pharynx) – **Proetz position** (head over bed edge – ethmoid/ sphenoid sinuses), **Parkinson position** (head turned to side – maxillary/ frontal sinuses); remain **5 min**; **decongestant drops max 3–5 days** (rebound congestion – rhinitis medicamentosa).
- **Inhalers (MDI)**: shake, exhale fully, inhale slowly with actuation, **hold breath 10 sec**, wait **1 min** between puffs; **bronchodilator before steroid**; **rinse mouth after steroid** (thrush); spacer. **DPI** –

don't shake, fast deep inhalation.

- **Transdermal patch** (NTG, fentanyl, nicotine): **hairless, clean, dry** area; **rotate sites**; **remove old patch first**; gloves; don't cut; fentanyl – no heat pads (↑ absorption).
- **Rectal suppository** – Sims' (left lateral), beyond internal sphincter (adult 7.5–10 cm), against rectal wall.
- **Vaginal** – lithotomy/ dorsal recumbent, applicator, **at bedtime**, lie **10–15 min**.
- **Topical** – gloves, clean area, thin layer; **steroid creams – thin, don't cover** unless ordered.

16.5 Storage, Controlled Drugs & Safety

- **Narcotics/ controlled drugs (NDPS Act 1985 – India): double-locked cupboard**, **separate register**, **count at every shift change by 2 nurses**, **wastage witnessed & signed by 2 nurses**, keys with in-charge; **empty ampoules returned** to pharmacy.
- **High-alert medications** (ISMP) – **concentrated KCl** (never IV push – dilute; stored separately with red label), insulin, heparin/ anticoagulants, opioids, chemotherapy, neuromuscular blockers, magnesium sulphate, hypertonic saline → **independent double-check**.
- **LASA (Look-Alike Sound-Alike)** drugs – Tall Man lettering (e.g. hydrOXYzine/ hydrALAZINE); separate storage.
- Storage: **cool, dry, dark** place (most **< 25 °C**); **refrigerator 2–8 °C** – **insulin, vaccines (cold chain), oxytocin, some antibiotics (reconstituted), suppositories, eye drops (some)**; light-sensitive – **nitroprusside, amber bottles**; check **expiry date** (FEFO – First Expiry First Out).
- **Cold chain** (vaccines) – **ILR (ice-lined refrigerator) +2 to +8 °C**; **deep freezer −15 to −25 °C**; **VVM (vaccine vial monitor)** – inner square lighter than outer circle = usable; **Shake test** – frozen DPT/ TT/ Hep B.
- **Paediatric doses** – Clark's rule & Young's rule (below).

16.6 Drug Calculations

Formulas

- **Dose required (tablet/ liquid) = (Desired dose ÷ Dose on hand) × Quantity (vehicle) → D/H × Q**
- **Young's rule** (child **1–12 yrs**): **Child dose = [Age (yrs) ÷ (Age + 12)] × Adult dose**
- **Clark's rule** (by weight): **Child dose = [Weight (lb) ÷ 150] × Adult dose** (or **wt kg ÷ 70**)
- **Fried's rule** (infant < 1–2 yr): **[Age (months) ÷ 150] × Adult dose**
- **BSA (Body surface area) rule** – most accurate: **(BSA m^2 ÷ 1.73) × Adult dose**
- **Percentage solution**: **1 % = 1 g in 100 ml** (w/v); **1 : 1000 = 1 g in 1000 ml = 1 mg/ml**; 1 : 10,000 = 0.1 mg/ml.
- **Dilution: $C_1V_1 = C_2V_2$**.
- **Units**: 1 g = 1000 mg; 1 mg = 1000 mcg; 1 kg = 2.2 lb; 1 L = 1000 ml; **1 tsp = 5 ml**, **1 tbsp = 15 ml**, **1 oz = 30 ml**, 1 cup = 240 ml; **1 grain (gr) = 60–65 mg**; 1 inch = 2.54 cm.

Solved calculations

1. Order **500 mg**, available **250 mg/tab** → 500/250 × 1 = **2 tablets**.
2. Order **75 mg**, available **50 mg in 2 ml** → 75/50 × 2 = **3 ml**.
3. Order **Inj. Gentamicin 60 mg**, vial **80 mg/2 ml** → 60/80 × 2 = **1.5 ml**.
4. **Paracetamol 15 mg/kg** for 20 kg child = **300 mg**; syrup 250 mg/5 ml → 300/250 × 5 = **6 ml**.
5. **Young's**: adult dose 500 mg, child 6 yrs → 6/(6+12) × 500 = **166.6 mg**.
6. **Clark's**: adult 300 mg, child 50 lb → 50/150 × 300 = **100 mg**.
7. Prepare **500 ml of 5 % solution** → 5 g/100 ml → **25 g in 500 ml**.
8. **Adrenaline 1 : 1000** – 0.5 ml = **0.5 mg**.
9. **Heparin 5000 units** ordered; vial **25,000 units/5 ml** → 5000/25000 × 5 = **1 ml**.
10. **Insulin 30 units** from U-100 → **0.3 ml**.
11. **Crystalline penicillin 10 lakh units** vial – add **5 ml** → 2 lakh units/ml; order 4 lakh → **2 ml**.

PYQ style

- "a.c." – **before meals**; "h.s." – **at bedtime**; "stat" – **immediately**.
- Intradermal angle – **10–15°**; SC – **45°**; IM – **90°**; IV – **25°**.
- Safest IM site – **ventrogluteal**; infant – **vastus lateralis**.
- Deltoid max volume – **1 ml**.
- Iron dextran – **Z-track**.
- Insulin mixing – **clear (regular) first**.
- Eye drop – **lower conjunctival sac**; press **inner canthus**.
- Ear drops adult – pinna **up & back**.
- Enteric-coated tablets – **not crushed**.
- Narcotics counted – **every shift by 2 nurses**.
- First action on medication error – **assess patient**.
- 1 tsp = **5 ml**; 1 grain = **60–65 mg**.

17 Hot & Cold Applications, Wound Care, Bandaging & Specimens

17.1 Hot & Cold Applications

	Heat	Cold
Blood vessels	**Vasodilation** → blood flow ↑	**Vasoconstriction** → blood flow ↓
Effects	↑ metabolism, ↑ tissue O_2, **muscle relaxation**, ↓ joint stiffness, **promotes suppuration (pus localisation)**, healing, ↓ pain (spasm)	**↓ swelling/ oedema**, **↓ bleeding**, **↓ pain (local anaesthetic – slows nerve conduction)**, ↓ metabolism, ↓ inflammation, ↓ temperature
Uses	Muscle spasm, **joint stiffness (arthritis, chronic)**, **abscess (localize pus)**, **IV infiltration (non-vesicant) / phlebitis**, menstrual cramps, retention of urine (warm water on perineum), **sitz bath**	**Acute injury (first 24–48 hrs) – sprain, strain**, haemorrhage/ epistaxis, **fever (cold sponge)**, **post-tonsillectomy (ice collar)**, insect bites, headache, burns (cool water), **joint swelling (acute)**
Contraindications	**First 24 hrs of injury/ trauma (↑ bleeding & swelling)**, **active bleeding**, **acute inflammation – appendicitis (rupture)**, **non-inflammatory oedema**, **localized malignancy (spread)**, open wound, **testes (spermatogenesis ↓)**, pregnant abdomen, impaired sensation (burns), metal implants (diathermy)	**Open wounds (↓ healing)**, **impaired circulation (PVD, Raynaud's, diabetes)**, shivering, cold allergy, **elderly/ infants** (hypothermia), oedematous tissue, neuropathy

- **Rebound phenomenon** – heat max effect at **20–30 min**; beyond **45–60 min** → **vasoconstriction (reverse)**; cold beyond ~**60 min** → **vasodilation**. So apply **20–30 min**, then rest.
- **Elderly, infants, diabetics, unconscious, paralysed, neuropathy** – **↓ sensation – burn/ frostbite risk** → extra care, lower temp.

Methods & temperatures

Application	Type	Temperature / time
Hot water bag (bottle)	Dry heat	**Adult 46–52 °C (115–125 °F)**; **children < 2 yrs, elderly, unconscious 40.5–46 °C**; fill **½–⅔**, **expel air**, check leak (invert), cover with flannel/ towel; never on unconscious/ diabetic; check skin every 5–10 min
Electric heating pad	Dry heat	Low setting; **don't lie on it**, no pins
Infrared/ heat lamp	Dry	**45–60 cm (18–24 inch)** distance, 20 min
Hot moist compress/ fomentation	Moist	**40.5–43 °C (105–110 °F)**; 15–20 min; change every 5 min; **moist heat penetrates deeper**
Sitz bath (hip bath)	Moist	**40–43 °C**, **15–20 min** – **perineal/ rectal**: haemorrhoids, episiotomy, fissure, post-delivery; watch for **faintness** (pooling of blood in pelvis)
Hot soak / immersion	Moist	40–43 °C, 15–20 min
Paraffin wax bath		Arthritis (hands) – ~52 °C

Application	Type	Temperature / time
Ice bag/ ice cap/ ice collar	Dry cold	Fill **½–⅔ with small ice pieces**, expel air, cover; **20–30 min**
Cold compress	Moist cold	**15 °C**; 20 min
Cold sponge/ tepid sponge	Moist	**Tepid 27–34 °C**; cold 18–27 °C; **15–20 min (20–30)**; **stop if shivering** (heat production); sponge with **long strokes**, ice bags at axilla/ groin; **recheck temp after 30 min**
Cooling blanket (hypothermia blanket)		Hyperthermia
Chemical hot/ cold packs		Squeeze to activate

17.2 Wounds

- **Wound** – break in continuity of skin/ tissue.
- **Classification**: **intentional** (surgical) vs **unintentional** (trauma); **open** vs **closed**; **acute** vs **chronic** (> 4–12 weeks – pressure ulcer, diabetic foot, venous ulcer).
- **Types**: **Incision** (clean cut – knife), **Laceration** (torn, irregular edges), **Abrasion** (scrape – superficial), **Puncture** (penetrating – nail; **tetanus risk**), **Contusion** (bruise – closed), **Avulsion** (tissue torn away), **Penetrating**, **Crush**.
- **Surgical wound classification (CDC)**: **Clean** (no inflamed/ GI/ GU/ respiratory tract entry – hernia, thyroid) – infection < 2 %; **Clean-contaminated** (GI/ respiratory entered under control – cholecystectomy); **Contaminated** (open fresh trauma, spillage); **Dirty/ infected** (old trauma, pus, perforated viscus).

Wound healing

- **Phases**: **Haemostasis** (immediate – platelet plug, clot) → **Inflammatory** (**0–4/ 3–6 days** – redness, heat, swelling, pain; neutrophils, macrophages) → **Proliferative/ Reconstructive** (**4–21 days** – **granulation tissue (beefy red, granular)**, collagen, angiogenesis, epithelialization, wound contraction – myofibroblasts) → **Maturation/ Remodelling** (**21 days – 1–2 years** – scar strengthens to ~**70–80 %** of original).
- **Primary intention** – edges approximated (sutured surgical) – minimal scar, fast. **Secondary intention** – wound left open, heals by **granulation & contraction** from bottom up – larger scar (pressure ulcer, burns, infected). **Tertiary intention (delayed primary closure)** – left open few days (infection control/ oedema) then sutured.
- **Factors delaying healing**: **age (elderly)**, **malnutrition (protein, Vit C, zinc)**, **diabetes**, **infection**, **smoking**, **steroids/ immunosuppressants/ chemo**, obesity, poor circulation/ anaemia, stress, radiation, foreign body, **excess moisture or dryness**.
- **Wound bed colour (RYB)**: **Red** (granulation – healthy → protect, keep moist), **Yellow** (slough/ exudate → cleanse, debride), **Black** (necrotic eschar → debride).
- **Exudate types**: **Serous** (clear, watery – plasma), **Sanguineous** (bright red – fresh bleeding), **Serosanguineous** (pale pink – normal early post-op), **Purulent** (pus – thick, yellow/ green, foul – infection).

Wound complications

- **Haemorrhage** (primary, reactionary – within 24 hrs, secondary – 7–14 days – infection).
- **Infection** – fever, ↑ pain, redness, warmth, swelling, pus, foul odour (usually **post-op day 3–7**).
- **Dehiscence** – **partial/ total separation of wound layers** (**post-op day 5–10**; obesity, coughing, malnutrition) – patient feels "**something gave way/ popped**", ↑ serosanguineous drainage.
- **Evisceration** – **protrusion of viscera through wound** → **EMERGENCY**: **stay with patient, call for help**, **cover with sterile towel/ dressing soaked in sterile NS (warm)**, **low Fowler's with knees flexed** (↓ tension), NPO, monitor shock, **never push organs back**.
- **Fistula**, **keloid** (excess scar beyond wound), **hypertrophic scar**, **adhesions**, **incisional hernia**.
- Prevention of dehiscence – **splint incision with pillow** when coughing, nutrition, abdominal binder.

17.3 Dressing

- **Purposes**: protect from contamination & trauma, **absorb drainage**, maintain **moist environment**, haemostasis (pressure), immobilize, debride, comfort, psychological.
- **Principles**: **surgical asepsis**, hand hygiene, **clean → dirty**: clean **from incision (cleanest) outward**, **top to bottom**, **one swab one stroke**; **drain site cleaned last (separately) in circular motion from drain outward**; old dressing removed with **clean gloves**, discard; **sterile gloves/ forceps** for new; **premedicate for pain 30 min before**; **assess wound** (size, colour, drainage, odour, approximation); date & initial on dressing; **dressing trolley – clean (upper) shelf for sterile, lower for unsterile**; infected wounds dressed **last** in ward.
- **Dry dressing** – clean, closed wounds; **wet-to-moist** – debridement; **wet-to-dry** (non-selective mechanical debridement – painful; less used); **pressure dressing** – bleeding.
- **Modern dressings**: **transparent film (Tegaderm)** – IV sites, stage 1–2 ulcers (see through, no absorb); **hydrocolloid (Duoderm)** – light–moderate exudate, autolytic debridement (change 3–5 days); **hydrogel** – **dry wounds – add moisture**; **alginate (seaweed)** – **heavy exudate**, haemostatic; **foam** – moderate–heavy; **silver/ antimicrobial**; **gauze**; **NPWT (VAC)** – **negative pressure (−125 mmHg)**.
- **Wound cleansing solution – Normal saline (best, physiological)**; avoid routine povidone-iodine, H_2O_2, Dakin's on granulating tissue (cytotoxic).

Drains

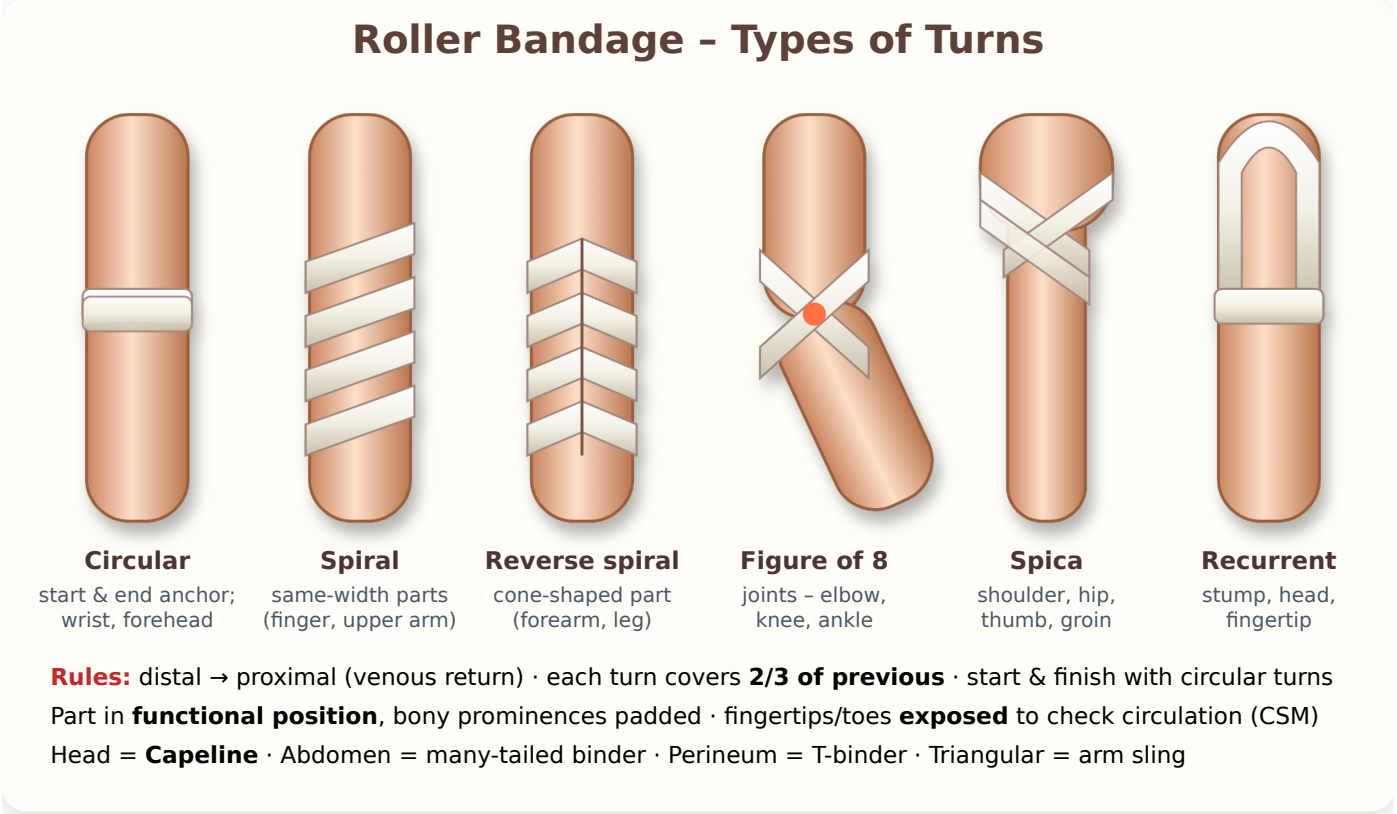
- **Passive** – **Penrose (open, soft rubber)** – gravity/ capillary; **corrugated drain**. **Active (closed suction)** – **Hemovac** (spring), **Jackson-Pratt (JP – bulb/ grenade)** – **compress bulb before closing cap** (creates suction), empty when **½ full**, measure.
- **T-tube** (bile – CBD), **chest tube**, **NG**. Report sudden ↑ bright red drainage.
- **Shortening a drain** – pull out by prescribed length & cut, re-pin (safety pin) (Penrose).

Sutures & staples

- **Absorbable** (catgut – plain/ chromic, **Vicryl (polyglactin)**, PDS) – internal; **non-absorbable** (**silk, nylon (Ethilon), polypropylene (Prolene)**, stainless steel, staples) – skin; removed.
- **Suture removal**: **face/ neck 3–5 days**, **scalp 7–10**, **trunk/ abdomen 7–10**, **arms 7–10**, **legs/ feet 10–14**, joints/ back 10–14. **Alternate sutures removed first** (check wound strength) → rest; **cut close to skin on one side, pull through from opposite (no outer part drawn through tissue)**; apply **Steri-strips**.

- **Staple removal** – staple remover (lower jaw under staple, squeeze).
- Suture types: **interrupted** (each separate – most common), **continuous**, **blanket (locking)**, **mattress (vertical/ horizontal)**, **subcuticular**, **retention** (deep – heavy).

17.4 Bandages & Binders



Roller bandage turns – kaunsa turn kahan

- **Purposes**: hold dressing/ splint, **support**, **pressure (bleeding/ oedema)**, immobilize, limit movement, correct deformity.
- **Types**: **Roller bandage** (gauze, crepe/ elastic, cotton, plaster), **Triangular bandage** (first aid – **arm sling**, head, hand, foot), **Many-tailed bandage/ binder** (abdomen, chest – post-op, **scultetus**), **T-bandage/ binder** (**perineum** – hold perineal dressing; double-T for male), **Breast binder**, **Abdominal binder**, **Tubular (stockinette)**, **Elastic (crepe) bandage**, **Plaster of Paris**.
- **Parts of roller bandage**: **initial (free) end, body, terminal end (drum/ roll)**; **head** = rolled part.
- **Turns**: **Circular** (anchor – start & finish; wrist, forehead), **Spiral** (cylindrical – finger, upper arm, trunk), **Reverse spiral** (cone-shaped – forearm, leg), **Figure-of-eight** (joints – elbow, knee, ankle), **Spica** (figure-8 around a **joint + body part** – shoulder, hip, thumb, groin), **Recurrent** (**stump, head (capeline), fingertip**).

Rules of bandaging

- Part in **functional/ natural position**, clean & dry, **pad bony prominences** & skin surfaces (axilla, between fingers).
- **Bandage roll held upwards (facing outward/ upwards)**, unroll **on the skin** (not in air) – even tension.
- **Start distal → proceed proximal** (venous return).
- **Begin & end with 2 circular turns** (anchor).
- **Each turn overlaps ⅔ (or ½) of previous** width.
- **Leave fingertips/ toes exposed** – check circulation (**CSM – colour, sensation, movement**; capillary refill) **every 1–2 hrs (15 min initially)**; numbness, tingling, pallor, cyanosis, coldness, pain → **loosen**.
- Secure with tape/ clip – **not over wound or bony prominence or inner aspect**; avoid knots on injured part.
- Elastic bandage – apply with **slight stretch**; remove & reapply every 8 hrs (or as ordered).
- Width: **finger 2.5 cm (1 inch)**, **hand/ arm 5 cm (2 inch)**, **leg 7.5–10 cm (3–4 inch)**, **trunk 10–15 cm**.

- **Triangular bandage**: base, apex, two ends; **arm sling (large/ broad sling)** – forearm fracture (hand slightly higher than elbow, **~10 cm/ 4 inch**); **St. John's (elevation) sling** – hand/ clavicle injury, hand **elevated toward opposite shoulder**; **collar & cuff** – humerus fracture; **reef knot** (flat, doesn't slip) – first aid.
- **Binders** – abdominal (post-op support, splint incision – applied **from bottom up**), **breast binder** (lactation suppression – older), **scultetus binder (many-tailed)** – applied **from bottom upward**, overlapping.

17.5 Specimen Collection – General Rules

- **Correct container**, **label at bedside** (name, ID, date, time, specimen type, ward), **requisition form**, **send immediately** (or refrigerate as directed), **standard precautions (gloves)**, sterile technique for culture, **collect before antibiotics** (cultures), adequate quantity, avoid contamination.

Specimen	Key points
Blood (venous)	Tourniquet **< 1 min**, cleanse, **order of draw: blood culture → citrate (light blue – PT/INR) → serum (red/ gold) → heparin (green) → EDTA (lavender/ purple – CBC, HbA1c) → fluoride/ oxalate (grey – glucose)**; **invert tubes gently** (don't shake – haemolysis)
Blood culture	**Before antibiotics**, **2 sets from 2 different sites**, during fever spike/ chills; **10 ml per bottle (adult)**; aerobic bottle first (with syringe); clean bottle top with alcohol
Capillary blood (glucose)	**Side of fingertip** (fewer nerve endings), **wipe first drop**; infant – **lateral/ medial heel**
ABG	**Radial artery**, **Allen's test** first, heparinized syringe, expel air bubbles, **ice**, **pressure 5 min (10 min if anticoagulants)**
Sputum	**Early morning**, **before breakfast**, **after rinsing mouth with plain water** (no toothpaste/ mouthwash), **deep cough** (not saliva), sterile container; **AFB – 2 samples (spot + early morning)**; 5–10 ml; send within 1–2 hrs

Specimen	Key points
Throat swab	Before antibiotics/ meals; **tongue depressor**, swab **tonsils & posterior pharynx** without touching tongue/ cheeks
Nasal/ nasopharyngeal swab	Head tilted back, swab to posterior nasopharynx, rotate
Wound swab	**Clean wound with NS first** (remove surface exudate), swab **viable tissue (centre) – Levine technique** (rotate 1 cm^2 with pressure), not pus/ necrotic
Urine	See Chapter 12 (MSU, catheter port, 24-hr)
Stool	See Chapter 13 (ova & parasites – warm, fresh; occult blood – diet restriction)
CSF	Lumbar puncture – 3 sterile tubes (biochem, micro, cell count) – send immediately (cells lyse)

PYQ style

- Heat causes – **vasodilation**; cold – **vasoconstriction**.
- Sprain first 24 hrs – **cold (ice)**; after 48 hrs – heat.
- Heat contraindicated – **appendicitis, first 24 hrs of injury, bleeding, malignancy**.
- Hot water bag temperature adult – **46–52 °C**; fill ⅔, expel air.
- Sitz bath – **40–43 °C, 15–20 min**.
- Tepid sponge – **27–34 °C**; stop if **shivering**.
- Wound cleaning direction – **from incision outward / clean to dirty**.
- Evisceration – **cover with sterile saline-soaked dressing**, low Fowler's knees flexed.
- Granulation tissue colour – **beefy red**; pus – **purulent**.
- Suture removal face – **3–5 days**; abdomen – **7–10 days**.
- Bandage applied – **distal to proximal**, each turn covers ⅔ of previous.
- Joint bandage – **figure of 8**; stump/ head – **recurrent**; cone-shaped limb – **reverse spiral**; shoulder/ hip – **spica**.
- Perineal dressing hold – **T-binder**.
- Blood culture – **before antibiotics, 2 sites**.
- Sputum for AFB – **early morning**, mouth rinsed with plain water.

18 First Aid & Emergency Care

18.1 Principles of First Aid

First aid

Immediate & temporary care given to an injured or suddenly ill person **before arrival of medical help**, using available resources. Aims – **3 P's: Preserve life, Prevent deterioration (worsening), Promote recovery**.

- **Golden rules**: **ensure scene safety first (self-safety)** → **DR ABC** – **D**anger, **R**esponse, **A**irway, **B**reathing, **C**irculation; call for help (**108 ambulance, 112 emergency**); **don't move casualty** unless danger (spinal injury suspicion); treat **life-threatening conditions first** (airway → breathing → bleeding → shock); reassure; **don't give anything by mouth** to unconscious/ abdominal injury/ surgery likely; keep warm; record & hand over.
- **Priority in casualty**: **(1) Airway obstruction/ not breathing**, **(2) severe bleeding**, **(3) shock**, (4) unconsciousness, (5) fractures, burns.
- **Recovery position (lateral/ semi-prone)** – **unconscious but breathing** casualty (no spinal injury) – keeps **airway open, drains vomit/ secretions**.
- **St. John Ambulance** (first aid training organization), **Indian Red Cross Society (1920)**. **World First Aid Day – 2nd Saturday of September**.

18.2 CPR / Basic Life Support (AHA)

Adult BLS – key numbers (AHA 2020/ 2025)

- **Check responsiveness** (tap & shout) & **breathing (look at chest) + carotid pulse simultaneously ≤ 10 sec**.
- **Call for help / 108, get AED**.
- Sequence **C-A-B** (Compressions → Airway → Breathing).
- **Compression**: **lower half of sternum (centre of chest, between nipples)**, **heel of one hand + other hand on top**, **arms straight (locked elbows)**, shoulders over hands.
- **Rate 100–120 /min**; **depth 5–6 cm (2–2.4 inch)**; **full chest recoil**; minimize interruptions (**< 10 sec**).
- **Ratio 30 : 2** (adult – 1 or 2 rescuers); **child/ infant 2 rescuers – 15 : 2**; 1 rescuer – 30 : 2.
- **Airway**: **head tilt–chin lift**; **jaw thrust** if spinal injury.
- **Rescue breath** – 1 sec each, visible chest rise (avoid excessive ventilation). With advanced airway – **1 breath every 6 sec (10/min)**.
- **AED** – as soon as available; **pads: right upper chest (below clavicle) & left side (below armpit)**; "**stand clear**" during analysis & shock; resume CPR immediately after shock.
- Rotate compressor **every 2 min (5 cycles)**.
- **Respiratory arrest only (pulse present)** – **1 breath every 6 sec (10/min)** adult; child/ infant – **every 2–3 sec (20–30/min)**.

	Adult	Child (1 yr – puberty)	Infant (< 1 yr)
Pulse check	**Carotid**	Carotid/ femoral	**Brachial**
Compression	2 hands	1 or 2 hands	**2 fingers / 2-thumb encircling** (just below nipple line)
Depth	**5–6 cm**	**~5 cm (⅓ AP diameter)**	**~4 cm (1.5 inch; ⅓ AP)**
Ratio (2 rescuers)	30 : 2	**15 : 2**	**15 : 2**
Lone rescuer – no phone	Call first then CPR	**CPR 2 min first then call**	**CPR 2 min first then call**

- **Chain of survival (OHCA)**: activate EMS → **high-quality CPR** → **defibrillation** → advanced resuscitation → post-arrest care → recovery.
- **Hands-only CPR** for untrained lay rescuers.
- Stop CPR when: **ROSC**, rescuer exhausted, scene unsafe, **qualified help takes over**, DNR/ death declared.

18.3 Choking (Foreign Body Airway Obstruction)

- **Universal choking sign** – **hands clutching throat**. **Mild** (can cough, speak) – **encourage coughing**, don't interfere. **Severe** (can't speak/ breathe/ cough, cyanosis, silent cough) – act.
- **Conscious adult/ child (> 1 yr)**: **Abdominal thrusts (Heimlich manoeuvre)** – stand behind, **fist (thumb side) above navel, below xiphoid**, quick **inward & upward** thrusts until object expelled or unconscious. (Some guidelines: 5 back blows + 5 abdominal thrusts.)
- **Pregnant/ obese**: **chest thrusts** (lower sternum).
- **Infant (< 1 yr)**: **5 back blows** (head lower, face down on forearm, between shoulder blades) **+ 5 chest thrusts** (2 fingers, lower sternum) – **NO abdominal thrusts** (liver injury); **no blind finger sweep**.
- **Unconscious** → lower to ground, **start CPR** (compressions); **look in mouth before breaths** – remove only if visible.
- **Self-Heimlich** – own fist or over back of chair.

18.4 Bleeding (Haemorrhage)

- Types: **Arterial** (**bright red, spurting/ pulsatile**, most dangerous), **Venous** (**dark red, steady flow**), **Capillary** (oozing, slow). **External/ internal**; **primary, reactionary (within 24 hrs), secondary (7–14 days)**.
- **Control of external bleeding**:
  1. **Direct pressure** with clean pad/ gloved hand – **most effective first step** (10–15 min continuous; **don't remove soaked pad – add more on top**)
  2. **Elevation** of injured part above heart (if no fracture)
  3. **Pressure points** (indirect – brachial, femoral) – adjunct
  4. **Tourniquet** – **life-threatening limb bleeding not controlled by direct pressure** – **5–7.5 cm (2–3 inch) above wound** (not on joint), **note time of application**, don't cover it, don't loosen (by first aider)
  5. **Haemostatic dressing**, pressure bandage
- **Embedded object – don't remove**; pad around & bandage (pressure around).

- **Epistaxis (nose bleed)**: **sit up & lean FORWARD** (not back), **pinch soft part of nose 10–15 min**, breathe through mouth, cold compress on bridge; don't blow nose.
- **Internal bleeding** signs – shock, pain, swelling, **haematemesis (coffee-ground), melena, haemoptysis (frothy bright red)**; treat for shock, urgent hospital.

18.5 Shock

- **Inadequate tissue perfusion**. Types: **hypovolaemic (haemorrhage, burns, dehydration – most common)**, cardiogenic, **distributive (septic, anaphylactic, neurogenic)**, obstructive.
- Signs: **restlessness/ anxiety (earliest)**, **pale, cold, clammy skin**, **rapid weak thready pulse**, rapid shallow breathing, thirst, **↓ BP (late)**, ↓ urine, confusion, unconsciousness.
- First aid: **lay flat, raise legs 20–30 cm (~12 inch) – shock position** (unless head/ chest/ spinal injury/ fracture of leg), **keep warm (blanket – not overheat)**, **loosen tight clothing**, control bleeding, **nothing by mouth** (moisten lips), reassure, monitor, **108**.
- **Anaphylaxis** – **adrenaline 1:1000 IM (0.5 mg adult, 0.3 mg child) in anterolateral thigh** – EpiPen; position – breathing difficulty → sitting; faint → lying with legs raised.
- **Fainting (syncope)** – temporary loss of consciousness (↓ brain blood flow): **lay flat, raise legs**, loosen clothes, fresh air; **don't sit up quickly**; if feeling faint – sit & put head between knees.

18.6 Fractures, Dislocations & Spinal Injury

- **Signs of fracture**: pain, **tenderness**, swelling, **deformity**, **loss of function**, **abnormal mobility**, **crepitus**, shortening, bruising.
- **First aid**: **immobilize "as it lies"** – **splint including joint above & below**; pad splint; check **CSM (circulation, sensation, movement) distal before & after**; **open fracture – cover with sterile dressing, don't push bone back**, control bleeding around (no direct pressure on bone end); elevate (after splinting); ice; treat shock; no food (surgery).
- **Splints** – rigid (wooden, SAM, inflatable/ air splint), soft (pillow, blanket), **anatomical splint (body part – injured leg tied to good leg)**, traction splint (femur – Thomas splint).
- **Specific**: **clavicle** – arm sling/ figure-of-8; **forearm/ wrist** – splint + arm sling; **humerus** – collar & cuff/ splint + sling; **ribs** – support arm on injured side, **no tight strapping**; **pelvis** – lie flat, legs together, padding; **femur** – traction/ long splint (heavy blood loss – shock); **skull** – head injury care.
- **Spinal injury** suspicion (fall from height, diving, RTA, neck pain, tingling/ weakness): **don't move**, **manual in-line stabilization of head**, **cervical collar**, **log roll** with 4 persons, **spine board**, **jaw thrust** for airway.
- **Dislocation** – **don't try to reduce**; immobilize in position found; ice.
- **Sprain/ strain** – **RICE** (Rest, Ice, Compression, Elevation) – first 24–48 hrs.

18.7 Burns, Scalds & Electric Shock

- **First aid for burns**: **Stop the burning** (**Stop, Drop & Roll**; smother with blanket), remove from source; **cool under cool running water for 20 min** (within 3 hrs) – **no ice, no butter/ oil/ toothpaste/ ghee/ ink**; **remove jewellery, watches, tight clothing early** (before swelling) – **don't pull off stuck clothes**; **don't break blisters**; **cover with clean non-fluffy material / cling film**; keep warm (hypothermia); fluids if conscious & minor; refer if large/ face/ hands/ genitals/ inhalation.

- **Chemical burns** – brush off dry chemical, **flush with copious water 20–30 min**; eyes – **irrigate 15–30 min (inner → outer canthus)**; remove contaminated clothing.
- **Electric shock** – **switch off power / break contact with dry non-conducting object (wood, rubber)** – **don't touch victim directly**; check ABC, **CPR**; look for **entry & exit burns**; **ECG monitoring** (arrhythmias); all electric injuries → hospital.
- **Lightning** – victim can be touched safely; "**reverse triage**" – treat apparently dead first.
- **Rule of nines & Parkland** – see MSN notes.

18.8 Poisoning, Bites & Stings

- **General**: ABC, **identify poison** (container, history) – save container/ vomitus, call **Poison Control/ 108**, **don't induce vomiting** (especially **corrosives – acids/ alkalis, kerosene/ petroleum – aspiration**), unconscious → recovery position; **activated charcoal** (hospital, within 1 hr – not for corrosives, hydrocarbons, iron, alcohol); skin contamination – remove clothes, wash; inhaled (CO/ gas) – **fresh air**, O_2.
- **Corrosive ingestion** – conscious → **sips of water/ milk** to dilute (some guidelines); no emesis.
- **Snake bite** – "**Do it R.I.G.H.T.**": **R**eassure, **I**mmobilize limb (splint like fracture, below heart level), **G**et to **H**ospital, **T**ell doctor symptoms; **DON'T: tourniquet, cut, suck, ice, electric shock, herbal remedy, alcohol**; remove rings/ tight items; **ASV (anti-snake venom) – polyvalent**.
- **Dog bite** – **wash wound with soap & running water 15 min** (most important), antiseptic (povidone-iodine), **don't suture immediately**, **anti-rabies vaccine ± RIG**, tetanus.
- **Bee/ wasp sting** – **scrape sting out** (credit card) – **don't squeeze with tweezers** (injects venom); ice, antihistamine; anaphylaxis watch.
- **Scorpion sting** – ice, immobilize, hospital (prazosin).
- **Insect in ear** – **tilt head, pour warm oil/ water** (floats out) – **not for TM perforation**; don't probe.
- **Foreign body in eye** – don't rub, **flush with water**, lift upper lid over lower; **embedded/ penetrating object – don't remove, cover both eyes** (paper cup over object).

18.9 Other Emergencies

- **Drowning** – remove safely (don't become victim), **5 initial rescue breaths** then CPR (**hypoxia main problem** – breaths important), don't try to drain water from lungs, keep warm, hospital (secondary drowning).
- **Heat stroke** – **core temp > 40 °C**, **hot dry skin**, confusion/ unconscious → **EMERGENCY: rapid cooling** (shade, remove clothes, **cold water immersion/ wet sheets + fan**, ice packs to **neck, axillae, groin**), **no antipyretics**. **Heat exhaustion** – sweating, weakness, cool clammy skin → rest in cool place, **ORS/ salt water**. **Heat cramps** – salt-water/ ORS, stretching.
- **Hypothermia** – remove wet clothes, **gradual rewarming** (blankets, warm drinks if conscious – **no alcohol**), handle gently. **Frostbite** – **rewarm in warm water 37–40 °C**, **don't rub/ massage**, don't rewarm if risk of refreezing.
- **Seizure** – **protect from injury**, cushion head, loosen clothing, **don't restrain**, **don't put anything in mouth**, time the seizure, **recovery position after**; > 5 min → emergency.
- **Hypoglycaemia** – conscious → **sugar/ glucose/ juice (15 g)**; unconscious → nothing by mouth, 108 (glucagon/ IV dextrose).
- **Asthma attack** – sit upright, reliever inhaler (salbutamol), calm.

- **Heart attack** – rest (W position/ half-sitting), loosen clothes, **aspirin 300–325 mg chewed** (if not allergic), call 108, CPR if arrest.
- **Stroke** – **BE FAST**, note time, 108, nothing by mouth.
- **Head injury** – observe consciousness (GCS/ AVPU), **assume neck injury**, recovery position if unconscious (with spinal care), bleeding from ear – **don't plug** (cover loosely – CSF), lean towards injured side.

18.10 Triage (Mass Casualty)

Colour	Priority	Example
Red	**Immediate (I)** – life-threatening but salvageable	Airway obstruction, severe bleeding, shock, tension pneumothorax
Yellow	**Delayed/ Urgent (II)** – serious but can wait	Fractures (stable), moderate burns
Green	**Minor (III) – "walking wounded"**	Sprains, minor cuts
Black	**Dead / expectant (IV)**	No breathing after airway opened, unsurvivable

- **START triage** – **R**espirations (> 30 → red), **P**erfusion (cap refill > 2 sec/ no radial pulse → red), **M**ental status (can't follow commands → red); walking → green. "**Do the greatest good for the greatest number**".

PYQ style

- Aim of first aid – **preserve life, prevent deterioration, promote recovery**.
- First step at accident scene – **ensure safety (danger)**.
- Unconscious breathing casualty – **recovery position**.
- CPR ratio adult – **30 : 2**; rate **100–120/min**; depth **5–6 cm**.
- Infant pulse check – **brachial**; infant choking – **back blows + chest thrusts**.
- Choking adult (conscious) – **Heimlich (abdominal thrusts)**.
- First method to control bleeding – **direct pressure**.
- Arterial bleeding – **bright red, spurting**.
- Shock position – **legs raised**; nosebleed – **lean forward**.
- Fracture splint – **joint above & below**.
- Burns first aid – **cool running water 20 min**.
- Kerosene/ acid poisoning – **don't induce vomiting**.
- Snake bite – **immobilize, no tourniquet/ cut/ suck**.
- Heat stroke – **rapid cooling**.
- Triage red tag – **immediate**.

19 Care of Unconscious, Peri-operative, Terminally Ill & Dead Patient

19.1 Care of the Unconscious Patient

- **Levels of consciousness**: **Alert/ conscious → Lethargic (drowsy) → Obtunded (arousable with stimulation) → Stupor/ semi-coma (only vigorous/ painful stimulus) → Coma (no response)**. **GCS 3–15** (≤ 8 = coma); **AVPU**.
- **Causes**: head injury, stroke, hypoglycaemia/ hyperglycaemia, poisoning, hypoxia, infections (meningitis), seizures, shock, uraemia, hepatic failure.

Nursing care – systems approach

- **Airway (priority 1)**: **lateral/ semi-prone (recovery) position**, head turned to side, **oropharyngeal airway (Guedel)**, **suction**, O_2; **remove dentures**; **no pillow under head** (if airway risk); jaw thrust.
- **Breathing & circulation**: monitor vitals, SpO_2, GCS, pupils (neuro observation chart – hourly/ 2-hourly).
- **Eye care**: **no blink/ corneal reflex → NS cleaning, artificial tears/ lubricant ointment, eye pads/ taping lids closed → prevent corneal ulceration/ exposure keratitis**.
- **Oral care**: every **2–4 hrs**, **side-lying**, suction ready, lip lubricant, padded tongue blade (no fingers).
- **Skin/ pressure care**: **turn every 2 hrs**, pressure-relieving mattress, skin clean & dry.
- **Positioning & ROM**: **passive ROM 2–3 times/day**, **footboard (footdrop)**, **trochanter roll**, **hand rolls**, splints – **contracture prevention**.
- **Nutrition & fluids**: **NG/ PEG feeding**, IV fluids, I/O, daily weight.
- **Elimination**: **catheter / condom catheter**, bowel care (stool softeners, suppository), monitor constipation & retention.
- **Safety**: **side rails up**, bed low, **seizure precautions**, no hot water bottles (burns), restraint only if needed.
- **Communication/ sensory stimulation**: **talk to patient, call by name, explain every procedure** (**hearing is last sense lost & first to return**), orient (date, time), touch, music, family involvement.
- **Infection prevention**: chest (pneumonia – turning, suction, chest physio), UTI, VAP bundle.
- **Eye, ear, nose, perineal, hair care**; maintain dignity & privacy.

19.2 Peri-operative Basics (FON level)

- **Pre-operative**: **informed consent** (before pre-medication), **NPO** (solids **6–8 hrs**, clear fluids **2 hrs**), **skin preparation (hair removal by clipper just before surgery)**, **bowel prep (enema)** as ordered, **bladder emptied** (void before OT), **remove jewellery, dentures, contact lenses, hairpins, nail polish, make-up, prosthesis**, hospital gown, **ID band**, allergy check, **vitals baseline**, investigations (CBC, BT/CT, blood grouping & cross-match, RFT, sugar, ECG, X-ray), **pre-medication (atropine/**

glycopyrrolate, sedative) – **side rails up after pre-med**; pre-op teaching (**deep breathing, coughing, leg exercises, incentive spirometry, early ambulation**); **pre-op checklist**.

- **Post-operative immediate**: **airway (lateral position till gag reflex returns)**, vitals **every 15 min** first hour, check **dressing & drains (bleeding)**, IV, pain, **first void within 6–8 hrs**, prevent complications – **atelectasis (first 24–48 hrs)**, shock/ haemorrhage, **DVT (early ambulation)**, paralytic ileus, wound infection (day 3–7), dehiscence. Post-spinal anaesthesia – **flat 6–8 hrs** (headache).
- **Post-op bed** – see Chapter 8.

19.3 Sensory Alterations

- **Sensory deprivation** (↓ stimuli – ICU isolation, blindness) – boredom, confusion, hallucinations. **Sensory overload** (excess stimuli – ICU noise, pain) – anxiety, confusion, **ICU psychosis/ delirium**.
- Nursing – meaningful stimulation (clock, calendar, radio, visitors), **reduce noise**, cluster care, **orientation**, glasses/ hearing aids.

19.4 Stress & Coping

- **Hans Selye – General Adaptation Syndrome (GAS)**: **(1) Alarm reaction** (fight or flight – sympathetic, adrenaline ↑) → **(2) Resistance** (adaptation) → **(3) Exhaustion** (if stress continues – illness/ death). **Local adaptation syndrome (LAS)** – inflammation, reflex pain response.
- **Eustress** (positive stress), **Distress** (negative).
- **Defence mechanisms** (Freud): denial, projection, rationalization, regression, repression (unconscious forgetting), suppression (conscious), displacement, sublimation (socially acceptable – healthiest), compensation, reaction formation, identification, intellectualization.

19.5 Loss, Grief & Dying

- **Loss** – actual/ perceived, maturational/ situational. **Grief** – emotional response to loss. **Bereavement** – state of having lost someone. **Mourning** – outward/ cultural expression.
- **Types of grief**: **Normal/ uncomplicated**, **Anticipatory** (before actual loss – terminal illness), **Complicated/ dysfunctional** (prolonged > 6–12 months, chronic, delayed, exaggerated, masked), **Disenfranchised** (not socially recognized – miscarriage, pet), **Ambiguous** (missing person).

> **Kübler-Ross – 5 Stages of Grief/ Dying (1969, "On Death and Dying")**
> **D-A-B-D-A**: **1. Denial** ("Nahi, ye mere saath nahi ho sakta" – shock, isolation) → **2. Anger** ("Mere saath hi kyun?" – blames staff/ God – **don't take personally**) → **3. Bargaining** ("Bas beti ki shaadi tak jee lun..." – deals with God) → **4. Depression** (sadness, withdrawal, crying – allow to express) → **5. Acceptance** (calm, peace, planning). Stages not always in order.
> **Nursing**: **active listening, presence, therapeutic silence, touch**, allow expression, don't argue in denial, honest information, spiritual support.

- Other models: **Engel** (shock & disbelief → developing awareness → restitution), **Bowlby** (numbness → yearning & searching → disorganization & despair → reorganization), **Worden's 4 tasks** of mourning.

Care of the dying patient

- **Palliative care (WHO)** – improves **quality of life** of patients & families facing life-threatening illness – **relief of pain & other symptoms, psychosocial & spiritual support**; neither hastens nor postpones death. **Hospice** – end-of-life (last ~6 months), home/ institution. **DNR (Do Not Resuscitate)** – documented; still give full comfort care.
- **Physical care**: pain relief (by the clock), mouth care (dry mouth), skin care, **position for comfort**, eye care, suction (secretions), elimination, nutrition (small, as desired – **don't force**), O_2 for comfort.
- **Psychosocial/ spiritual**: be present, allow family, **religious rituals** (Gangajal, Tulsi, Quran recitation, last rites – priest), honest answers, life review, fulfil wishes.
- **"Death rattle"** – noisy breathing from secretions → **side position, glycopyrrolate/ hyoscine**, gentle oral suction.

Signs of approaching death (impending death)

- **↓ level of consciousness**, increasing sleep, confusion/ restlessness (terminal agitation)
- **Cheyne-Stokes respiration**, apnoea periods, **death rattle**, mouth breathing
- **Skin cold, mottled (knees & feet first), cyanosed, clammy**; **"Hippocratic facies"** (pinched nose, sunken eyes)
- **Pulse weak, rapid & irregular; BP falls**
- **↓ urine output**, incontinence; loss of swallowing & gag reflex
- **Pupils dilated, fixed**; blurred vision; jaw sags
- **Hearing is the last sense to go** – keep talking gently

Signs of death (clinical)

- **Cessation of heartbeat (no pulse, no heart sounds)** & **respiration**, **fixed dilated pupils (no light reflex)**, **flat ECG (asystole)**, no reflexes, loss of muscle tone.
- **Brain death** – irreversible loss of all brain functions including **brainstem** (no brainstem reflexes, **apnoea test +ve**) – certified by **board of doctors** (India – **Transplantation of Human Organs & Tissues Act (THOTA) 1994** – **4 doctors** panel, tests repeated after 6 hrs).
- **Post-mortem changes**:
  - **Pallor mortis** – paleness (15–30 min)
  - **Algor mortis** – **cooling of body** (~1 °C/hr, ~1.5 °F/hr) till room temp
  - **Livor mortis (post-mortem lividity/ hypostasis)** – **purplish discolouration in dependent parts** (blood settles by gravity) – starts **20 min–2 hrs**, **fixed after 6–8 hrs (12 hrs)** → **place body supine with pillow under head** (prevent face discolouration)
  - **Rigor mortis** – **stiffening of muscles** (ATP depletion) – starts **2–4 hrs** after death, **begins in small muscles – eyelids/ jaw/ face/ neck** (involuntary first – **heart**), spreads downward, **complete in ~12 hrs**, disappears in **24–48 (36) hrs** (same order) → **position body before rigor sets in** (close eyes, mouth, limbs straight)
  - **Putrefaction/ decomposition** – after 24–48 hrs (first sign – **greenish discolouration of right iliac fossa**)

19.6 Last Office (Care of the Dead Body)

Steps of last office

1. **Death declared & certified by doctor** (time noted); inform relatives (doctor) & nursing in-charge; **respect religious/ cultural beliefs** – ask family.
2. Hand hygiene, **standard precautions (gloves, apron)**; screen/ privacy.
3. **Place body supine, flat**, arms by sides (or across abdomen), **one pillow under head & shoulders** (prevent pooling of blood in face – **livor mortis**).
4. **Close eyes** (gently press lids **for 30 sec**; moist cotton pads if needed); **close mouth** – **insert dentures** (shape of face) – support jaw with **rolled towel/ bandage under chin**.
5. **Remove tubes, drains, catheters, IV lines** (cut & cap or as per policy) & dress wounds – **EXCEPT in medico-legal cases (MLC)/ post-mortem cases** – **leave all tubes in place**, cut close & tape.
6. **Plug orifices** (mouth, nose, ears, rectum, vagina) with **cotton** as per policy to prevent leakage (some policies now discourage – follow hospital).
7. **Clean the body** (sponge), comb hair, remove jewellery (**hand over valuables to relatives with witness & document**; wedding ring may be left – taped, as per family wish).
8. **Clean gown/ shroud**; **identification tags** – **wrist, ankle (big toe) & outside shroud** (name, age, sex, ID, date & time of death, ward).
9. Wrap in **shroud/ mortuary sheet**; **infectious cases – "biohazard/ danger of infection" label** & body bag.
10. Allow family to view & grieve before shifting; **shift to mortuary** (or hand over to relatives) **within 1–2 hrs** (before rigor complete); **death certificate (Form 4/ 4A – MCCD)**, **death report/ register**, documentation; **terminal cleaning of unit**.

- **Medico-legal case (MLC)** – unnatural death (accident, poisoning, burns, suicide, assault, sudden unexplained death, unknown/ unidentified body) → **inform police**, **body for post-mortem (autopsy)**, **don't remove tubes/ lines**, preserve clothes/ evidence, no last office cleaning beyond minimal.
- **Autopsy (post-mortem)** – clinical (consent of relatives) vs medico-legal (no consent needed – police).
- **Organ & tissue donation**: **eye (cornea) donation – within 6 hrs** of death (keep **eyes closed with moist cotton, head raised on pillow, fans off, AC on**, ice packs on lids); **skin – within 6 hrs**; organs (heart, lungs, liver, kidneys) – only from **brain-dead (heart-beating) donors**; **NOTTO** (National Organ & Tissue Transplant Organisation); **3 August = Indian Organ Donation Day** (2023 se; pehle 27 November); **13 August = World Organ Donation Day**; eye donation fortnight **25 Aug – 8 Sept**.
- **Euthanasia** – active (illegal in India), passive (allowed with safeguards – SC 2011, 2018); **living will** valid.

PYQ style

- Unconscious patient position – **lateral/ semi-prone (recovery)**.
- Last sense to be lost in dying patient – **hearing**.
- First stage of Kübler-Ross – **denial**; last – **acceptance**; "if only..." – **bargaining**.
- Rigor mortis starts – **2–4 hrs** after death, begins in **jaw/ eyelids (small muscles)**.
- Livor mortis – **purple discolouration of dependent parts** → pillow under head.
- Algor mortis – **cooling**.
- Dentures in dead body – **insert (maintain face shape)**.
- MLC body – **tubes not removed**.
- Identification tags on dead body – **wrist, toe & shroud**.
- Eye donation – **within 6 hrs**.
- GAS – **Hans Selye**; stages – **alarm, resistance, exhaustion**.

20 Quick Revision – Master Tables & 150 One-Liners

20.1 Master Table – Numbers yaad karo

Topic	Value
Normal oral temperature	**37 °C / 98.6 °F** (36.5–37.5)
Hyperpyrexia / Hypothermia	**> 41 °C / < 35 °C**
Pulse / RR / BP (adult)	**60–100 / 12–20 / < 120/80**
Newborn pulse / RR	**120–160 / 30–60**
Oral / Axillary / Rectal thermometer time	**3 / 5–10 / 2–3 min**
Rectal thermometer insertion (adult)	**3.5 cm (1.5 inch)**
BP cuff bladder width	**40 % arm circumference**; 2.5 cm above antecubital fossa
Bath water temperature	**43–46 °C (110–115 °F)**
Hot water bag (adult/ child-elderly)	**46–52 °C / 40.5–46 °C**, fill ⅔
Sitz bath	**40–43 °C, 15–20 min**
Tepid sponge	**27–34 °C**
Enema solution temperature	**40.5–43 °C** (child 37.7 °C)
Enema can height	**30–45 cm (12–18 inch)**
Rectal tube insertion adult / child / infant	**7.5–10 / 5–7.5 / 2.5–3.75 cm**
Enema volume adult (cleansing)	**750–1000 ml**
Oil retention / Fleet enema	**90–150 ml / 120 ml**
Flatus tube max time	**20–30 min**
Female / Male catheter insertion	**5–7.5 cm / 17–22.5 cm**
Foley balloon	**5–10 ml sterile water**
Catheter size female / male	**12–14 / 14–16 Fr**
Normal urine output	**1–1.5 L/day; ≥ 30 ml/hr**
Oliguria / Anuria / Polyuria	**< 400 / < 100 ml/day / > 3 L/day**
Urine specific gravity	**1.005–1.030**
Bladder capacity	**300–600 ml**
NG tube length (NEX)	**~50–55 cm (45–55)**
NG aspirate pH	**≤ 5**
Feed syringe height	**15–30 cm**
Flush NG tube	**30 ml water**
Nasal cannula flow / FiO_2	**1–6 L / 24–44 %**

Topic	Value
Simple mask / NRBM / Venturi	**5–10 L / 10–15 L / precise 24–60 %**
Suction pressure adult / child / infant	**100–150 / 80–100 / 60–80 mmHg**
Suction time per pass	**10–15 sec, max 3 passes**
Tracheostomy cuff pressure	**20–25 mmHg**
Steam inhalation water	**~60–70 °C**, distance 23–30 cm, 15–20 min
Injection angles ID / SC / IM / IV	**10–15° / 45–90° / 90° / 25°**
Deltoid max volume	**1 ml** (2–3 finger-widths below acromion)
Z-track skin pull	**2.5–3.5 cm**, wait 10 sec
IV tourniquet	**10–15 cm above site, < 1–2 min**
Peripheral cannula change	**72–96 hrs**
Macro / micro drip factor	**15 or 20 / 60 gtt/ml**
Blood transfusion – complete within	**4 hrs**; first **15 min** observe
Blood storage / platelets / FFP	**2–6 °C / 20–24 °C (5 days) / −30 °C (1 yr)**
Autoclave	**121 °C, 15 psi, 15–20 min**
Hot air oven	**160 °C, 2 hrs**
Boiling (disinfection)	**100 °C, 10–20 min**
Glutaraldehyde 2 %	**HLD 20 min; sterilization 10 hrs**
Hand wash / hand rub	**40–60 sec / 20–30 sec**
Surgical scrub	**2–6 min** (traditional 5 min)
Sodium hypochlorite – blood spill	**1 %**
Alcohol – antiseptic	**70 %**
Turning schedule	**every 2 hrs (chair – 1 hr)**
Braden score at risk / range	**≤ 18 / 6–23**
Crutch – below axilla	**2–3 finger (5 cm)**, elbow 20–30°
Base of support (feet apart)	**~30 cm**
CPR rate / depth / ratio	**100–120 / 5–6 cm / 30 : 2**
Infant CPR depth	**4 cm**; 2 rescuers **15 : 2**
Pulse check (CPR)	**≤ 10 sec**
Suture removal face / abdomen / legs	**3–5 / 7–10 / 10–14 days**
Rigor mortis	Starts **2–4 hrs**, full **12 hrs**, passes **24–48 hrs**
Eye donation	**within 6 hrs**
BMI normal (Asian)	**18.5–22.9**
Waist risk (Asian)	**M > 90, F > 80 cm**
Distance between beds	**3–6 ft (1–2 m)**
Ward temperature	**20–24 °C**

Topic	Value
Calories – CHO / protein / fat	**4 / 4 / 9 kcal/g**
Paracetamol max adult	**4 g/day**

20.2 Master Table – Positions

Procedure / Condition	Position
Enema, suppository, rectal temp	**Left lateral (Sims')**
Female catheterization, perineal care	**Dorsal recumbent**
Vaginal exam, Pap smear, delivery, cystoscopy	**Lithotomy**
Sigmoidoscopy, rectal/ prostate exam	**Knee-chest**
Dyspnoea, heart failure, NG insertion & feeding	**Fowler's / High Fowler's**
Severe dyspnoea/ asthma/ COPD	**Orthopnoeic / tripod**
↑ICP, after head surgery (supratentorial)	**Semi-Fowler's (30°)**
Shock	**Modified Trendelenburg (legs up)**
Air embolism	**Left lateral + Trendelenburg**
Unconscious patient, oral care of unconscious	**Lateral/ recovery (semi-prone)**
Post-tonsillectomy	**Semi-prone/ prone, head to side**
Post spinal anaesthesia	**Flat/ supine 6–8 hrs**
After lumbar puncture	**Flat/ supine 4–12 hrs**
Lumbar puncture	**Lateral, knees to chest (fetal)** / sitting bent
Thoracentesis	**Sitting, leaning forward on over-bed table**
Liver biopsy (after)	**Right lateral**
Abdominal examination	**Supine, knees slightly flexed**
Postural drainage – lower lobes	**Trendelenburg**
Spinal injury moving	**Log rolling**
Wound evisceration	**Low Fowler's, knees flexed**
Epistaxis	**Sitting, leaning forward**
Hip replacement	**Abduction, no flexion > 90°**
Above-knee amputation (after 24 hrs)	**Prone periods** (hip flexion contracture prevention)
Back care	**Prone / lateral**
Pregnancy (late), supine hypotension	**Left lateral**

20.3 Master Table – Persons & Years

Person / Event	Remember
Florence Nightingale	Born **12 May 1820**; Crimean war **1854**; school **1860** St. Thomas'; "Notes on Nursing" **1859**
Virginia Henderson	**14 basic needs**, definition of nursing (ICN)
Hippocrates	Father of medicine
Semmelweis	Hand washing (**1847**)
Joseph Lister	Antiseptic surgery – **carbolic acid** (1867)
Louis Pasteur	Germ theory, rabies vaccine, pasteurization
Robert Koch	TB bacillus (**1882**), postulates
Henry Dunant	Red Cross (**1863**)
Clara Barton	American Red Cross (1881)
Lydia Hall	"Nursing process" (1955), Care-Core-Cure
Maslow	Hierarchy of needs (**1943**)
Peplau	Interpersonal relations (1952)
Orem	Self-care deficit
Roy	Adaptation model
Leininger	Transcultural nursing
Benner	Novice to expert
Kübler-Ross	5 stages of grief (**1969**)
Hans Selye	General Adaptation Syndrome
Melzack & Wall	Gate control theory (**1965**)
Lawrence Weed	POMR/ SOAP
Korotkoff	BP sounds (1905)
Riva-Rocci	Mercury sphygmomanometer
INC	Act **1947**
TNAI	**1908**
ICN	**1899** (Geneva)
First hospital India	**Fort St. George, Madras (1664)**
First school of nursing India	**Govt. General Hospital, Madras (1871)**
First B.Sc Nursing	**1946** – Delhi & Vellore
BMW Rules	**2016** (amended 2018, 2019)

20.4 Master Table – Colours

Item	Colour
O_2 cylinder (India)	**Black body, white shoulder**
N_2O cylinder	**Blue**
CO_2 cylinder	**Grey**
BMW – anatomical/ soiled	**Yellow**
BMW – contaminated plastic	**Red**
BMW – sharps	**White translucent**
BMW – glassware	**Blue**
IV cannula 14 / 16 / 18 / 20 / 22 / 24 G	**Orange / Grey / Green / Pink / Blue / Yellow**
Allergy band	**Red**
Fall risk band	**Yellow**
EDTA tube (CBC) / Citrate (PT) / Fluoride (sugar) / Plain (serum)	**Lavender / Light blue / Grey / Red**
Triage	**Red – immediate, Yellow – delayed, Green – minor, Black – dead**
Granulation tissue	**Beefy red**
Melena	**Black tarry**
Obstructive jaundice stool	**Clay**
Rifampicin urine	**Orange-red**

20.5 150 One-Liners – Exam se pehle zaroor padho

One-liners 1–50

1. Nursing process first step – **Assessment**.
2. Best/ primary source of data – **patient**; patient ki feelings – **subjective data**.
3. "Related to" in nursing diagnosis – **etiology**.
4. Risk nursing diagnosis has – **2 parts**.
5. Highest Maslow need – **self-actualization**.
6. First priority in any patient – **airway**.
7. Goal must be – **SMART**.
8. Independent nursing action example – **turning/ positioning, health teaching**.
9. Founder of modern nursing – **Florence Nightingale**.
10. Lady with the lamp – **Florence Nightingale**.
11. International Nurses Day – **12 May**.
12. Nightingale Pledge written by – **Lystra Gretter (1893)**.
13. INC established under Act of – **1947**.
14. Nurse's role protecting patient's rights – **advocate**.
15. Doing no harm – **non-maleficence**.
16. Telling truth – **veracity**; keeping promise – **fidelity**.
17. Patient's right to decide – **autonomy**.
18. Touching without consent – **battery**; threat – **assault**.
19. Unnecessary restraint – **false imprisonment**.
20. Written defamation – **libel**; spoken – **slander**.
21. "Thing speaks for itself" – **res ipsa loquitur**.
22. Consent explained by – **doctor**; nurse is – **witness**.
23. Minimum age for consent – **18 years**.
24. Error in record – **single line, "error", sign**.
25. Incident report – **not in patient's chart**.
26. Best way to teach a skill – **demonstration & return demonstration**.
27. Non-therapeutic response – **"Don't worry" (false reassurance)**, "Why...?"
28. Intimate zone – **0–18 inch**.
29. Discharge planning begins – **at admission**.
30. Patient identification – **2 identifiers (not room number)**.
31. Restraint check – **every 15–30 min**; release – **every 2 hrs**.
32. Fire safety – **RACE**; extinguisher – **PASS**.
33. Electrical fire – **CO_2/ dry powder, no water**.
34. Thermoregulatory centre – **hypothalamus**.
35. Most accurate temperature – **rectal**; safest – **axillary**.
36. Rectal temp contraindicated – **MI, diarrhoea, rectal surgery, neonates**.
37. Oral temp wait after hot/ cold drink/ smoking – **15–30 min**.
38. Fever returning to normal daily – **intermittent (malaria)**.
39. Sudden fall of fever – **crisis**; gradual – **lysis**.
40. Heat loss by fan – **convection**; ice pack – **conduction**; sweating – **evaporation**.

41. Apical pulse site – **left 5th ICS, MCL**.
42. Pulse deficit – **apical − radial (AF)**.
43. Cheyne-Stokes – **dying/ HF/ ↑ICP**.
44. Kussmaul – **DKA (metabolic acidosis)**.
45. Stridor – **upper airway obstruction (inspiratory)**.
46. Cuff too narrow – **false high BP**.
47. 1st Korotkoff – **systolic**; 5th – **diastolic**.
48. Orthostatic hypotension – **SBP fall ≥ 20 / DBP ≥ 10** on standing.
49. SpO_2 falsely high – **CO poisoning**.
50. Most reliable pain indicator – **patient's self-report**.

One-liners 51–100

51. Physical exam first technique – **inspection**.
52. Abdomen exam order – **I, A, P, P**.
53. Temperature palpated with – **dorsum of hand**.
54. Percussion over normal lung – **resonant**; liver – **dull**; stomach – **tympanic**.
55. Bell of stethoscope – **low-pitched sounds**.
56. Muscle power against gravity only – **grade 3**.
57. Normal capillary refill – **< 2–3 sec**.
58. Clubbing angle – **> 180°**.
59. Most effective HAI prevention – **hand hygiene**.
60. Most common HAI – **UTI**.
61. Sterilization – kills **spores** also; disinfection – **not spores**.
62. Autoclave biological indicator – **Geobacillus stearothermophilus**.
63. Heat-sensitive items – **ethylene oxide**.
64. Disposable syringes – **gamma radiation**.
65. Endoscope – **2 % glutaraldehyde**.
66. Sharps – **white translucent puncture-proof** container.
67. Needle recapping – **never (one-hand scoop if must)**.
68. Needle-stick first action – **wash with soap & water** (don't squeeze).
69. HIV PEP start – **within 2 hrs, max 72 hrs, 28 days**.
70. Airborne precaution – **TB, measles, chickenpox (N95, negative pressure)**.
71. PPE donning last – **gloves**; doffing first – **gloves**.
72. Sterile field border – **1 inch unsterile**.
73. Below waist level – **unsterile**.
74. Sterile pack first flap – **away from nurse**.
75. Eye cleaning – **inner to outer canthus**.
76. Perineal care (female) – **front to back**.
77. Unconscious oral care – **lateral position, suction ready**.
78. Long gliding massage stroke – **effleurage**.
79. Bed after discharge – **closed bed**.
80. Bed cradle – **keeps linen off body**.
81. Footdrop prevention – **footboard**.
82. External hip rotation prevention – **trochanter roll**.
83. Head lice – **permethrin 1 %**; repeat **7–10 days**.
84. Deep sleep with GH release – **N3 (delta)**.
85. Dreaming sleep – **REM**.
86. Gate control theory – **Melzack & Wall**.
87. Opioid side effect without tolerance – **constipation**.
88. Pain scale 3–7 yr child – **Wong-Baker FACES**.
89. Centre of gravity – **S2**.
90. Lifting – **bend knees, not back**; load **close to body**.
91. Most common site bed sore – **sacrum**.
92. Stage 2 pressure ulcer – **blister/ partial thickness**.

93. Isometric exercise – **no joint movement** (quadriceps setting).
94. Passive ROM – **maintains joint mobility** (not strength).
95. Footdrop = **plantar flexion**.
96. 4-point gait – **most stable**.
97. Crutch weight bearing – **hands**.
98. Cane held – **opposite to affected leg**.
99. Stairs up – **good leg first**; down – **bad leg (with crutches) first**.
100. Wheelchair transfer – **unaffected side, wheels locked**.

One-liners 101–150

101. Richest Vit C – **amla**; night blindness – **Vit A**.
102. Pellagra 3 D's – **niacin (B3)**.
103. Beriberi – **thiamine (B1)**; scurvy – **Vit C**.
104. Kwashiorkor – **protein deficiency with oedema**.
105. First post-op diet – **clear liquid**.
106. Hepatic encephalopathy – **low protein**; burns – **high protein**.
107. NG length – **NEX**; placement – **X-ray**.
108. NG feeding position – **semi/ high Fowler's**, 30–60 min after.
109. Tube feeding most common complication – **diarrhoea**.
110. TPN abrupt stop – **hypoglycaemia** → D10W.
111. Normal urine output – **≥ 30 ml/hr**.
112. 24-hr urine – **discard first morning sample**.
113. Culture from catheter – **sampling port, sterile syringe**.
114. Foley balloon – **sterile water**.
115. Catheter in vagina by mistake – **leave it, insert new**.
116. Urine bag – **below bladder**.
117. Retention first measure – **privacy, running water, warm water on perineum**.
118. Stress incontinence – **Kegel**.
119. Enema position – **left lateral (Sims')**.
120. Cramps during enema – **lower the can/ stop, deep breath**.
121. Oil retention enema – **soften stool, retain 30 min–3 hrs**.
122. Melena – **upper GI bleed**; clay stool – **obstructive jaundice**.
123. Valsalva avoided in – **cardiac, ↑ICP, eye surgery**.
124. Earliest hypoxia sign – **restlessness**; late – **cyanosis**.
125. Cyanosis when reduced Hb > **5 g/dL**.
126. COPD O_2 – **low flow/ Venturi, SpO_2 88–92 %**.
127. O_2 supports – **combustion** (no smoking).
128. Suction applied – **while withdrawing**.
129. Incentive spirometry – **inhale slowly, hold 3–5 sec**.
130. Postural drainage – **before meals**.
131. Blood compatible fluid – **0.9 % NS**.
132. Infiltration – **cool, pale, swollen**; phlebitis – **red, warm, tender**.
133. Air embolism – **left side, head down**.
134. Fluid overload – **high Fowler's, slow IV**.
135. Drops/min – **volume × DF ÷ minutes**.
136. Transfusion reaction first – **stop transfusion**.
137. "p.c." – **after meals**; "h.s." – **bedtime**.
138. 5 rights – **patient, drug, dose, route, time**.
139. ID angle – **10–15°**; Mantoux read – **48–72 hrs**.
140. Safest IM site – **ventrogluteal**; infant – **vastus lateralis**.
141. Iron dextran – **Z-track**.
142. Insulin mix – **clear before cloudy**.

143. Eye drops – **lower conjunctival sac**.
144. Ear drops adult – **up & back**; child < 3 – **down & back**.
145. Young's rule – **Age ÷ (Age + 12) × adult dose**.
146. Heat in first 24 hrs of injury – **contraindicated** (use cold).
147. Bandage – **distal to proximal**; joint – **figure-of-8**.
148. Evisceration – **sterile saline-soaked dressing**.
149. CPR – **30 : 2, 100–120/min, 5–6 cm**; infant choking – **back blows + chest thrusts**.
150. Kübler-Ross first stage – **denial**; last sense lost – **hearing**.



Exam strategy – Asha Didi ki last advice

- **GSSSB paper me Fundamentals se direct + priority dono type ke questions aate hain.** Numbers (temperature, angles, lengths, volumes) ki tables **roz 1 baar** padho.
- Priority question: **Safety → ABC → Assessment pehle → Least invasive pehle → Patient ko akela mat chhodo**.
- "**First / immediate / priority / best / most appropriate**" words underline karo.
- **"Except / NOT / Incorrect"** wale questions dhyan se – ek line me galat option dhoondo.
- MCQ Bank PDF ke saare questions **2 baar** solve karo; galti wale questions alag likho aur unke notes is PDF me dobara padho.
- Negative marking (0.25) – 50:50 tak pahunch ke hi guess karo.

— Fundamentals of Nursing · GSSSB Staff Nurse Preparation · Padhte raho, revise karte raho! —